

The Viable But Non-Culturable (VBNC) State in Bacteria: A Comprehensive Guide to Mechanisms, Detection, and Clinical Implications

Generated at: 2026-03-31
Source URL: <https://www.bacteriastudy.com/posts/the-viable-but-nonculturable-vbnc-state-in-bacteria-a-comprehensive-guide-to-mechanisms-detection-and-clinical-implications>

The Viable But Non-Culturable (VBNC) State in Bacteria: A Comprehensive Guide to Mechanisms, Detection, and Clinical Implications

Abstract

This article provides a comprehensive analysis of the bacterial Viable But Non-Culturable (VBNC) state, a dormant survival strategy with profound implications for public health, clinical diagnostics, and drug development. We explore the fundamental biological characteristics and induction mechanisms of VBNC cells, detail advanced methodological approaches for their detection and quantification, address critical challenges in distinguishing VBNC states from similar phenomena, and evaluate their validated role in antibiotic resistance and virulence retention. Synthesizing current research, this review aims to equip researchers and pharmaceutical professionals with the knowledge to develop novel strategies to counteract the risks posed by these elusive bacterial populations.

Unveiling the Stealth Survival Strategy: Fundamentals of the VBNC State

The **Viable but Non-Culturable (VBNC) state** represents a fundamental survival strategy adopted by numerous bacterial species when confronted with unfavorable environmental conditions. Cells in this state are characterized by a profound contradiction: they are **metabolically active** and maintain membrane integrity, yet **cannot form colonies** on conventional growth media typically used for their cultivation [1] [2]. This phenomenon challenges the traditional microbiological paradigm that equates culturability with viability, necessitating a revision of how we define and detect living bacteria.

First proposed in 1982, the VBNC concept has since been identified in over 100 bacterial species across diverse genera [3] [4]. This state is not a precursor to cell death but rather a **dormancy strategy** that allows bacteria to persist through stressful conditions while retaining the potential to resuscitate when conditions improve [5]. The study of the VBNC state has significant implications for public health, food safety, clinical microbiology, and environmental science, as standard culture-based methods fail to detect these potentially pathogenic organisms [2].

Defining Characteristics and Molecular Basis

Diagnostic Criteria for the VBNC State

The VBNC state is defined by three primary characteristics that distinguish it from other physiological states such as dormancy, persistence, or cell death:

- Loss of Culturability:** The inability to form colonies on standard laboratory media that normally support growth of the bacterium, despite maintaining viability [4].
- Metabolic Activity:** Continued, albeit reduced, metabolic processes evidenced by ATP production, nutrient transport, respiration, and gene expression [1] [2].
- Resuscitation Capability:** The capacity to return to a culturable state upon removal of the inducing stressor or application of specific resuscitation signals [5] [2].

Distinguishing VBNC from Related States

The VBNC state is often confused with other non-growing bacterial states. The table below clarifies the key distinctions:

Table 1: Differentiation Between VBNC State and Related Physiological States

Physiological State	Culturability	Metabolic Activity	Resuscitation Conditions	Primary Inducers
VBNC	Lost	Measurably active	Requires specific stimuli (often different from original growth conditions)	Multiple environmental stresses

Physiological State	Culturability	Metabolic Activity	Resuscitation Conditions	Primary Inducers
Dormancy	Variable	Below detection limit	Restoration of permissive conditions	Nutrient limitation
Persister Cells	Retained	Reduced	Removal of antibiotic	Antibiotic treatment
Dead Cells	Lost	None	Not possible	Lethal damage

Cellular and Molecular Adaptations

Upon entering the VBNC state, bacteria undergo significant morphological and physiological transformations:

- **Reduced Cell Size:** Cells typically become smaller and spherical in shape [1].
- **Metabolic Downregulation:** Drastic reduction in nutrient transport, respiration rates, and macromolecular synthesis [1].
- **Membrane Modifications:** Changes in fatty acid composition and increased cross-linking in peptidoglycan layers [2].
- **Gene Expression Continuity:** Despite metabolic reduction, VBNC cells maintain selective gene expression, including continued transcription of virulence genes in pathogens [2] [4].
- **Enhanced Stress Resistance:** VBNC cells demonstrate increased resistance to antibiotics, disinfectants, and additional environmental challenges [6].

The following diagram illustrates the transition between cellular states and the defining characteristics of the VBNC state:

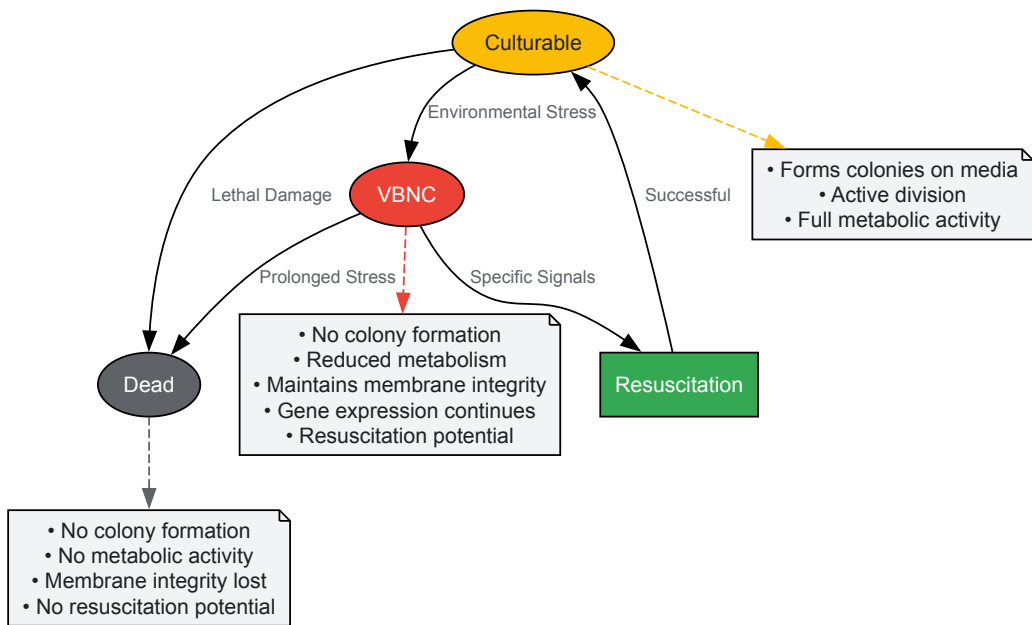


Diagram 1: Bacterial State Transitions and VBNC Characteristics

Environmental Inducers of the VBNC State

Multiple environmental stressors can trigger the transition into the VBNC state. The table below summarizes major inducing factors and their effects on various bacterial species:

Table 2: Environmental Factors Inducing the VBNC State in Various Bacterial Species

Inducing Factor	Example Organisms	Experimental Conditions	Time to VBNC
Nutrient Starvation	<i>Vibrio cholerae</i> , <i>E. coli</i>	Artificial seawater, minimal media	Days to weeks
Temperature Extremes	<i>Campylobacter jejuni</i> , <i>Listeria monocytogenes</i>	4°C incubation	15-30 days
Oxidative Stress	<i>Pseudomonas aeruginosa</i>	Chlorine (1-3 mg/L)	5-30 minutes
Osmotic Stress	<i>Salmonella</i> Oranienburg	High NaCl concentrations	48 hours
Acidic Conditions	<i>Campylobacter jejuni</i>	pH 4.0	2 hours

Inducing Factor	Example Organisms	Experimental Conditions	Time to VBNC
UV Radiation	<i>Pseudomonas aeruginosa</i>	245 nm wavelength	1-30 minutes
Food Preservatives	<i>Listeria monocytogenes</i>	Potassium sorbate (pH 4.0)	24 hours
Heavy Metals	<i>E. coli</i>	Copper, cadmium solutions	Hours to days

Resuscitation from the VBNC state occurs when appropriate conditions are restored or specific signals are received:

- **Temperature Upshift:** Increasing temperature from stressful to permissive levels can trigger resuscitation [5].
- **Nutrient Supplementation:** Addition of specific nutrients, sometimes including reactive oxygen species neutralizers like sodium pyruvate [5].
- **Host Passage:** VBNC pathogens often resuscitate upon entering appropriate host environments [2].
- **Quorum Sensing Signals:** Some evidence suggests cell-to-cell communication molecules may facilitate resuscitation [4].
- **Removal of Stressors:** Simply eliminating the inducing stressor may allow resuscitation in some cases [7].

The complexity of resuscitation is highlighted by findings that VBNC cells induced by different stressors may exhibit varying resuscitation capabilities. For example, UV-induced VBNC *P. aeruginosa* resuscitated more quickly than those induced by chlorine disinfection [7].

Detection Methodologies and Technical Approaches

Limitations of Conventional Culture Methods

Traditional microbiological detection relying on colony formation on solid media fails completely to detect VBNC cells, leading to significant underestimation of viable bacterial populations in clinical, environmental, and food samples [2]. This limitation poses serious challenges for public health protection, as pathogens in the VBNC state retain virulence and can resuscitate to cause infections [2] [4].

Advanced Detection Techniques

Multiple culture-independent approaches have been developed to detect and quantify VBNC cells:

Viability Staining and Flow Cytometry

This approach utilizes fluorescent dyes that differentiate between cells with intact and compromised membranes:

- **SYTO 9 and Propidium Iodide (PI):** The LIVE/DEAD BacLight bacterial viability kit stains all cells with SYTO 9 (green fluorescence), while PI (red fluorescence) penetrates only cells with damaged membranes [8].
- **Methodology:** Stained cells are analyzed by flow cytometry to distinguish between viable (SYTO 9 positive only) and dead (PI positive) populations.
- **Limitations:** This method can overestimate viability in complex matrices like food samples or process wash water due to interference from particulate matter [8].

Viability PCR (v-PCR) Techniques

Molecular methods combined with viability markers represent the most advanced approach for VBNC detection:

- **PMA (Propidium Monoazide) and PMAx:** These DNA-intercalating dyes penetrate cells with compromised membranes and covalently bind to DNA upon photoactivation, preventing PCR amplification [8] [9].
- **EMA (Ethidium Monoazide):** Similar to PMA but with higher cytotoxicity toward viable cells, potentially leading to false positives [8].
- **Optimized Protocol:** Recent studies demonstrate that combining EMA (10 µM) and PMAx (75 µM) with incubation at 40°C for 40 minutes followed by 15-minute light exposure effectively inhibits PCR amplification from dead cells while allowing detection of VBNC cells [8].

Molecular Quantification Methods

- **qPCR (Quantitative PCR):** Following PMA treatment, qPCR targets specific genes to quantify viable cells [9] [7].
- **ddPCR (Droplet Digital PCR):** This emerging technology provides absolute quantification without standard curves by partitioning samples into thousands of nanoliter droplets [9] [10]. Recent studies have optimized PMA concentration (5-200 µM) and incubation time (5-30 minutes) for accurate VBNC cell quantification [9].

The following diagram illustrates the integrated experimental workflow for VBNC state detection and quantification:

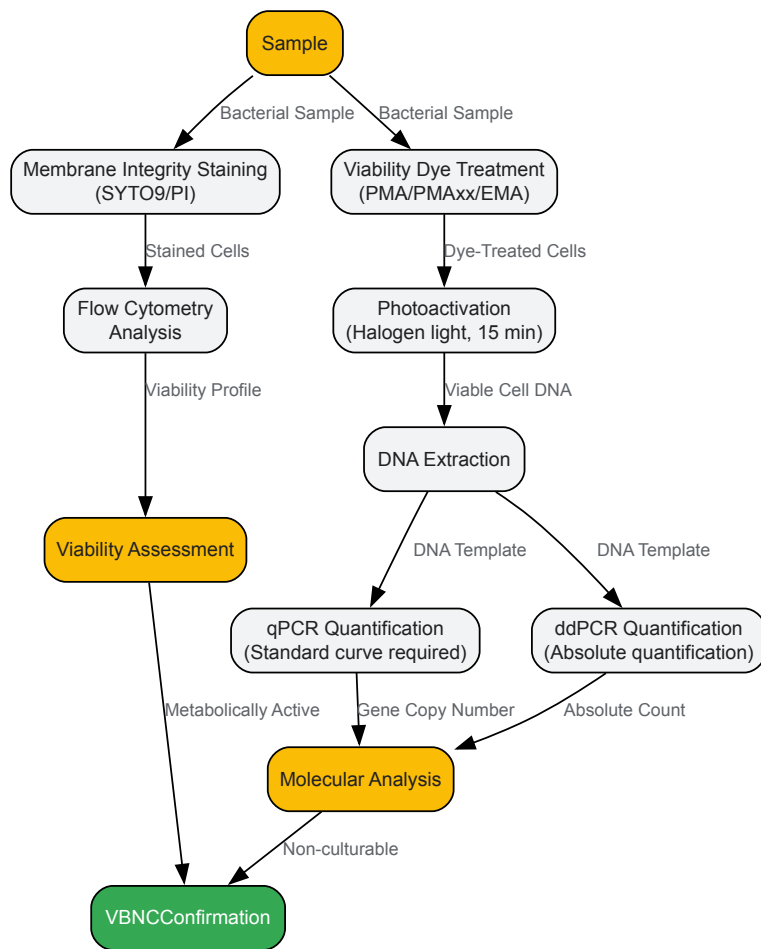


Diagram 2: VBNC Detection Methodological Workflow

The Scientist's Toolkit: Essential Research Reagents

Table 3: Essential Research Reagents for VBNC State Investigation

Reagent/Category	Specific Examples	Function/Application	Technical Considerations
Viability Dyes	PMA, PMAxx, EMA	Differentiate cells based on membrane integrity	PMAxx shows improved performance over PMA; EMA may cause false positives
Nucleic Acid Stains	SYTO 9, Propidium Iodide (PI)	Membrane integrity assessment via fluorescence	Used in combination for flow cytometry
Culture Media	LB broth/agar, BHI broth, ASW	Culturability assessment and resuscitation studies	Artificial Sea Water (ASW) used for VBNC induction
PCR Reagents	Primers (rpoB, adhE, KP), polymerases	Gene-targeted quantification of viable cells	Target single-copy genes for accurate quantification
Disinfection Agents	Sodium hypochlorite, peracetic acid	Inducing VBNC state in experimental systems	Sodium thiosulfate used to quench disinfectants
Antibiotics	Ciprofloxacin, polymyxin	Studying VBNC state resistance and resuscitation inhibition	Ciprofloxacin inhibits VBNC cell recovery

Research Implications and Future Directions

The recognition of the VBNC state has profound implications across multiple scientific disciplines:

- **Clinical Microbiology:** VBNC pathogens may explain recurrent infections and culture-negative diagnostic results, particularly in cases like urinary tract infections where VBNC *E. coli* demonstrates antibiotic resistance [2].

- **Food Safety:** Routine food testing may fail to detect VBNC pathogens, creating false assurances of safety, as demonstrated in outbreaks linked to VBNC *E. coli* and *Salmonella* [5] [2].
- **Water Treatment:** Disinfection processes using chlorine, UV, or peracetic acid may induce VBNC states in pathogens rather than eliminating them, potentially leading to post-treatment resuscitation [7].
- **Antimicrobial Development:** VBNC cells exhibit enhanced resistance to antimicrobial agents, necessitating novel approaches that target these persistent populations [6].

Future research priorities include standardizing detection methodologies across different bacterial species and matrices, elucidating the genetic regulation of VBNC entry and exit, and developing interventions that specifically target VBNC cells to prevent resuscitation and disease transmission. The integration of advanced molecular techniques like ddPCR with optimized viability staining provides promising avenues for more accurate detection and quantification of these elusive bacterial populations.

Key Inducing Stresses in Clinical and Environmental Settings

The **viable but non-culturable (VBNC) state** is a unique survival strategy adopted by numerous bacterial species when confronted with unfavorable environmental conditions [11]. In this state, cells lose the ability to form colonies on conventional culture media—the standard method for detecting viability in microbiology laboratories—while maintaining metabolic activity and cellular integrity [12] [11]. This phenomenon presents substantial challenges for public health, clinical diagnostics, and food safety, as VBNC pathogens can evade detection while retaining virulence potential and resuscitating when conditions improve [13] [12].

Understanding the specific stressors that trigger this dormant state is crucial for multiple disciplines. In clinical settings, VBNC formation may contribute to persistent and recurrent infections, while in industrial and environmental contexts, it compromises the efficacy of sterilization and monitoring protocols [14] [7]. This technical guide synthesizes current research on key inducing stresses across clinical and environmental settings, providing detailed experimental data and methodologies to advance VBNC state research.

Key Inducing Stresses for the VBNC State

Bacteria enter the VBNC state as a survival response to various physicochemical stresses. The table below summarizes the primary inducing factors identified in recent research, along with their effective thresholds for representative bacterial species.

Table 1: Key Stressors Inducing the VBNC State in Various Bacterial Pathogens

Stress Category	Specific Stressor	Bacterial Species	Effective Concentration/Duration	Experimental Conditions
Chemical Disinfectants	Sodium hypochlorite (Chlorine)	<i>Listeria monocytogenes</i>	37.5 ppm for 10 min [13]	20°C in laboratory medium [13]
	Sodium hypochlorite	<i>Pseudomonas aeruginosa</i>	1-3 mg/L for 5-30 min [7]	Drinking water, room temperature [7]
	Sodium dichloroisocyanurate	<i>Listeria monocytogenes</i>	50-75 ppm for 10 min [13]	20°C in laboratory medium [13]
	Hydrogen Peroxide	<i>Listeria monocytogenes</i>	12,000 ppm for 10 min [13]	20°C in laboratory medium [13]
	Peracetic Acid (PAA)	<i>Pseudomonas aeruginosa</i>	40 µM for 30 min [7]	Drinking water, room temperature [7]
Food Preservatives	Potassium Sorbate	<i>Listeria monocytogenes</i>	pH 2.0 for 10 min [13]	20°C in laboratory medium [13]
Physical Stresses	UV Radiation	<i>Pseudomonas aeruginosa</i>	245 nm wavelength for 30 min [7]	Drinking water, room temperature [7]
	Low Temperature	<i>Shigella flexneri</i>	4°C for 19-24 days [15]	Distilled water with varying NaCl concentrations [15]
Osmotic Stress	Sodium Chloride (NaCl)	<i>Shigella flexneri</i>	0-30% solutions at 4°C [15]	Incubation until loss of cultivability (19-24 days) [15]
Nutritional Stress	Starvation in Distilled Water	<i>Shigella flexneri</i>	4°C for 19 days [15]	Distilled water without nutrients [15]

Analysis of Key Stress Categories

Chemical Disinfectants: Oxidizing agents like chlorine and hydrogen peroxide are particularly effective at inducing the VBNC state. These compounds cause damage to cellular components, leading to a loss of culturability while membrane integrity may be preserved [13] [7]. Different serotypes of *Listeria monocytogenes* (1/2a, 4b, 1/2c, 4d) showed varying resilience but ultimately entered the VBNC state when exposed to chlorine-based disinfectants [13].

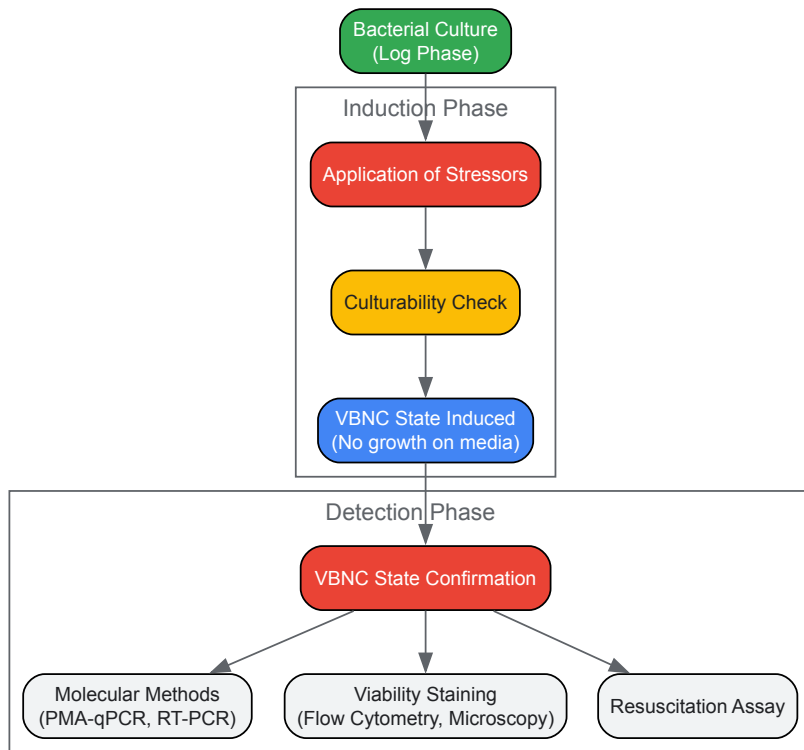
Physical Stresses: UV radiation effectively damages bacterial DNA, leading to a loss of replicative capacity while maintaining metabolic activity [7]. Low temperature, especially refrigeration temperatures (4°C), significantly prolongs bacterial survival in the VBNC state, particularly when combined with other stresses like osmotic pressure [15].

Osmotic and Nutritional Stress: The combination of low temperature and high salt concentrations creates a potent inducing environment, as demonstrated in *Shigella flexneri* [15]. Nutrient deprivation in minimal aqueous environments forces bacteria to dramatically reduce their metabolic activity to survive extended periods [15].

Experimental Protocols for VBNC Induction and Detection

- **Bacterial Strain and Preparation:** *Shigella flexneri* ATCC 12022 is prepared to 0.5 McFarland standard turbidity.
- **Stress Application:**
 - Prepare NaCl solutions across a concentration gradient (0%, 5%, 10%, 15%, 20%, 25%, 30%).
 - Inoculate 100 µL of bacterial suspension into 4.9 mL of each salt solution.
 - Incubate at 4°C. Perform six biological replicates for statistical rigor.
- **Culturability Assessment:**
 - Daily subculturing onto Brain Heart Infusion (BHI) agar enriched with 0.6% yeast extract and 1% sodium pyruvate.
 - Incubate subcultured plates at 37°C for 16-24 hours.
 - The VBNC state is considered induced when no bacterial colonies are observed on the BHI medium after incubation.
- **VBNC State Confirmation:**
 - **RNA Extraction and RT-PCR:** Extract RNA from pelleted cells and perform reverse transcription. Amplify virulence genes (*ipaH* and *ipaD*) via PCR to confirm metabolic activity and viability.
 - **Fluorescence Microscopy:** Centrifuge non-culturable samples, wash with PBS, and stain. Use untreated bacteria and ethanol-killed bacteria as positive and negative controls, respectively.
- **Bacterial Cultivation and Disinfection:**
 - Culture *P. aeruginosa* in Luria-Bertani (LB) medium to logarithmic phase.
 - Centrifuge, wash, and resuspend in phosphate-buffered saline (PBS).
 - Apply disinfectant stresses:
 - **UV:** Expose in Petri dish to 245 nm UV light for timed intervals.
 - **Chlorine:** Add NaClO (0.5-3 mg/L) and quench with sodium thiosulfate at timed intervals.
 - **Peracetic Acid:** Add PAA (40 µM) and quench with sodium thiosulfate.
- **Viable Cell Quantification via PMA-qPCR:**
 - **PMA Treatment:** Add PMA dye to samples, incubate in the dark for 5 minutes, then expose to a 650 W halogen light source on ice for 5 minutes.
 - **DNA Extraction and qPCR:** Filter treated samples, extract DNA, and perform qPCR targeting specific genes.
 - **Calculation of VBNC Cells:** Subtract the number of culturable cells (CFU/mL) from the number of viable cells (determined by PMA-qPCR) to quantify the VBNC population.
- **Resuscitation Potential Assessment:**
 - Incubate VBNC cell suspensions in LB medium at 37°C with shaking.
 - Sample bihourly and plate on nutrient agar to monitor the return of culturability.

The following workflow diagram illustrates the core experimental and detection process for the VBNC state:



Advanced Detection Methodologies

Accurate detection of VBNC cells is technically challenging as they are invisible to standard plating methods. The following table compares key modern detection techniques.

Table 2: Methodologies for Detecting and Confirming the VBNC State

Method Category	Specific Technique	Principle	Key Applications	Considerations
Molecular Viability Testing	PMA/Dye-based qPCR (v-qPCR)	PMA penetrates dead cells with compromised membranes, binding to DNA and inhibiting its amplification in PCR. Primers target specific virulence or housekeeping genes [16] [7] [17].	Detection in complex water matrices (PWW) [17], fecal samples [16], drinking water [7].	Can overestimate if dead cells have intact membranes; requires optimization for each matrix [17].
	Droplet Digital PCR (ddPCR)	Partitions sample into thousands of droplets for absolute quantification of target genes without a standard curve. Often combined with PMA (PMA-ddPCR) [16].	Absolute quantification of viable cells, especially with low concentrations (e.g., in fecal samples) [16].	Highly precise, does not require external standard curve; uses single-copy genes (e.g., <i>rpoB</i> , <i>adhE</i>) [16].
Cell Staining & Cytometry	Live/Dead Fluorescence Staining & Microscopy	Uses fluorescent dyes to distinguish cells with intact (live/VBNC) vs. damaged (dead) membranes [15].	Direct visualization and confirmation of VBNC state in laboratory cultures [15].	Can be subjective; may not be suitable for complex, particulate-laden samples like PWW [17].
	Flow Cytometry	Automates the analysis of fluorescently stained cells, providing population-level data on viability [17].	High-throughput analysis of bacterial viability in pure suspensions.	Can overestimate dead cells in complex matrices like PWW due to interference [17].
Metabolic & Virulence Activity	Reverse Transcriptase PCR (RT-PCR)	Detects messenger RNA (mRNA), which has a short half-life, confirming ongoing metabolic activity and gene expression in VBNC cells [15].	Confirming viability and expression of virulence genes (e.g., <i>ipaH</i> , <i>ipaD</i> in <i>S. flexneri</i>) [15].	Technically demanding due to RNA instability; confirms metabolic activity.
	Adenosine Triphosphate	Measures intracellular ATP levels as an indicator of metabolic activity [13] [7].	Assessing metabolic activity of VBNC cells post-	Correlates with metabolic activity but does not confirm

Method Category	Specific Technique	Principle	Key Applications	Considerations
	(ATP) Assay		disinfection [7].	virulence.

Optimized Detection Protocol for Complex Matrices

For accurate detection in complex matrices like process wash water (PWW) from the food industry, an optimized v-qPCR protocol has been validated [17]:

- **Sample Treatment:** Combine EMA (10 μM) and PMAxx (75 μM) dyes.
- **Incubation:** Incubate at 40°C for 40 minutes, followed by a 15-minute light exposure.
- **Rationale:** This dual-dye approach at elevated temperature more effectively penetrates and inhibits DNA amplification from dead cells, reducing false positives and providing a more reliable count of VBNC cells in challenging environments.

The Scientist's Toolkit: Essential Research Reagents

Table 3: Essential Reagents and Materials for VBNC State Research

Item Name	Specification/Example	Primary Function in VBNC Research
Viability Dyes	Propidium Monoazide (PMA/PMAxx) [17]	DNA intercalator that penetrates only dead cells with compromised membranes, inhibiting PCR amplification.
	Ethidium Monoazide (EMA) [17]	DNA intercalator often used in combination with PMA for enhanced exclusion of dead cell DNA.
Nucleic Acid Kits	RNA Extraction Kit (e.g., DENA Zist Asia) [15]	Isolates high-quality RNA for RT-PCR to confirm metabolic activity via mRNA detection.
	cDNA Synthesis Kit [15]	Converts extracted RNA to complementary DNA (cDNA) for subsequent PCR amplification.
Culture Media	Brain Heart Infusion (BHI) with supplements [15]	Enriched medium used for checking cultivability; yeast extract and sodium pyruvate can aid recovery of stressed cells.
	Luria-Bertani (LB) Broth/Agar [16] [7]	Standard medium for culturing and resuscitating bacteria like <i>P. aeruginosa</i> and <i>K. pneumoniae</i> .
Chemical Stressors	Sodium Hypochlorite (NaClO) [13] [7]	Common disinfectant used to induce the VBNC state in laboratory experiments.
	Hydrogen Peroxide (H2O2) [13]	Oxidative stressor used to trigger the VBNC state.
Buffers & Solutions	Phosphate Buffered Saline (PBS) [7] [15]	Used for washing and resuspending bacterial cells without causing osmotic shock.
	Artificial Seawater (ASW) [16]	Defined medium used for VBNC induction under low-nutrient conditions.
qPCR/ddPCR Reagents	Primer Sets for target genes (e.g., <i>rpoB</i> , <i>adhE</i> , <i>KP</i>) [16]	Target specific single-copy genes in the bacterial genome for precise quantification.
	Premix Ex Taq or similar Master Mix [16] [15]	Optimized cocktail for efficient and specific PCR amplification.

The induction of the VBNC state by common clinical and environmental stresses represents a significant and underappreciated challenge in microbiology. Chemical disinfectants, physical processes, and natural stressors can trigger this survival state, allowing pathogens to evade detection and potentially resuscitate to cause infections. Moving forward, research must focus on elucidating the genetic and molecular mechanisms governing the VBNC transition and resuscitation. Furthermore, the development of robust, standardized detection methods that can be widely implemented in clinical, industrial, and regulatory settings is paramount to accurately assess risk and develop effective strategies to combat persistent VBNC pathogens.

Cellular and Molecular Hallmarks of VBNC Entry

The viable but nonculturable (VBNC) state is a unique survival strategy employed by many bacteria when confronted with adverse environmental conditions [1] [18]. In this dormant state, bacteria lose the ability to form colonies on routine culture media—the cornerstone of conventional

microbiology—yet remain alive with measurable metabolic activity and intact cell membranes [1] [2]. The VBNC state represents a significant public health concern, as pathogenic bacteria in this state cannot be detected by standard plating methods but retain virulence and can resuscitate under favorable conditions, leading to potential disease outbreaks [18] [2]. Since its initial discovery in *Escherichia coli* and *Vibrio cholerae* in 1982, over 100 bacterial species have been documented to enter this physiological state [18]. Understanding the cellular triggers and molecular mechanisms governing entry into the VBNC state is therefore crucial for multiple fields, including clinical microbiology, food safety, and public health. This technical guide synthesizes current knowledge on the hallmarks of VBNC entry, providing researchers with a comprehensive framework for studying this complex bacterial adaptation.

Inducing Conditions and Quantitative Thresholds

Bacteria enter the VBNC state in response to a diverse array of environmental stresses. The table below summarizes the primary inducing factors and their documented effects on various bacterial species.

Table 1: Environmental Conditions Inducing the VBNC State and Their Documented Effects

Inducing Condition	Specific Stressors	Example Affected Species	Quantifiable Effects	Research Context
Physical Stresses	Low temperature (e.g., 4°C)	<i>Vibrio vulnificus</i> , <i>E. coli</i> O157:H7, <i>Staphylococcus aureus</i>	Induced after 18 days in <i>S. aureus</i> with citric acid and low temperature [18].	Artificial seawater, food models [18] [19].
	UV Radiation	<i>Pseudomonas aeruginosa</i>	>99.9% reduction in culturability; majority enter VBNC state [7].	Drinking water disinfection [7].
	Desiccation	<i>Pseudomonas putida</i> KT2440	Induced as a survival strategy [1].	Laboratory microcosm [1].
Chemical Stresses	Nutrient Starvation	<i>E. coli</i> , <i>Shigella dysenteriae</i> , <i>Klebsiella pneumoniae</i>	Common in oligotrophic environments like water systems [18] [2].	Artificial seawater, tap water microcosms [18].
	Chlorination (NaClO)	<i>P. aeruginosa</i> , <i>Listeria monocytogenes</i>	>99.9% reduction in culturability; evident membrane disruption [18] [7].	Drinking water disinfection, food processing [18] [7].
	Extreme pH (acidic/alkaline)	<i>E. coli</i> O157:H7, <i>S. aureus</i>	Induced under both acidic and alkaline pH conditions [18].	Food models (e.g., fruit juices) [18] [2].
	Food Preservatives & Essential Oils	<i>Listeria monocytogenes</i> , <i>Vibrio vulnificus</i>	Induced by citral at 4MIC (160 µL/L) treatment for 4.5h [20].	Seafood safety research [20].
	Heavy Metals (e.g., Copper)	<i>E. coli</i> O104:H4, <i>Acidovorax citrulli</i>	Induced into VBNC state; resuscitated by chelating agents [19].	Laboratory studies [19].
Other Stresses	High Osmolarity/Salinity	<i>E. coli</i> O157:H7	Induced alongside low temperature and osmotic stress [18].	Laboratory studies [18].
	Oxygen Deprivation	<i>E. coli</i> , <i>Pasteurella piscicida</i>	Induced under low oxygen content [18] [19].	Laboratory microcosms [18] [19].

Cellular and Molecular Mechanisms of Entry

The transition from a culturable state to a VBNC state is not a passive degradation but an actively regulated process with distinct cellular and molecular hallmarks. The following diagram synthesizes the current understanding of this complex process into a coherent signaling pathway.

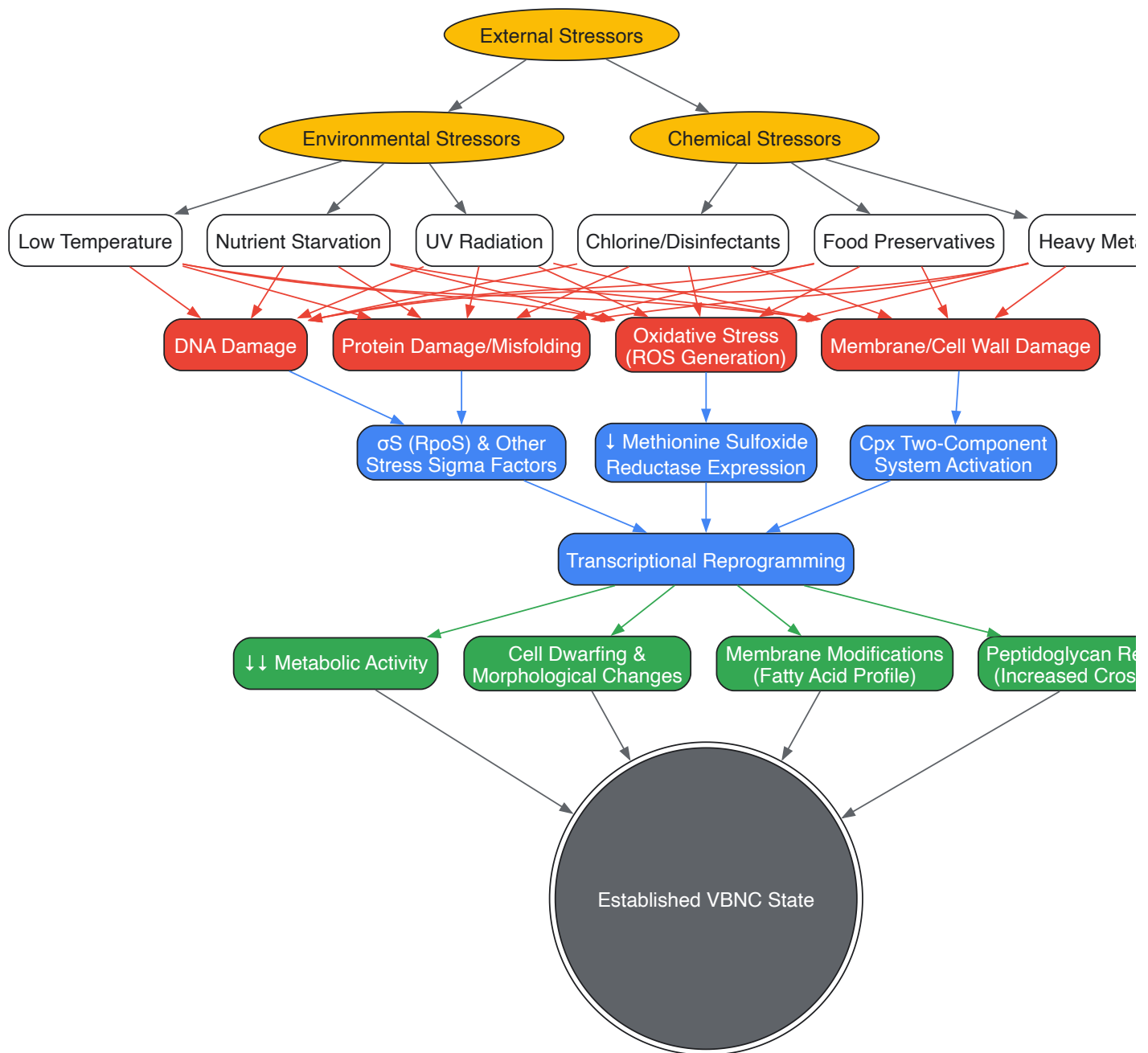


Diagram 1: Integrated Signaling Pathway for VBNC Entry. The diagram illustrates how diverse environmental and chemical stressors converge on core cellular damage pathways, triggering a regulated transcriptional response that culminates in the hallmarks of the VBNC state.

Interpretation of the Molecular Pathway

The entry into the VBNC state is a multifaceted response initiated by the perception of stress. As shown in Diagram 1, various stressors lead to fundamental cellular damage, including oxidative stress, macromolecular damage, and envelope stress. A key molecular event is the downregulation of critical repair enzymes, such as **methionine sulfoxide reductase**, which was identified via transcriptomics in *Vibrio vulnificus* as being central to the loss of culturability induced by citral [20]. This compromises the cell's ability to repair oxidative damage, locking it into a stress state.

This damage is sensed and transduced into a transcriptional response largely governed by the induction of alternative sigma factors like **RpoS (σS)** and the activation of two-component systems such as the **Cpx pathway** [20]. The Cpx system, which responds to envelope stress, is critically involved in reshaping the cell envelope [20]. This coordinated genetic reprogramming drives the phenotypic hallmarks of VBNC entry: a dramatic reduction in metabolic activity, cell dwarving, extensive modifications to the cytoplasmic membrane fatty acid composition, and increased cross-linking within the peptidoglycan layer [1] [2]. These changes collectively enable survival by minimizing energy expenditure and reinforcing cellular integrity against further environmental insults.

Essential Research Methodologies

Studying the VBNC state requires a combination of classic microbiological techniques and modern molecular biology tools, as the target cells evade standard culture-based detection.

A critical challenge in VBNC research is conclusively demonstrating that colony growth following a stress event results from the resuscitation of non-culturable cells, rather than the mere regrowth of a small number of persistent culturable cells. The following workflow outlines the stringent controls required to confirm true resuscitation.



Diagram 2: Experimental Workflow to Confirm VBNC Resuscitation. This multi-step process uses dilution, antibiotics, and scavengers to rule out the regrowth of residual culturable cells.

The validity of the VBNC state hinges on demonstrating resuscitation. Key strategies include:

- **Serial Dilution:** The induced VBNC-state bacterial suspension is serially diluted to a point where any remaining culturable cells are statistically eliminated [19].
- **Antibiotic Treatment:** Adding antibiotics like ampicillin to the medium *after* VBNC induction inhibits the proliferation of any remaining culturable cells, ensuring that subsequent growth is from resuscitated VBNC cells [19].
- **H₂O₂ Scavengers:** The addition of sodium pyruvate or catalase to the resuscitation medium quenches residual hydrogen peroxide, excluding the possibility that recovery is due solely to the regrowth of H₂O₂-sensitive culturable cells [19].

Detection and Quantification Techniques

Table 2: Key Methods for Detecting and Quantifying VBNC Cells

Method Category	Specific Technique	Principle	Key Application & Consideration
Viability Staining & Microscopy	Direct Viable Count (DVC)	Staining + antibiotic-induced elongation distinguishes viable (elongated) from dead (small) cells [18].	Initial, microscopy-based method; can be subjective [18].
	LIVE/DEAD Staining (e.g., SYTO-9/PI)	Fluorescent dyes distinguish intact (green) vs. damaged (red) membranes [18].	Must be combined with plate count to distinguish VBNC from culturable live cells [18].
	CTC-DAPI Staining	Measures respiratory activity via tetrazolium salt reduction [18].	Indicator of metabolic activity in non-culturable cells.
Molecular Methods (Viability-PCR)	PMA-/PMAxx-qPCR	Dye (PMA) penetrates dead cells, binds DNA, and blocks PCR amplification; only viable cells (intact membranes) are quantified [7] [9] [10].	Gold standard; requires optimization of dye concentration and incubation [9] [10]. Target long gene segments (>500 bp) for higher reliability [7] [21].
	PMA-droplet digital PCR (ddPCR)	Same principle as PMA-qPCR, but uses microdroplet partitioning for absolute quantification without a standard curve [9] [10].	Higher precision and resistance to PCR inhibitors; ideal for complex samples (e.g., feces) [10].
Activity Assays	Intracellular ATP Measurement	Quantifies ATP levels as a marker of metabolic activity [7].	Correlates with viability status; VBNC cells maintain moderate ATP levels [7].

Method Category	Specific Technique	Principle	Key Application & Consideration
	Electron Microscopy (SEM/TEM)	Visualizes cell morphology and membrane integrity at high resolution [7] [10].	Identifies hallmarks like dwarfing and membrane changes [7] [10].

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Reagents and Materials for VBNC Research

Reagent/Material	Function in VBNC Research	Example Usage & Notes
Propidium Monoazide (PMA) / PMAxx	Viability dye for molecular detection; selectively enters dead cells with compromised membranes and covalently binds DNA upon light exposure, preventing its PCR amplification [7] [9] [10].	Essential for vPCR (viability PCR) methods like PMA-qPCR and PMA-ddPCR. Concentration (e.g., 5-200 μM) and incubation time (5-30 min) require optimization [9] [10].
Ferrioxamine E	A siderophore that provides the essential micronutrient iron (III); acts as a resuscitation-promoting factor and growth supplement [22].	Supplementing media (5-200 ng/mL) improves recovery of VBNC cells of <i>Salmonella</i> , <i>Cronobacter</i> spp., and <i>Staphylococcus aureus</i> from food matrices [22].
Chlorine-Based Disinfectants (e.g., NaClO)	A standard chemical stressor for inducing the VBNC state in laboratory settings, mimicking water treatment processes [18] [7].	Used at concentrations of 0.5-3 mg/L to induce the VBNC state in bacteria like <i>P. aeruginosa</i> ; requires quenching with sodium thiosulfate [7].
Specific Gene Primers (Long Amplicon)	Targeting long gene fragments (>500 bp) for PCR detection after PMA treatment increases the likelihood of amplification failure from damaged DNA in dead cells, improving selectivity for viable cells [7] [21].	e.g., <i>groEL</i> (510 bp) for <i>V. parahaemolyticus</i> ; <i>ompW</i> (588 bp) for <i>V. cholerae</i> [21].
Sodium Pyruvate	Acts as a scavenger for reactive oxygen species (ROS) like hydrogen peroxide (H2O2) in culture media [19].	Critical control in resuscitation experiments; eliminates the confounding effect of H2O2-sensitive persister cells that might regrow, ensuring true resuscitation is observed [19].
Artificial Sea Water (ASW)	A defined, nutrient-limited medium used to induce the VBNC state via starvation and/or low-temperature incubation [10].	Standardized medium for VBNC induction studies, e.g., for <i>Klebsiella pneumoniae</i> and <i>Vibrio</i> species [10].

The entry of bacteria into the VBNC state is a finely orchestrated survival response, characterized by specific cellular and molecular hallmarks. Key indicators include a dramatic downscaling of metabolism, morphological reshaping, and a robust molecular response involving sigma factors and two-component systems that rewire gene expression to ensure protection and persistence. Advancements in detection methodologies, particularly viability-PCR techniques, have been instrumental in illuminating this once-elusive state. The experimental frameworks and tools detailed in this guide provide a foundation for continued research. A deeper understanding of the triggers and mechanisms of VBNC entry is not only fundamental to bacterial physiology but also critical for mitigating the significant risks these dormant cells pose to public health, clinical practice, and food safety. Future work should focus on identifying universal molecular markers for VBNC entry and developing targeted strategies to either prevent its induction or safely eliminate VBNC cells.

The viable but nonculturable (VBNC) state is a unique survival strategy adopted by numerous bacteria when confronted with adverse environmental conditions. In this physiological state, cells maintain metabolic activity and viability but cannot form colonies on conventional culture media, the standard method for detecting living bacteria [18]. This dormancy state presents significant challenges for public health, food safety, and clinical medicine, as VBNC pathogens remain potentially pathogenic and can resuscitate when conditions improve [18] [14]. Understanding the molecular mechanisms governing entry into, maintenance within, and resuscitation from the VBNC state is therefore crucial for managing bacterial infections and controlling microbial contamination. Among the complex network of molecular pathways involved, three core triggers have emerged as particularly significant: the stringent response, toxin-antitoxin (TA) systems, and oxidative stress. This review synthesizes current knowledge on these fundamental molecular mechanisms, their interconnectedness, and their roles in the VBNC state, providing researchers with a comprehensive technical guide to these critical processes.

The Stringent Response: A Master Regulator of Bacterial Stress Adaptation

The stringent response is a global bacterial stress adaptation mechanism primarily mediated by the alarmones (p)ppGpp (guanosine tetra- or pentaphosphate). This response is catalyzed by RelA and SpoT enzymes in many bacteria, with RelA primarily synthesizing (p)ppGpp in response to

nutrient starvation, particularly uncharged tRNA during amino acid limitation [23]. Under stressful conditions, (p)ppGpp accumulates and triggers a massive reprogramming of cellular physiology, shifting resources from growth-related processes to stress survival pathways.

Molecular Mechanisms and Genetic Regulation

The stringent response functions through several interconnected mechanisms. (p)ppGpp directly binds to RNA polymerase, altering gene expression patterns by favoring the transcription of stress response and virulence genes while repressing growth-related genes for ribosome biogenesis and DNA replication [23]. This regulatory network is further fine-tuned through interactions with other stress response regulators, such as the sigma factor RpoS (σS), which controls the expression of numerous stress resistance genes [23]. The hipAB toxin-antitoxin system also interfaces with the stringent response; HipA phosphorylates glutamate-tRNA ligase, leading to uncharged tRNA accumulation that stimulates (p)ppGpp production via RelA [23].

Role in VBNC State Formation and Maintenance

The stringent response serves as a critical bridge between environmental stress perception and VBNC state induction. Recent research demonstrates that knockout strains with disrupted stringent response pathways, such as ΔrelA E. coli, exhibit diminished growth rates and reduced formation of reactive oxygen species (ROS), consequently impairing their ability to develop antibiotic resistance and potentially enter dormancy states [23]. In VBNC *Cronobacter sakazakii*, both *relA* and *spoT* genes are significantly upregulated, indicating that the stringent response regulates antioxidant defense mechanisms, enabling bacterial survival within macrophages by resisting oxidative killing [24]. This enhanced oxidative stress tolerance facilitates pathogen persistence in hostile environments, including within host immune cells.

Table 1: Key Genes in the Stringent Response and Their Roles in the VBNC State

Gene/Protein	Function	Effect on VBNC State	Experimental Evidence
RelA	(p)ppGpp synthase; triggers stringent response	Central regulator; ΔrelA strains show impaired resistance development and reduced ROS [23]	Knockout E. coli ΔrelA had lower growth rates during antibiotic resistance evolution [23]
SpoT	Bifunctional: synthesizes and degrades (p)ppGpp	Works with RelA to modulate (p)ppGpp levels	Upregulated in VBNC <i>Cronobacter sakazakii</i> [24]
RpoS (σS)	RNA polymerase sigma factor; stress response regulator	Downstream effector of (p)ppGpp; activates stress resistance genes	ΔrpoS knockout showed clear growth rate decrease during kanamycin resistance evolution [23]
HipA	Toxin kinase; phosphorylates Glu-tRNA ligase	Induces (p)ppGpp production via RelA by creating uncharged tRNA [23]	Part of the type II toxin-antitoxin system [23]

Toxin-Antitoxin Systems: Conditional Growth Inhibitors

Type II toxin-antitoxin (TA) systems are genetic modules composed of two elements: a stable toxin protein that inhibits essential cellular processes (e.g., translation, DNA replication) and a labile antitoxin protein or RNA that neutralizes the toxin under normal conditions [25] [26]. These systems are ubiquitously encoded in bacterial chromosomes and have been historically implicated in stress survival, persister cell formation, and biofilm maintenance, though their exact functions remain an active area of research and debate [25].

Diversity and Mechanisms of Action

TA systems are classified based on the nature of the antitoxin and its mechanism of toxin neutralization. Type II systems, the best-characterized class, feature protein antitoxins that directly bind to and inhibit protein toxins [25]. Well-studied examples include:

- **RelBE**: A ribonuclease toxin that cleaves mRNA; transcription increases following nutritional stress [26].
- **MazEF**: An endoribonuclease toxin implicated in stress-induced stasis [25].
- **CcdAB**: The first discovered TA system; originally found on plasmids where it performs post-segregational killing [26].
- **VapBC**: One of the most common TA families; VapC toxins are ribonucleases [25].

These systems often exhibit conditional cooperativity, where low toxin concentrations enhance antitoxin-binding to DNA promoters, while high concentrations disrupt this binding, providing autoregulatory feedback [26].

Controversial Role in VBNC State

The involvement of TA systems in the VBNC state presents a complex and somewhat controversial picture. While TA systems are frequently postulated to regulate cell growth following stress, and their transcription is strongly induced by diverse stress conditions, recent evidence questions

whether this transcriptional induction translates to toxin activation [26].

Multiple studies report substantial increases (often >10-fold) in TA system transcription following stresses like amino acid starvation (induced by serine hydroxamate), translation inhibition (chloramphenicol), oxidative stress (hydrogen peroxide), and DNA damage [26]. However, despite this strong transcriptional response, a strain lacking 10 chromosomal TA systems ($\Delta 10TA$) showed no growth disadvantage compared to wild-type *E. coli* following these stresses, suggesting TA systems are not critical for surviving these individual stresses [26].

The mechanistic explanation for this discrepancy lies in protein stability: although free antitoxin is rapidly degraded during stress, leading to increased TA transcription due to relief of transcriptional autorepression, antitoxin bound to toxin is protected from proteolysis. This protection prevents toxin liberation and activity, meaning transcriptional induction does not necessarily indicate toxin activation [26]. Nonetheless, a study on *Salmonella enteritidis* found that deletion of the *yeaZ* gene, which promotes VBNC cell resuscitation, led to upregulation of TA system-related genes, suggesting a potential indirect role in VBNC biology [27].

Table 2: Experimental Responses of Type II TA Systems to Stress Conditions

Stress Condition	Inducing Agent	Example TA System Response	Evidence of Toxin Activation?
Amino Acid Starvation	Serine Hydroxamate (SHX)	>6-fold increase in <i>mqsRA</i> , <i>relBE</i> , <i>yefM-yoeB</i> mRNA [26]	No; $\Delta 10TA$ strain showed no growth defect [26]
Translation Inhibition	Chloramphenicol	>6-fold increase in <i>mqsRA</i> , <i>relBE</i> , <i>yefM-yoeB</i> mRNA [26]	No; $\Delta 10TA$ strain showed no growth defect [26]
Oxidative Stress	Hydrogen Peroxide	Transcriptional induction of multiple TA systems [26]	No evidence from RNA sequencing [26]
DNA Synthesis Inhibition	Trimethoprim	Transcriptional induction of multiple TA systems [26]	Not detected [26]
Heat Shock	Temperature shift (30°C to 45°C)	Substantial increases in TA transcription [26]	Not detected [26]

Oxidative Stress: A Common Inducer of Bacterial Dormancy

Oxidative stress occurs when bacteria encounter an imbalance between reactive oxygen species (ROS) production and their detoxification, leading to accumulation of damaging molecules like superoxide anions (O_2^-), hydrogen peroxide (H_2O_2), and hydroxyl radicals ($OH\bullet$). This stressor is particularly relevant as it is both a common environmental challenge and a key weapon used by host immune systems to kill pathogens.

Bacteria face oxidative stress from multiple sources:

- **Environmental exposures:** UV radiation, hydrogen peroxide, and other oxidizing agents [18] [28].
- **Metabolic byproducts:** Endogenous ROS production from electron transport during respiration [23].
- **Host immune response:** Phagocytes like macrophages generate an oxidative burst to kill internalized bacteria [24].
- **Antibiotic exposure:** Bactericidal antibiotics can induce ROS formation as part of their killing mechanism [23].

To combat oxidative damage, bacteria employ sophisticated defense systems including superoxide dismutases (SodA, SodB) that convert O_2^- to H_2O_2 , catalases (KatG) that break down H_2O_2 to water and oxygen, and thioredoxin (TrxA) systems that repair oxidative damage [23] [24].

Oxidative Stress and VBNC State Interconnections

Oxidative stress serves as a direct trigger for VBNC state entry while also interacting with other molecular pathways. Sublethal ROS levels can cause DNA damage that activates stress responses and increases mutation rates, potentially accelerating resistance development [23]. UV disinfection, which induces oxidative damage, can trigger *E. coli* to enter the VBNC state at doses as low as 4.5 mJ/cm², with the DNA damage repair protein RecA playing a key role in this process [28].

Once in the VBNC state, bacteria exhibit enhanced resistance to oxidative stress. VBNC *Cronobacter sakazakii* demonstrates higher survival rates under oxidative conditions compared to culturable cells, with corresponding upregulation of antioxidant genes (*sodA*, *katG*, *trxA*) and stringent response genes (*relA*, *spoT*) [24]. This suggests that VBNC cells activate a coordinated defense program, potentially regulated by the stringent response, to survive within hostile environments like macrophages.

Table 3: Key Oxidative Stress Defense Components and Their Roles in VBNC Cells

Gene/Protein	Function in Oxidative Stress Defense	Role in VBNC State
SodA, SodB	Superoxide dismutase; destroys superoxide anion radicals [23]	Upregulated in VBNC <i>Cronobacter sakazakii</i> ; essential for survival in macrophages [24]
KatG	Catalase-peroxidase; breaks down hydrogen peroxide [24]	Upregulated in VBNC <i>Cronobacter sakazakii</i> [24]
TrxA	Thioredoxin; involved in redox homeostasis and damage repair [24]	Upregulated in VBNC <i>Cronobacter sakazakii</i> [24]
SoxR, SoxS	Transcriptional regulators of superoxide response [23]	Knockout strains (Δ soxR, Δ soxS) showed faster growth rate decreases during resistance evolution [23]
RecA	DNA repair protein; responds to oxidative DNA damage [28]	Expression indicates recovery capacity in UV-induced VBNC <i>E. coli</i> [28]

Integrated Molecular Pathways and Technical Approaches

The three core molecular triggers do not function in isolation but rather form an integrated network that regulates entry into and maintenance of the VBNC state. Understanding these interconnected pathways and the methods to study them is essential for researchers investigating bacterial dormancy.

Pathway Interconnections

The stringent response, TA systems, and oxidative stress pathways intersect at multiple regulatory points. The stringent response alarmone (p)ppGpp can stimulate ROS formation, creating a potential link between nutrient stress and oxidative damage [23]. Additionally, the stringent response regulates antioxidant defense mechanisms in VBNC cells, as demonstrated by the coordinated upregulation of *relA/spoT* and antioxidant genes in VBNC *Cronobacter sakazakii* [24]. While TA systems may not be directly activated by stress, their transcription is influenced by these master regulators, and they may play supporting roles in the overall stress response network.

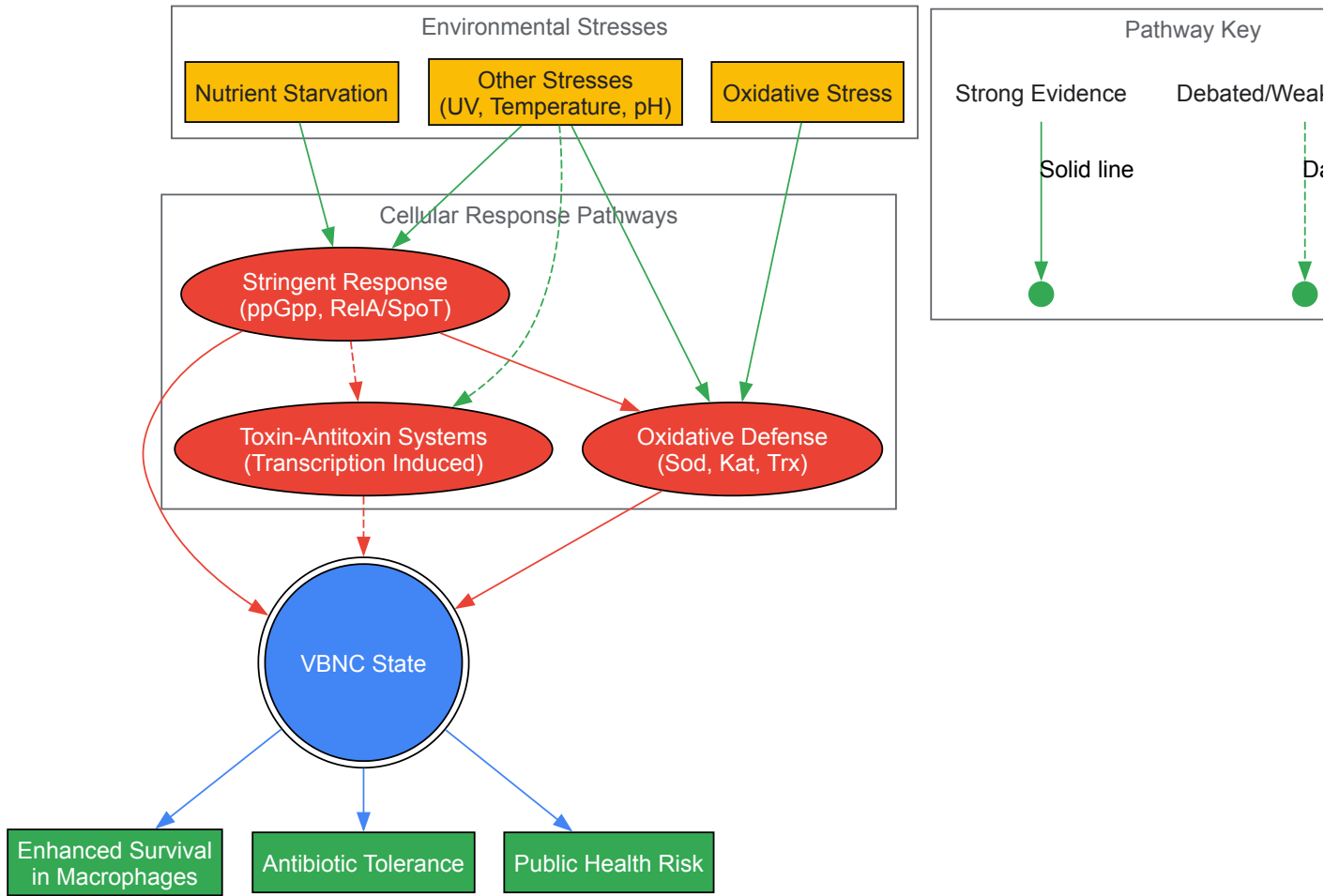


Diagram 1: Integrated Molecular Pathways to VBNC State. This diagram illustrates the interconnected network of environmental stresses and cellular responses that lead to VBNC state formation, highlighting both well-established connections (solid lines) and debated/weaker evidence (dashed lines).

Essential Research Reagents and Methodologies

Studying the VBNC state and its molecular triggers requires specialized approaches that combine classical microbiology with modern molecular techniques. The following toolkit represents essential reagents and methodologies used in this field.

Table 4: Research Reagent Solutions for Investigating VBNC Molecular Triggers

Category	Specific Reagents/Methods	Application/Function	Example Use
VBNC Induction	Ampicillin (400 µg/mL in saline) [24], UV light (4.5 mJ/cm²) [28], Low temperature + osmotic stress [15]	Induce VBNC state under controlled conditions	Inducing VBNC in <i>Cronobacter sakazakii</i> [24] and <i>E. coli</i> [28]
Viability Staining	LIVE/DEAD BacLight fluorescent stain (SYTO-9/PI) [18] [29], CTC-DAPI staining [18]	Distinguish live/dead cells based on membrane integrity and metabolic activity	Monitoring <i>P. syringae</i> viability after acetosyringone treatment [29]
Molecular Detection	PMAXx-qPCR [24], RT-PCR for virulence genes (<i>ipaH</i> , <i>ipaD</i>) [15], RNA sequencing [29]	Detect and quantify VBNC cells; measure gene expression	Detecting VBNC <i>Salmonella</i> [15] and gene expression in VBNC <i>P. syringae</i> [29]
Mutant Construction	Lambda Red homologous recombination [27], Single-gene knockouts (Δ relA, Δ sodA, etc.) [23]	Create specific gene deletions to study function	Constructing Δ yeaZ <i>Salmonella</i> mutant [27]
Stress Application	Serine hydroxamate (SHX) [26], Hydrogen peroxide [26] [24], Chloramphenicol [26]	Induce specific stress responses (stringent, oxidative, translation inhibition)	Studying TA system transcription after stress [26]

Experimental Workflow for VBNC Research

A systematic approach is essential for rigorously investigating the molecular triggers of the VBNC state. The following workflow, visualized in Diagram 2, outlines key methodological stages from strain preparation through data interpretation.

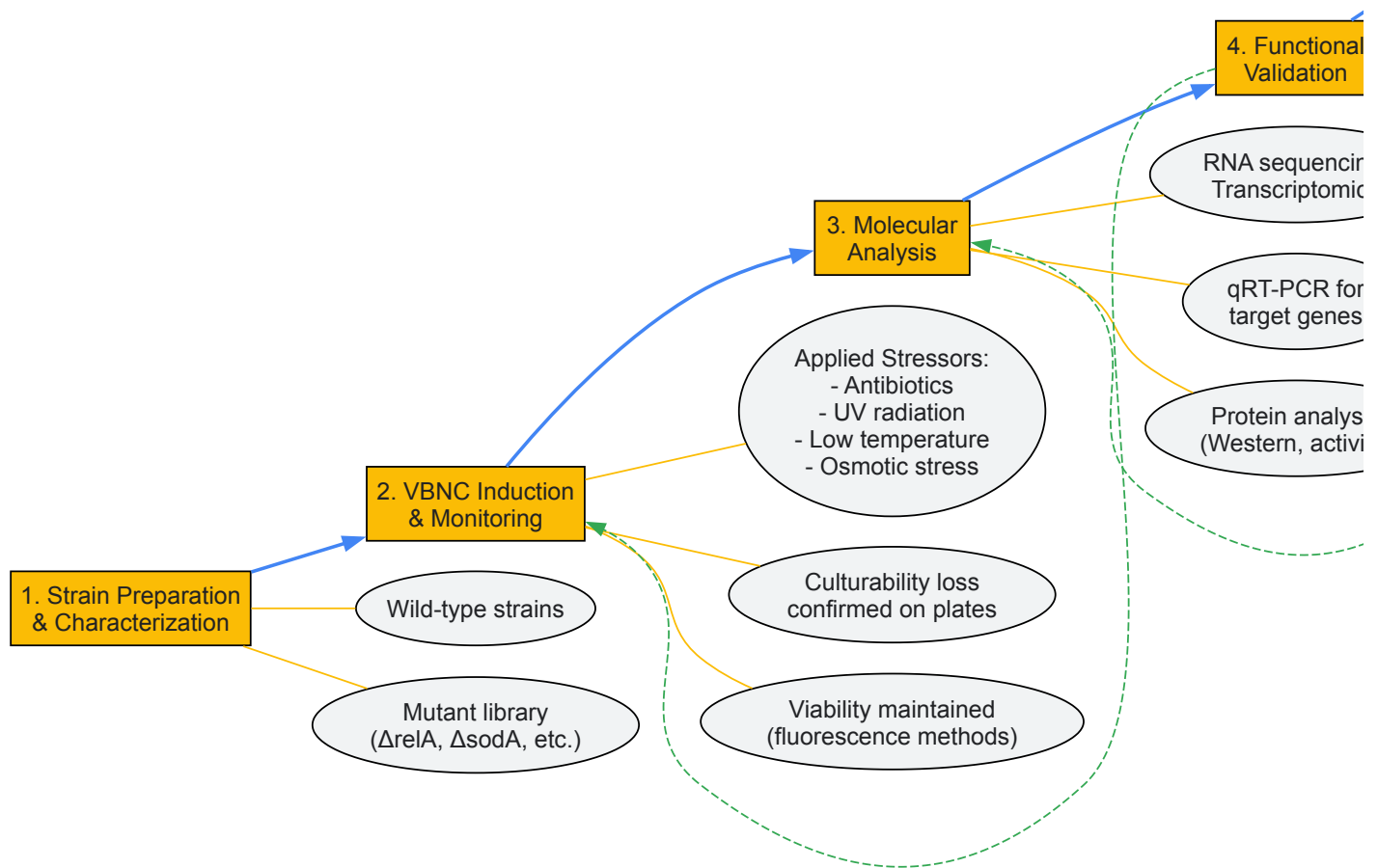


Diagram 2: Experimental Workflow for VBNC Research. This diagram outlines a systematic approach for investigating molecular triggers of the VBNC state, from initial strain preparation through functional validation and resuscitation studies. Dashed lines represent potential iterative research cycles.

The stringent response, toxin-antitoxin systems, and oxidative stress represent three core molecular triggers that collectively regulate entry into and maintenance of the VBNC state. While the stringent response and oxidative stress demonstrate well-established roles in this dormancy state, with clear evidence of their activation and contribution to stress adaptation and survival, the involvement of TA systems appears more nuanced, with strong transcriptional induction but limited evidence of toxin activation during stress. The interconnectedness of these pathways creates a robust network that allows bacteria to sense environmental threats and implement survival strategies, including transition to the VBNC state.

Significant challenges remain in fully elucidating these mechanisms, particularly in distinguishing between correlation and causation in pathway activation, understanding the precise sequence of molecular events during VBNC induction, and developing strategies to prevent VBNC formation or eradicate VBNC cells. Future research should focus on temporal analyses of pathway activation, single-cell studies to address population heterogeneity, and interventional approaches targeting these core pathways to control VBNC bacteria. As our understanding of these fundamental molecular mechanisms deepens, so too will our ability to combat the significant public health challenges posed by VBNC pathogens in clinical, food safety, and environmental contexts.

Distinguishing VBNC from Persister Cells and True Cell Death

Within the realm of bacterial stress response and survival, the **viable but non-culturable (VBNC) state** and **persister cells** represent two critical dormant phenotypes that confound traditional microbiological detection and clinical treatment. Both states allow bacteria to withstand lethal environmental insults and antibiotic challenges, yet they are fundamentally distinct in their physiological characteristics and clinical implications. The accurate differentiation between VBNC cells, persister cells, and truly dead cells is not merely academic—it represents a pressing challenge in

clinical microbiology, pharmaceutical development, and public health. Failure to distinguish these states can lead to misinterpretation of treatment efficacy, unexplained disease recurrence, and underestimation of bacterial contamination risks. This guide provides a comprehensive technical framework for researchers and drug development professionals to correctly identify and characterize these distinct cellular states within the broader context of bacterial VBNC research.

Defining the States: Theoretical Framework and Key Characteristics

Conceptual Definitions and Distinctions

The **Viable But Non-Culturable (VBNC) state** is a survival strategy wherein bacteria, in response to environmental stress, lose the ability to form colonies on conventional growth media that would normally support their proliferation, while maintaining viability and metabolic activity [4]. These cells are not dead; they retain an intact cell membrane, demonstrate low-level metabolic activity, continue gene expression, and possess the potential to resuscitate under appropriate conditions [30] [31]. The VBNC state is now recognized as a major public health concern because these cells evade standard culture-based detection methods yet remain capable of causing infections.

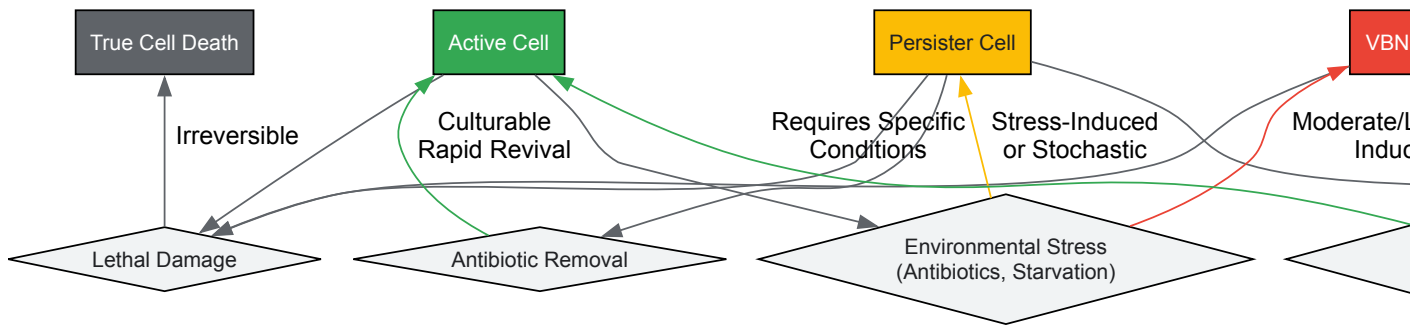
Persister cells represent a small subpopulation of dormant, non-growing cells within an otherwise susceptible bacterial population that exhibit extreme tolerance to high-dose antibiotic treatment without undergoing genetic resistance mutations [30] [32]. Unlike VBNC cells, persisters remain culturable—they can resume growth on standard media shortly after the removal of the antibiotic stress [30] [4]. Their formation can be stochastic or induced by environmental cues, and they are increasingly implicated in recurrent and chronic bacterial infections.

True Cell Death represents the irreversible loss of bacterial viability, characterized by complete and permanent cessation of all metabolic functions and irreversible damage to cellular integrity, particularly the cell membrane [8] [4].

The Dormancy Continuum Hypothesis

Emerging evidence suggests that VBNC cells and persisters may not represent completely separate phenomena but rather exist along a **dormancy continuum** [30]. This model proposes that active cells under stress can transition into persisters, which may subsequently develop into VBNC cells in response to prolonged or intensified stress [30] [14]. Supporting this hypothesis, research on *Vibrio vulnificus* demonstrated that persister cells isolated through antibiotic treatment transitioned into the VBNC state more rapidly (4-5 days) compared to log-phase cells (7-10 days), suggesting that persisters are primed for deeper dormancy [30] [14].

The following diagram illustrates this theoretical relationship and the key differentiating features along the dormancy spectrum:



Comparative Analysis: Key Differentiating Parameters

The accurate discrimination between VBNC cells, persister cells, and dead cells requires assessment across multiple cellular parameters. The table below summarizes the defining characteristics of each state:

Table 1: Key Differentiating Features of VBNC Cells, Persister Cells, and True Cell Death

Parameter	VBNC Cells	Persister Cells	True Cell Death
Culturability on Standard Media	Lost (CFU = 0) [4]	Retained (able to form colonies after stress removal) [30] [4]	Lost (irreversible)
Metabolic Activity	Low but measurable; maintains membrane potential & ATP [4]	Greatly reduced but detectable [32]	Absent (irreversible loss)
Membrane Integrity	Intact [8] [4]	Intact [30]	Compromised/damaged [8]

Parameter	VBNC Cells	Persister Cells	True Cell Death
Gene Expression/Translation	Active (continuous but reduced) [31] [4]	Active but altered [30]	Absent
Induction Triggers	Moderate, long-term stresses: starvation, extreme T°, salinity, chlorine, food preservatives [31] [4] [17]	Specific, often antibiotic-mediated stress [4] [32]	Lethal insults (extreme heat, high doses of disinfectants)
Resuscitation	Requires specific stimuli (e.g., temperature upshift, nutrient addition, Rpf); conditions differ from original culture [30] [4]	Rapid upon removal of inducing stress (e.g., antibiotic removal) [30] [4]	Not possible
Antibiotic Tolerance	High (due to low metabolic activity) [30] [33]	High tolerance (antibiotic-specific) [30] [32]	N/A
Detection Methods	Viability qPCR (PMA/EMA-qPCR), Live/Dead staining combined with culturability assessment [8] [4] [34]	Culture after antibiotic removal & washing [30]	Standard plate count (no growth), permeability to viability dyes [8]

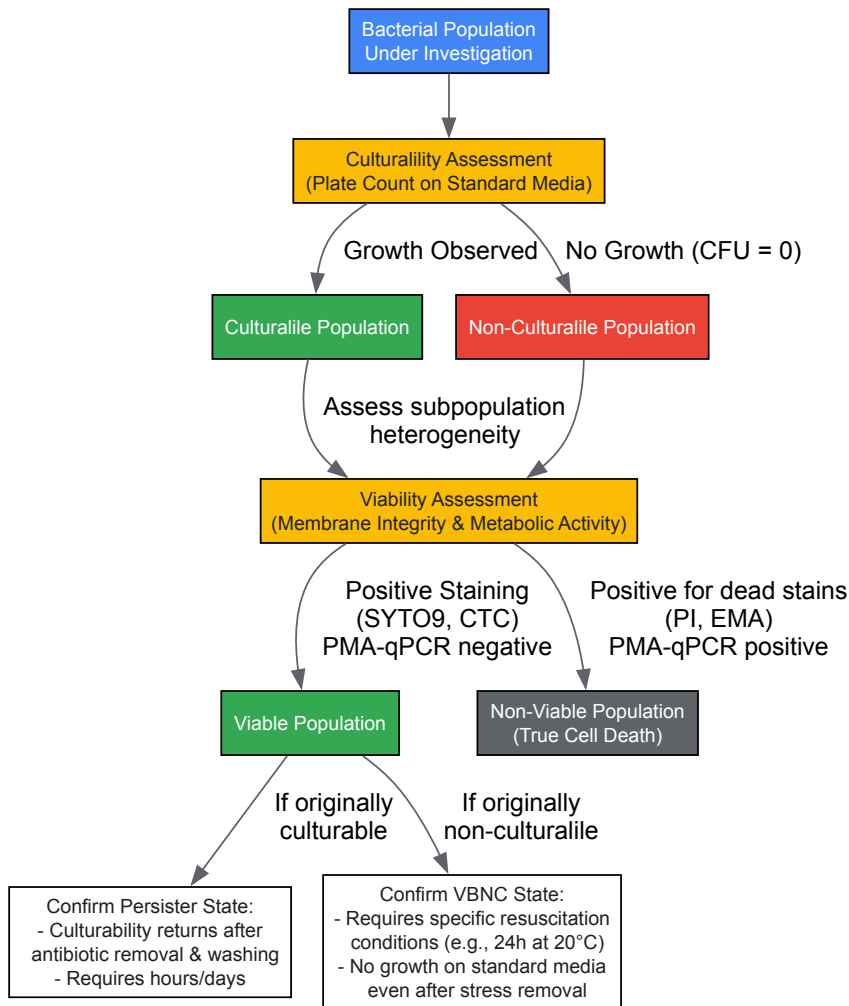
Beyond the fundamental differences outlined above, several additional distinctions are critical for researchers:

- Virulence Potential:** VBNC cells often retain pathogenicity. Pathogenic *E. coli* O157 and *Campylobacter jejuni* in the VBNC state have been linked to food poisoning incidents, and VBNC cells can resuscitate *in vivo* to regain full infectivity [30] [34]. Persister cells, while tolerant to antibiotics, do not necessarily exhibit altered virulence upon resuscitation.
- Molecular Mechanisms:** Toxin-antitoxin (TAS) systems are classically implicated in persister formation but have also been shown to play a role in VBNC induction, particularly in response to human serum, suggesting overlapping but distinct regulatory pathways [30].

Methodological Guide: Detection and Differentiation Protocols

Core Experimental Workflow

A robust approach to differentiating these states requires a combination of cultural and molecular methods. The following workflow provides a systematic protocol for characterization:



Detailed Experimental Protocols

Protocol for Isolating and Confirming Persister Cells

This protocol is adapted from studies on *Vibrio vulnificus* and *E. coli* [30]:

- **Culture Preparation:** Grow the bacterial strain to log phase ($OD_{610} \sim 0.15-0.25$) in appropriate broth (e.g., Heart Infusion broth).
- **Antibiotic Treatment:** Expose the culture to a high concentration of a bactericidal antibiotic (e.g., 100 $\mu\text{g/mL}$ ampicillin) for 4 hours at the optimal growth temperature with aeration.
- **Wash and Remove Antibiotic:** Centrifuge the antibiotic-treated culture and wash the pellet four times with a buffer (e.g., PBS, 1/2 Artificial Seawater for vibrios, or 0.85% NaCl for *E. coli*) to thoroughly remove the antibiotic.
- **Assess Culturability:** Perform serial dilutions of the washed cell suspension and plate on appropriate non-selective solid media (e.g., HI agar). Incubate under optimal conditions and enumerate colony-forming units (CFU/mL).
- **Interpretation:** The surviving cells that regrow on the plates after this process are defined as **persister cells**. Their culturability returns rapidly (within the standard incubation time of the organism) once the antibiotic is removed [30] [4].

Protocol for Inducing and Confirming the VBNC State

This protocol uses nutrient starvation at low temperature, a common VBNC-inducing condition [30]:

- **Induction:** Take a log-phase culture, wash twice with a nutrient-free buffer (e.g., 1/2 Artificial Seawater for marine vibrios) to remove nutrients. Dilute 1:100 in the same buffer and incubate statically at 4°C.
- **Monitor Culturability:** Quantify culturable cells daily by plate counts. The population is considered to have entered the VBNC state when culturable counts drop below the detection limit (<10 CFU/mL) on standard media that normally supports growth [30].
- **Assess Viability of Non-Culturable Cells:** Use a viability assay to confirm the cells are not dead.
 - **BacLight Live/Dead Kit:** Stain cells with SYTO 9 (green, penetrates all cells) and propidium iodide (red, penetrates only membrane-compromised cells). Viable cells with intact membranes will fluoresce green. Count green-fluorescent cells using epifluorescence microscopy or flow cytometry [30].

- **Viability qPCR (v-qPCR):** Treat samples with propidium monoazide (PMA) or PMAx, which penetrates only dead cells with compromised membranes and binds to DNA, inhibiting its amplification in PCR. After light exposure and DNA extraction, perform qPCR. The signal corresponds to DNA from viable (including VBNC) cells with intact membranes [8] [34].
- **Resuscitation Test:** To confirm the VBNC state, attempt to resuscitate the non-culturable population. Incubate the VBNC-inducing culture at a permissive temperature (e.g., 20°C for 24 hours) without adding rich nutrients. After this incubation, re-plate on standard media. An increase in CFU indicates resuscitation from the VBNC state [30]. A key differentiator is that resuscitation of VBNC cells often requires specific conditions different from their original growth conditions, unlike persisters [4].

Protocol for Viability qPCR (v-qPCR) to Detect VBNC Cells

This method is crucial for detecting and quantifying VBNC cells in complex matrices [8] [34] [17]:

- **Sample Preparation:** Prepare bacterial samples containing a mixture of live, dead, and potentially VBNC cells.
- **Dye Treatment:** Add a combination of EMA (10 µM) and PMAx (75 µM) to the sample. EMA penetrates cells with slightly damaged membranes and is effluxed by active cells, while PMAx is excluded from cells with intact membranes. Incubate the sample in the dark at 40°C for 40 minutes with occasional mixing.
- **Photoactivation:** Expose the tube to a bright halogen light source (e.g., 500-W halogen lamp at 20 cm distance) for 15 minutes to crosslink the dyes to DNA from dead cells.
- **DNA Extraction:** Wash the cells to remove residual dye and extract genomic DNA using a commercial kit.
- **qPCR Analysis:** Perform quantitative PCR using species-specific primers (e.g., targeting *rpoB* for *C. jejuni* [34]). The resulting qPCR signal will predominantly originate from viable cells (including VBNC), as the DNA from dead cells is covalently modified and cannot be amplified.
- **Quantification:** The number of VBNC cells can be estimated by subtracting the number of culturable cells (from plate counts) from the total viable cell count obtained via v-qPCR [34].

The Scientist's Toolkit: Essential Research Reagents and Materials

Table 2: Key Research Reagent Solutions for Studying Dormant Bacterial States

Reagent/Material	Function/Application	Specific Example
Propidium Monoazide (PMA)/PMAxx	Viability dye for v-qPCR; enters dead cells with compromised membranes, binding DNA and inhibiting PCR amplification [8] [34].	Used in v-qPCR to differentiate viable VBNC cells (PMA-negative) from dead cells (PMA-positive) in <i>Listeria monocytogenes</i> and <i>Campylobacter jejuni</i> [8] [34] [17].
Ethidium Monoazide (EMA)	Viability dye often used in combination with PMA for v-qPCR; can penetrate slightly damaged membranes [8] [17].	A combination of EMA (10 µM) and PMAx (75 µM) was optimal for detecting VBNC <i>L. monocytogenes</i> in process wash water [8] [17].
BacLight Live/Dead Viability Kit	Fluorescent staining for microscopy/flow cytometry; SYTO9 stains all cells, PI stains only dead cells with damaged membranes [30] [8].	Used to confirm membrane integrity of <i>Vibrio vulnificus</i> VBNC cells by flow cytometry; viable cells fluoresce green [30].
Resuscitation-Promoting Factor (Rpf)	Bacterial cytokine that stimulates the resuscitation of VBNC cells [35].	Identified in <i>Micrococcus luteus</i> and other Actinobacteria; can be used to resuscitate VBNC cells in bioremediation studies [35].
Toxin-Antitoxin System Mutants	Genetic tools to study molecular mechanisms of dormancy [30].	Used in <i>E. coli</i> and <i>V. vulnificus</i> to demonstrate the role of TAS in the formation of both persister and VBNC cells [30].
Artificial Seawater (ASW)/Saline	Nutrient-free buffer for inducing VBNC state in aquatic bacteria [30].	Used to induce the VBNC state in <i>V. vulnificus</i> via nutrient starvation at low temperatures [30].
Chlorine-based Sanitizers	Chemical stressor to induce VBNC state in foodborne pathogens [17].	Sodium hypochlorite (10 mg/L free chlorine) induces VBNC state in <i>L. monocytogenes</i> and <i>Salmonella enterica</i> in produce wash water [17].

The precise differentiation between VBNC cells, persister cells, and true cellular death is a cornerstone of advanced microbiological research with profound implications for clinical diagnostics, therapeutic development, and public health microbiology. While both VBNC and persister cells represent dormant, stress-tolerant phenotypes, they are defined by critical differences in **culturability** and **resuscitation requirements**. Persisters remain culturable on standard media upon stress removal, whereas VBNC cells lose this ability and require specific resuscitation signals. The emerging "dormancy continuum" model provides a valuable theoretical framework, suggesting these states may be interconnected points on a spectrum of microbial dormancy.

Moving forward, researchers must employ a multi-faceted approach, combining traditional culturability assays with modern molecular techniques like viability qPCR and advanced staining methods. Standardizing these protocols across laboratories will be essential for generating comparable data and driving the field toward a unified understanding of these elusive bacterial states. Furthermore, the development of novel therapeutic agents that either prevent entry into dormancy or effectively eradicate dormant cells represents one of the most promising yet challenging frontiers in combating persistent and recurrent bacterial infections.

Beyond the Plate: Advanced Methodologies for Detecting and Quantifying VBNC Cells

Limitations of Conventional Culture and the Need for Viability Testing

For over a century, the gold standard for detecting and quantifying viable bacteria has been their ability to form visible colonies on nutrient media. However, this conventional culture approach fundamentally underestimates true microbial viability. The "great plate count anomaly" describes the phenomenon where microscopic counts of viable cells far exceed the numbers capable of growing on standard culture media [36]. A key explanation for this discrepancy is the **viable but non-culturable (VBNC) state**, a dormant survival strategy adopted by numerous bacterial species in response to environmental stress [5] [37]. In the VBNC state, bacteria fail to grow on routine media but maintain metabolic activity, membrane integrity, and potential pathogenicity, presenting a formidable challenge to public health, food safety, and pharmaceutical sterility testing [37] [38]. This whitepaper details the limitations of traditional culture methods, explores the VBNC state's characteristics, and advocates for the adoption of advanced viability testing to mitigate the risks posed by these undetectable pathogens.

The Viable But Non-Culturable State: A Survival Strategy

Defining the VBNC State

The VBNC state is a unique physiological condition induced by various environmental stresses. Cells in this state are characterized by a loss of culturability on standard media that normally support their growth, while they retain viability markers such as an intact membrane, metabolic activity, and genetic potential for resuscitation [4]. First identified in 1982 in *Escherichia coli* and *Vibrio cholerae* [5], the list of bacteria known to enter the VBNC state has grown to encompass over 100 species, including significant human pathogens such as *E. coli* O157:H7, *Salmonella enterica*, *Listeria monocytogenes*, and *Campylobacter jejuni* [5] [3] [38].

Key Characteristics and Differences from Other States

It is crucial to differentiate VBNC cells from both culturable cells and dead cells. The table below summarizes the defining characteristics.

Table 1: Key Characteristics of Different Bacterial Physiological States

Feature	Culturable Cells	VBNC Cells	Dead Cells
Culturability	Forms colonies on standard media	Cannot form colonies on standard media	Cannot form colonies on any media
Metabolic Activity	High	Low but measurable [37]	Absent
Membrane Integrity	Intact	Intact [37]	Damaged
Genetic Material	Intact	Intact and retained [37]	Degraded
Gene Expression	Active	Continued transcription & translation [37] [3]	Absent
Respiratory Activity	Present	Present [37]	Absent
Potential for Resuscitation	Not applicable	Yes, under specific conditions	No

VBNC cells also differ from **persister cells**, another dormant subpopulation. While both are tolerant to antibiotics, persister cells remain culturable once the antibiotic is removed, whereas VBNC cells are non-culturable and require specific resuscitation signals to regain culturability [37] [4].

Limitations of Conventional Culture-Based Methods

Reliance on plate counting and other growth-based assays for viability assessment presents several critical limitations:

- Inability to Detect VBNC Pathogens:** This is the most significant drawback. Conventional methods yield false negatives when pathogens enter the VBNC state, creating a dangerous blind spot in safety monitoring [5] [38]. For instance, *E. coli* O157:H7 can enter the VBNC state on lettuce and spinach, evading detection before consumption [5].

- **Underestimation of Microbial Load and Risk:** The failure to detect VBNC cells leads to a significant underestimation of the total viable bacterial population in clinical, environmental, and product samples, resulting in a false sense of security [39].
- **Inability to Inform on True Metabolic State:** Culture methods confirm only the ability to replicate under specific conditions. They provide no direct information on the metabolic state, membrane integrity, or virulence potential of cells that do not form colonies [37].
- **Link to Unexplained Disease Outbreaks:** The VBNC state is suspected to be a factor in foodborne outbreaks where routine testing fails to identify a causative agent. For example, VBNC *Salmonella* Oranienburg was linked to an outbreak from dried squid, and VBNC *E. coli* O157 was implicated in an outbreak from salted salmon roe [5].

Conditions Inducing the VBNC State and Associated Risks

A wide array of common stresses can induce the VBNC state, many of which are standard practices in food processing, water treatment, and pharmaceutical manufacturing.

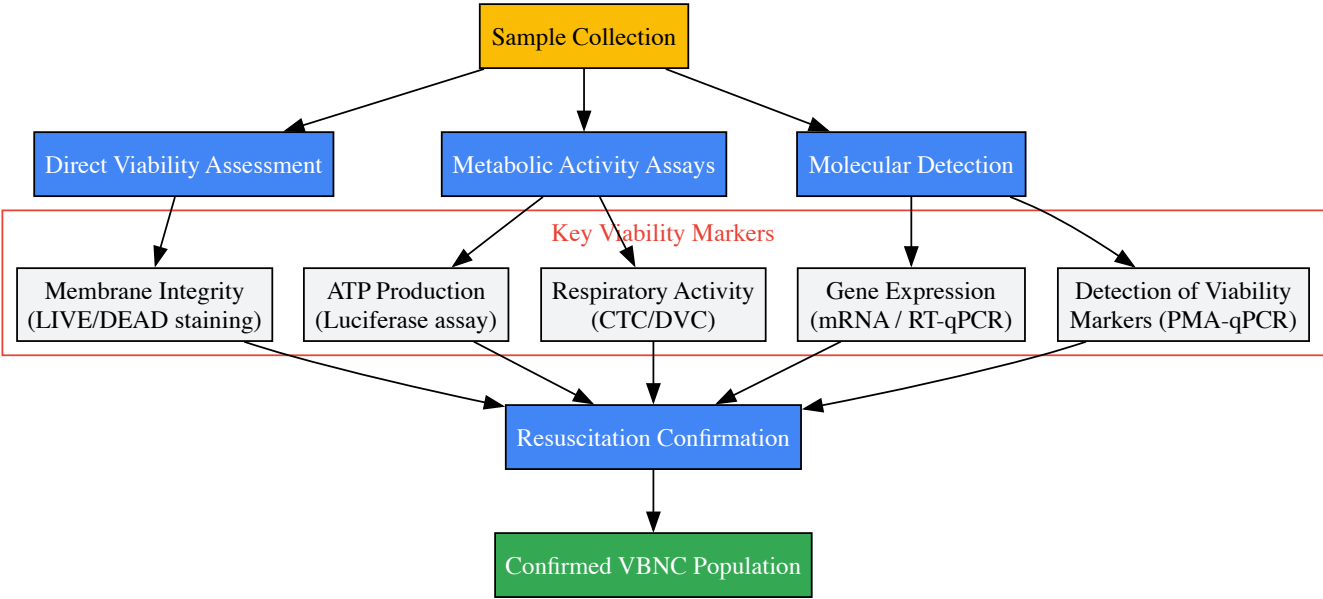
Table 2: Common Inducers of the VBNC State in Foodborne and Waterborne Pathogens

Inducing Condition	Example Pathogens	Relevant Context
Low Temperature	<i>Listeria monocytogenes</i> , <i>E. coli</i> O157:H7 [5]	Refrigerated food storage
Nutrient Starvation	<i>Vibrio cholerae</i> , <i>Campylobacter jejuni</i> [5] [37]	Water systems, low-nutrient foods
High Osmolarity/Salinity	<i>Salmonella</i> Oranienburg [5]	Salted foods, cheese
Oxidative Stress	<i>Campylobacter jejuni</i> [38]	Exposure to disinfectants
Low pH	<i>Listeria monocytogenes</i> [5]	Acidic foods, beverages
Chlorination	Various waterborne pathogens [5] [3]	Water treatment
UV Light/Irradiation	<i>E. coli</i> O157:H7 [5] [38]	Food and surface sterilization
High Pressure	<i>Listeria monocytogenes</i> , <i>Bacillus cereus</i> [5]	Food pasteurization (e.g., HPP)
Preservatives	Potassium sorbate, sodium benzoate [5] [3]	Processed foods, cosmetics
Household Cleaners	<i>L. monocytogenes</i> , <i>S. aureus</i> , <i>E. coli</i> [39]	Sanitation protocols

The major risk associated with VBNC cells is their **potential for resuscitation**. When the inducing stress is removed or conditions become favorable, VBNC cells can resuscitate and resume full metabolic activity and virulence. Resuscitation can occur in food products [38], water systems, or inside a human host, potentially leading to disease outbreaks long after a product was deemed "safe" by conventional testing [5] [40]. Furthermore, some VBNC pathogens, such as certain strains of *E. coli*, retain the ability to produce toxins while in the non-culturable state, posing a direct health threat even without resuscitation [39] [40].

Advanced Viability Testing Methodologies

To overcome the limitations of culture-based methods, a suite of growth-independent techniques is required. The following workflow outlines a comprehensive strategy for detecting VBNC cells.



Detailed Experimental Protocols

Propidium Monoazide (PMA) coupled with qPCR

This method selectively detects cells with intact membranes, a key characteristic of VBNC cells [4].

- **Sample Preparation:** Suspend bacterial cells in PBS.
- **PMA Treatment:** Add PMA dye to the sample to a final concentration of 50-100 µM. Incubate in the dark for 5-10 minutes.
- **Photoactivation:** Expose the sample to bright light (e.g., a 500-W halogen lamp) for 15-20 minutes. PMA crosses compromised membranes of dead cells and intercalates into DNA, forming a covalent crosslink upon light exposure. This crosslink inhibits PCR amplification. In viable/VBNC cells with intact membranes, PMA cannot enter and thus DNA remains amplifiable.
- **DNA Extraction:** Proceed with standard genomic DNA extraction.
- **qPCR Amplification:** Perform qPCR targeting a species-specific gene. A positive signal indicates the presence of viable/VBNC cells with intact membranes.

Direct Viable Count (DVC) and Fluorescence Microscopy

This method identifies cells that are metabolically active and capable of elongation [37].

- **Sample Incubation:** Incubate the sample in a nutrient broth (e.g., yeast extract broth) containing nalidixic acid (0.002%) and yeast extract. Nalidixic acid inhibits DNA division without affecting RNA or protein synthesis.
- **Incubation Conditions:** Incubate for 6-8 hours at an appropriate growth temperature.
- **Staining:** Filter the sample and stain with a fluorescent dye such as acridine orange.
- **Enumeration:** Examine under an epifluorescence microscope. Metabolically active VBNC cells will appear as elongated, fluorescent filaments due to continued growth without division. Compare counts to those from a control without nalidixic acid.

Adenosine Triphosphate (ATP) Bioluminescence Assay

This assay measures metabolic activity by detecting ATP, present in all living cells [39].

- **Cell Lysis:** Lyse bacterial cells to release intracellular ATP.
- **Reaction:** Mix the lysate with a luciferin-luciferase enzyme substrate.
- **Detection:** Measure the resulting bioluminescent light output with a luminometer. The light intensity is directly proportional to the ATP concentration, which correlates with the number of viable/VBNC cells.

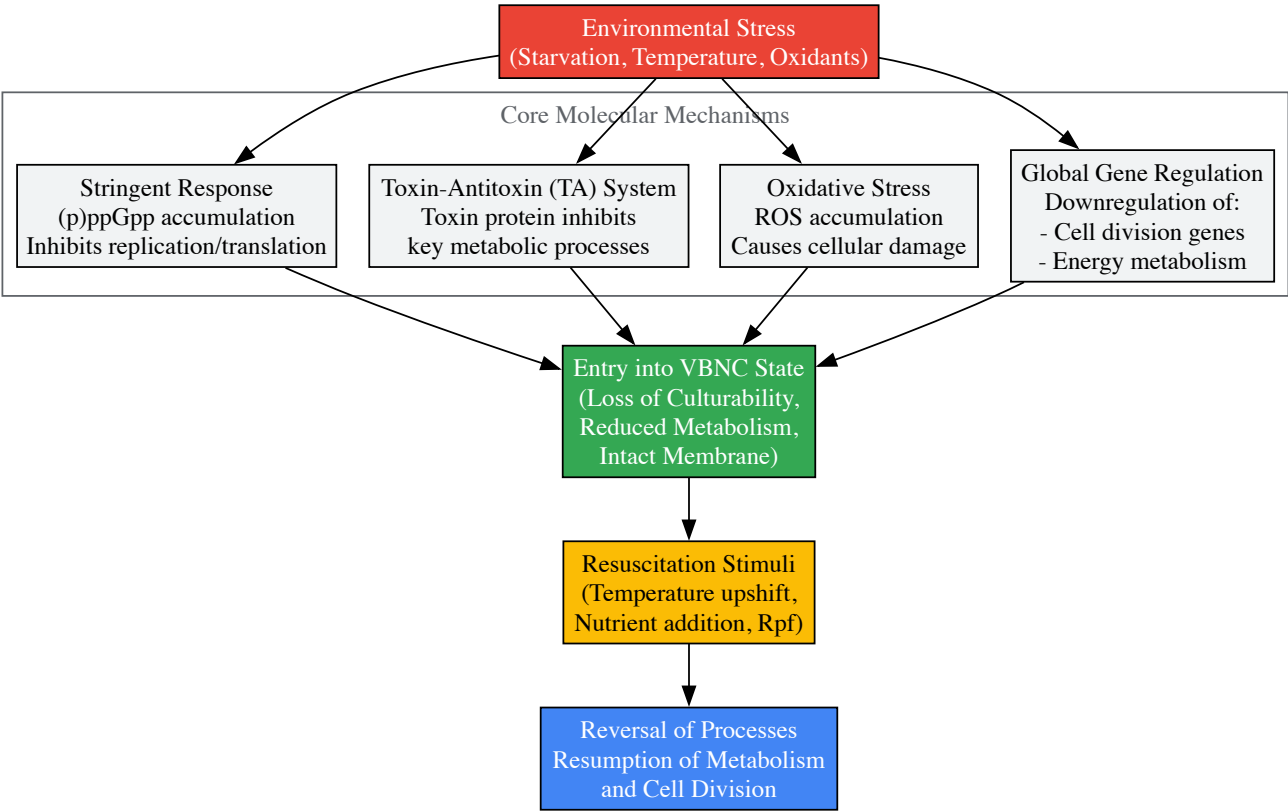
The Scientist's Toolkit: Key Research Reagent Solutions

Table 3: Essential Reagents and Kits for VBNC Research

Reagent/Kits	Function	Key Application in VBNC Research
PMA/Dye Treatment Kits (e.g., PMAxx, EMA)	Selective DNA modification of dead cells	Differentiating VBNC (PMA-negative) from dead (PMA-positive) cells in PCR assays [4]

Reagent/Kits	Function	Key Application in VBNC Research
LIVE/DEAD BacLight Bacterial Viability Kits	Dual fluorescent staining of cells	Simultaneously visualizing cells with intact (green) and damaged (red) membranes via microscopy [39]
ATP Bioluminescence Assay Kits	Quantification of cellular ATP	Rapidly measuring metabolic activity as a marker of viability in VBNC cells [39]
Bacterial RNA Extraction Kits	Isolation of high-quality RNA	Detecting gene expression (mRNA) in metabolically active VBNC cells via RT-qPCR [37] [40]
Resuscitation-Promoting Factor (Rpf)	Bacterial cytokine	Stimulating the resuscitation of VBNC cells in specific Gram-positive bacteria for experimental confirmation [40]
CTC (5-Cyano-2,3-Ditolyl Tetrazolium Chloride)	Tetrazolium dye	Detecting respiratory activity in VBNC cells by measuring electron transport system activity [37]

Understanding the molecular basis of the VBNC state is key to developing targeted detection and control methods. Several interconnected regulatory pathways are involved.



The **stringent response** is a key global regulatory mechanism. Under stress, alarmones (p)ppGpp accumulate, redirecting cellular resources away from growth and division and towards maintenance and survival, facilitating the transition to the VBNC state [38]. The **Toxin-Antitoxin (TA) system** further promotes dormancy; under stress, unstable antitoxins are degraded, allowing stable toxin proteins to inhibit essential processes like translation and ATP synthesis [38]. Additionally, **oxidative stress** with reactive oxygen species (ROS) accumulation contributes to VBNC induction by damaging cellular components, a process mitigated by pre-treatment with radical scavengers like sodium pyruvate [38].

The limitations of conventional culture methods are no longer a theoretical concern but a practical and significant risk across multiple industries. The existence of the VBNC state fundamentally undermines the principle that "what cannot be cultured is not a threat." The failure to detect these dormant, yet viable and potentially virulent, pathogens can have severe consequences for public health and product safety. A paradigm shift from reliance on culturability to the assessment of true viability is urgently needed. The integration of advanced, growth-independent methodologies—such as PMA-qPCR, ATP bioluminescence, and sophisticated fluorescent assays—into routine testing frameworks is essential to close this critical detection gap. For researchers and drug developers, acknowledging and addressing the challenge of the VBNC state is a crucial step towards developing more robust safety protocols and therapeutic strategies against resilient bacterial pathogens.

The **viable but non-culturable (VBNC) state** represents a dormant condition in which bacteria fail to grow on routine culture media yet remain metabolically active and retain pathogenicity. This survival strategy, adopted by numerous pathogens in response to environmental stresses, poses a significant challenge to public health and food safety, as these cells escape detection by conventional culture-based methods while maintaining the

potential for virulence and resuscitation [41]. Within this context, nucleic acid-based detection methods combined with viability dyes have emerged as powerful tools for differentiating between live VBNC cells and dead bacteria.

Propidium monoazide (PMA), a DNA-intercalating dye that selectively penetrates cells with compromised membranes, enables this distinction by inhibiting PCR amplification from dead cells [42] [43]. When integrated with quantitative PCR (qPCR) or the more advanced droplet digital PCR (ddPCR), PMA provides a means to specifically detect and quantify viable pathogens. This technical guide details optimized protocols for both PMA-qPCR and PMA-ddPCR, providing researchers with robust methodologies for investigating the VBNC state in bacterial pathogens.

Core Principles: PMA Mechanism and PCR Platforms

The Propidium Monoazide (PMA) Mechanism

PMA functions as a **viability dye** based on its selective permeability characteristics. The molecule penetrates only bacteria with damaged or compromised cytoplasmic membranes – a definitive characteristic of dead cells. Once inside, PMA intercalates with DNA and, upon exposure to intense visible light, forms stable covalent bonds that permanently modify the DNA backbone. This modification **inhibits PCR amplification**, effectively silencing the signal from dead cells [43] [44]. In contrast, viable cells (including those in the VBNC state) with intact membranes exclude the dye, allowing their DNA to be amplified and detected normally. This principle forms the foundation for all PMA-coupled molecular detection methods.

Quantitative PCR (qPCR) measures PCR amplification in real-time using fluorescence, quantifying the target DNA based on the cycle threshold (Cq) at which fluorescence crosses a defined threshold. This requires a standard curve of known concentrations for absolute quantification [45] [46]. Its limitations include susceptibility to inhibition from sample matrices and dependence on amplification efficiency.

Droplet Digital PCR (ddPCR) represents a third-generation PCR technology that provides absolute quantification without standard curves. The reaction mixture is partitioned into thousands of nanoliter-sized water-in-oil droplets, each functioning as an individual PCR reactor. After endpoint amplification, droplets are counted as positive or negative based on fluorescence, and the absolute concentration of target DNA is calculated using Poisson statistics [42] [47]. This partitioning enhances resistance to PCR inhibitors and improves precision for low-abundance targets – critical advantages when working with stressed VBNC cells that may be present in low numbers.

Table 1: Comparison of qPCR and ddPCR Characteristics

Feature	PMA-qPCR	PMA-ddPCR
Quantification Method	Relative (requires standard curve)	Absolute (no standard curve needed)
Principle	Real-time fluorescence monitoring	End-point counting of positive partitions
Tolerance to Inhibitors	Moderate	High
Sensitivity	High	Very High
Precision	Good	Excellent
Throughput	High	Moderate to High
Cost	Lower	Higher

Experimental Protocols

Sample Preparation and PMA Treatment

Bacterial Strain and VBNC Induction

- Bacterial Strains:** Protocols have been successfully applied to various pathogens including *Escherichia coli* O157:H7 [42], *Vibrio cholerae* [47] [46], *Pseudomonas aeruginosa* [45], and *Brucella suis* [43].
- VBNC Induction:** For *V. cholerae*, grow to mid-exponential phase in LB broth, wash twice with artificial seawater (ASW), and resuspend in ASW at approximately 1×10^7 CFU/mL. Incubate at 4°C with oxygen limitation (fill vials completely to exclude air) for several weeks until culturability is lost on routine media while maintaining membrane integrity [46].

PMA Treatment Optimization

The efficiency of PMA treatment is critical for accurate viability assessment and must be optimized for each bacterial strain and sample matrix.

Table 2: Optimized PMA Treatment Conditions for Various Bacterial Species

Bacterial Species	Optimal PMA Concentration	Incubation Time	Light Exposure	Citation
<i>Vibrio cholerae</i>	20 µM	20 min (in dark)	15 min on ice, 650-W halogen lamp	[46]
<i>Brucella suis</i>	15 µg/mL	10 min (in dark)	Determined empirically, 650-W halogen lamp	[43]
<i>Escherichia coli</i> O157:H7	50 µM (suggested)	Not specified	15 min on ice, strong visible light	[42]
<i>Erwinia amylovora</i>	Up to 500 µM tested	Not specified	Determined empirically	[44]

Procedure:

- Prepare PMA stock solution (e.g., 1-20 mM in water or DMSO) and store at -20°C protected from light.
- Add appropriate volume of PMA to 200 µL of bacterial suspension to achieve desired working concentration.
- Incubate in the dark for the specified time with occasional mixing.
- Place samples on ice and expose to strong visible light (500-700 W halogen lamp) for 15-20 minutes at a distance of 15-20 cm to photo-activate the dye.
- Proceed to DNA extraction or direct PCR setup.

DNA Extraction and Quality Control

While commercial DNA extraction kits (e.g., Wizard Genomic DNA Purification Kit) are commonly used [\[46\]](#), the **direct oil-enveloped bacterial method** offers a rapid alternative for ddPCR that eliminates extraction steps. This method involves mixing bacterial cells directly with PCR premix, which are then encapsulated in oil droplets with cell lysis occurring during the initial PCR heating steps [\[47\]](#).

For traditional extraction:

- Use kit-specific protocols with optional double elution to maximize DNA yield [\[46\]](#).
- Quantify DNA concentration using spectrophotometry (NanoDrop) or fluorometry.
- Assess DNA purity (A260/A280 ratio ~1.8-2.0).

Primer and Probe Design Considerations

Target Selection:

- **Single-copy chromosomal genes** are essential for accurate cell enumeration. Examples include:
 - *gyrB* (DNA gyrase subunit B) for *P. aeruginosa* [\[45\]](#)
 - *VC1376* (GGDEF family protein), *thyA* (thymidylate synthase), and *recA* (ATP-binding protein) for *V. cholerae* [\[46\]](#)
 - *BCSP31* (cell surface protein) for *Brucella* species [\[43\]](#)
- **Amplicon length:** Longer amplicons (≥500 bp) improve PMA efficiency due to increased probability of dye binding, reducing false positives from dead cells [\[44\]](#).

Validation:

- Verify specificity against target species and closely related non-target organisms.
- Determine amplification efficiency (90-110% for qPCR).
- For ddPCR, optimize primer concentrations and annealing temperatures to maximize separation between positive and negative droplets.

PMA-qPCR Protocol

Reaction Setup:

- Prepare 20 µL reactions containing:
 - 1× FastStart Universal SYBR Green Master Mix or TaqMan Universal PCR Master Mix
 - 0.25-0.5 µM each forward and reverse primer
 - 1-5 µL template DNA (or 10-100 ng total DNA)
 - Nuclease-free water to volume
- For *Brucella* detection using BCSP31 gene: 94°C for 2 min; 40 cycles of 94°C for 45 s, 55°C for 30 s, 72°C for 32 s [\[43\]](#).

Quantification:

- Generate standard curve using serial dilutions of known concentrations of target DNA (genomic DNA or cloned target fragment).
- Calculate copy numbers or cell equivalents from standard curve.
- Include negative controls (no template) and positive controls in each run.

PMA-ddPCR Protocol

Reaction Setup and Partitioning:

- Prepare reaction mixture similar to qPCR but using ddPCR Supermix.
- For *V. cholerae* quantification: 20 µL reactions containing 1× EvaGreen Supermix, 0.25 µM primers, and template DNA [47].
- Generate droplets using appropriate droplet generator (e.g., QX200 Droplet Generator).
- Transfer droplets to 96-well PCR plate and seal properly.

Amplification and Reading:

- Perform PCR amplification with optimized cycling conditions.
- For *E. coli* O157:H7: 95°C for 10 min; 40 cycles of 94°C for 30 s and 60°C for 60 s; 98°C for 10 min; then 4°C hold [42].
- Read plate using droplet reader (e.g., QX200 Droplet Reader).
- Analyze data using companion software (QuantaSoft) to determine copies/µL in the original reaction.

Absolute Quantification Calculation:
$$[\text{Concentration (cells/}\mu\text{L)}] = \frac{\text{Copies/}\mu\text{L (from ddPCR)}}{\text{Number of target gene copies per cell}}$$
 For single-copy genes, the copy number equals cell number.

Critical Factors for Protocol Success

Optimization and Validation

- **PMA Concentration Titration:** Test increasing PMA concentrations (e.g., 10-100 µM) against samples containing known ratios of live:dead cells to determine the minimum concentration that completely suppresses dead cell signal [43] [44].
- **Killing Method Validation:** Use appropriate killing methods (heat, ethanol, antibiotics) that sufficiently compromise membrane integrity without fragmenting DNA. Validate complete killing by culture.
- **Inhibition Testing:** For complex samples (food, fecal, environmental), include internal controls or sample dilution to detect and overcome PCR inhibition.

Troubleshooting Common Issues

- **High Background from Dead Cells:** Increase PMA concentration; extend light exposure time; increase amplicon length; reduce thermal cycles in ddPCR [44].
- **Incomplete VBNC Induction:** Verify loss of culturability on multiple media types; confirm viability through membrane integrity staining (SYTO9).
- **Poor ddPCR Droplet Generation:** Ensure samples are free of particulates; vortex and spin down reaction mix before partitioning; check oil and cartridge storage conditions.

The Scientist's Toolkit: Essential Research Reagents

Table 3: Key Reagents and Materials for PMA-PCR Viability Detection

Reagent/Equipment	Function/Application	Examples/Notes
Propidium Monoazide (PMA)	Viability dye; selectively inhibits DNA amplification from membrane-compromised cells	Available from Biotium; prepare fresh stock solutions or aliquot stored at -20°C protected from light
Halogen Lamp	Photo-activation of PMA after incubation	500-700 W; maintain samples on ice during exposure to prevent overheating
Single-Copy Gene Primers/Probes	Target amplification for accurate cell counting	Must be validated for specificity and amplification efficiency; avoid multi-copy targets
Digital PCR System	Absolute quantification of target DNA	QX200 system (Bio-Rad); QuantStudio 3D (Thermo Fisher); Naica system (Stilla Technologies)
Droplet Generation Oil/Reagents	Creating water-in-oil partitions for ddPCR	Use manufacturer-recommended oils and consumables for optimal droplet generation
DNA Extraction Kits	Nucleic acid purification from bacterial samples	Wizard Genomic DNA Purification Kit (Promega); DNeasy Blood & Tissue Kit (Qiagen)

Reagent/Equipment	Function/Application	Examples/Notes
Viability Stains	Independent validation of membrane integrity	SYTO9 (component of LIVE/DEAD BacLight kit); propidium iodide (PI)

Application Workflows

The following diagram illustrates the complete experimental workflow for VBNC detection using PMA-ddPCR, from sample preparation to data analysis:

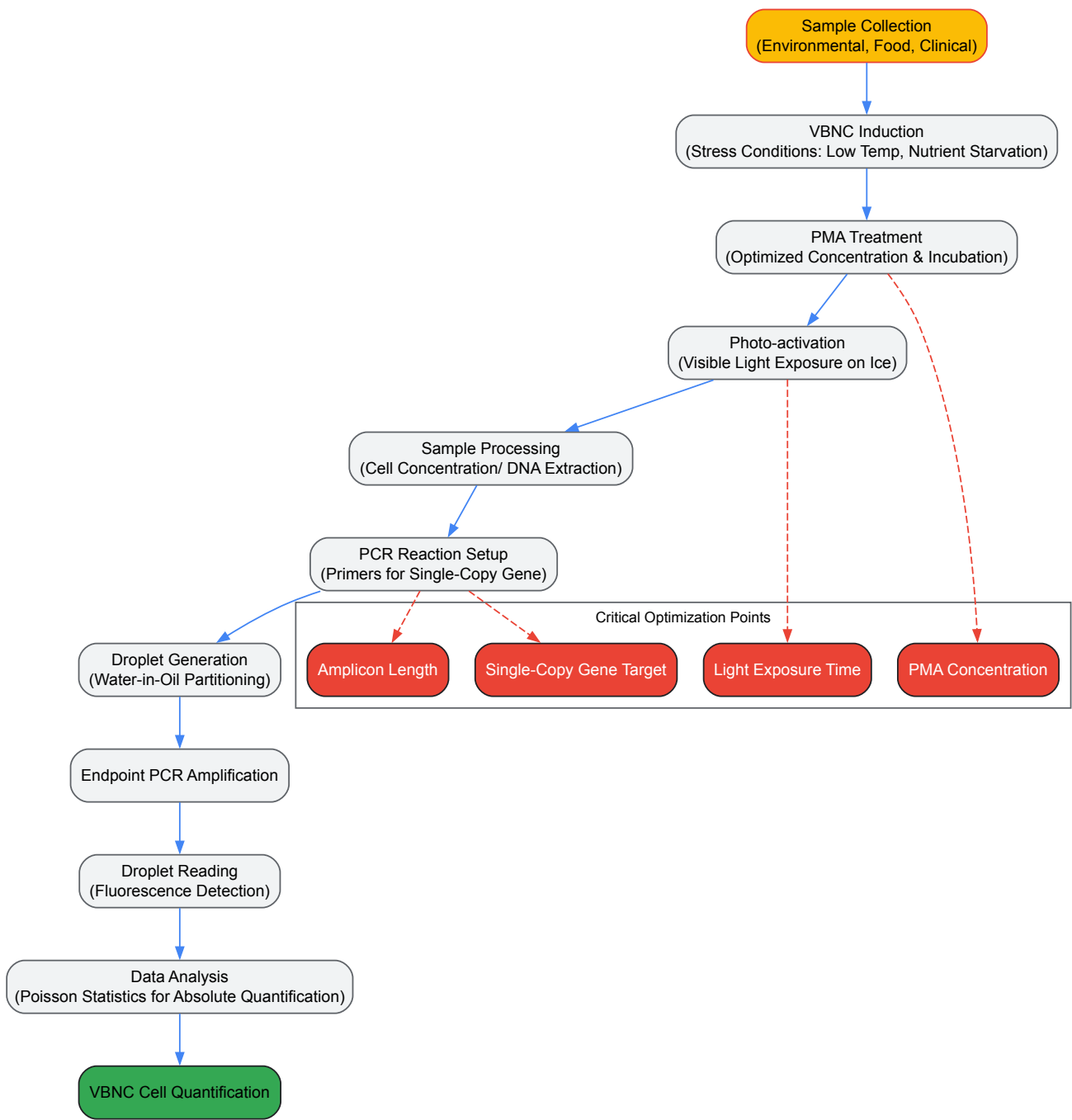


Diagram 1: Complete workflow for VBNC cell detection and quantification using PMA-ddPCR

The mechanism of PMA action and its integration with PCR detection is illustrated below:

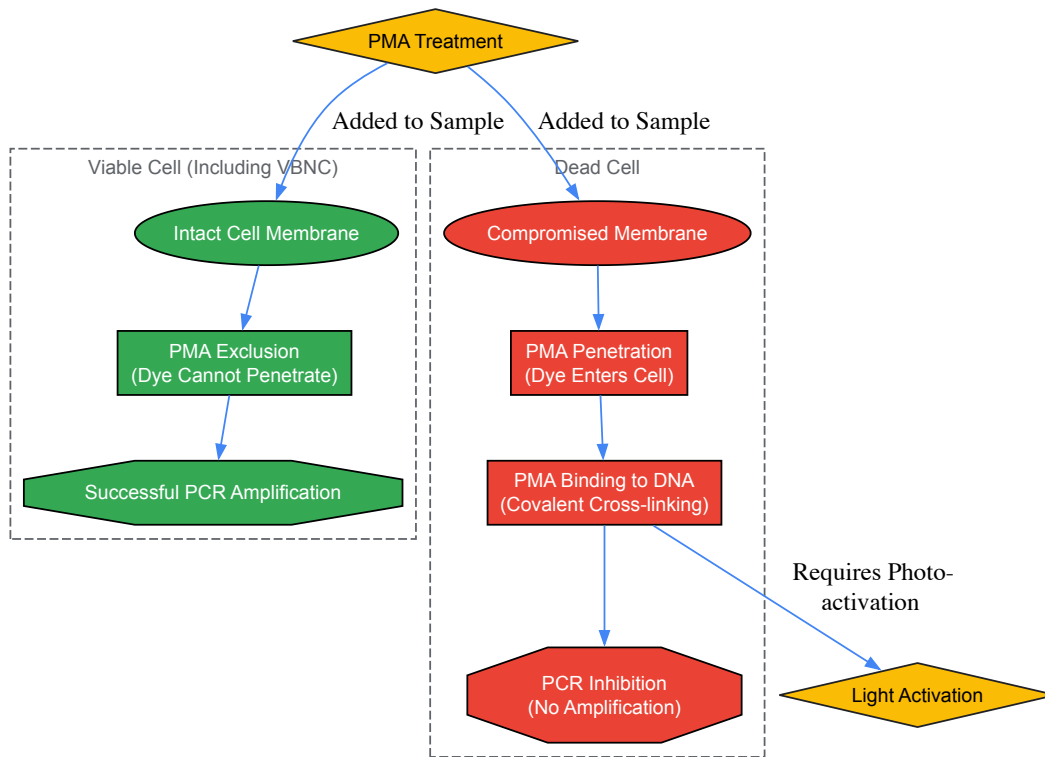


Diagram 2: Mechanism of PMA selection for viable cell detection

PMA-qPCR and PMA-ddPCR represent significant advancements in our ability to study the VBNC state in bacterial pathogens. While PMA-qPCR offers a more accessible platform for routine viability testing, PMA-ddPCR provides superior quantification accuracy, particularly for complex samples and low-abundance targets. The protocols detailed in this guide, when properly optimized and validated, enable researchers to accurately quantify VBNC populations, ultimately enhancing our understanding of bacterial dormancy and its implications for public health, food safety, and clinical microbiology.

The viable but non-culturable (VBNC) state represents a critical survival strategy for bacteria facing environmental stress, enabling persistence despite an inability to grow on standard culture media. This state poses significant challenges for public health, food safety, and pharmaceutical development, as VBNC cells maintain metabolic activity and membrane integrity while evading conventional detection methods. This technical guide comprehensively details flow cytometry and staining methodologies for assessing metabolic activity and membrane integrity in VBNC bacteria. We provide experimental protocols, quantitative data comparisons, and technical specifications for research reagents essential for investigating this dormant bacterial population. Within the broader context of VBNC research, these assays provide crucial tools for understanding bacterial dormancy, resuscitation mechanisms, and developing interventions against persistent infections.

The viable but non-culturable (VBNC) state is a dormant condition induced by various stressors, including nutrient deprivation, extreme temperatures, and exposure to antibiotics or disinfectants [48] [3]. In this state, bacteria cannot proliferate on routine culture media but remain alive with intact membranes and reduced metabolic activity [17] [14]. The significance of VBNC cells extends across multiple fields, from clinical microbiology to food safety, as they contribute to chronic infections, antimicrobial treatment failures, and product contamination despite negative culture tests [48] [14]. Notably, pathogens including *Listeria monocytogenes*, *Salmonella enterica*, and *Vibrio cholerae* can enter the VBNC state, posing hidden risks to public health [17] [49].

A primary challenge in VBNC research lies in accurately distinguishing these dormant cells from dead bacteria. Conventional culture-based methods are inadequate, necessitating alternative viability assessments based on membrane integrity and metabolic function [48]. Flow cytometry combined with fluorescent staining has emerged as a powerful tool for multiparametric analysis of bacterial physiology at single-cell resolution, enabling researchers to characterize VBNC populations without relying on culturability [50] [51].

Core Principles of Viability Assessment in VBNC Research

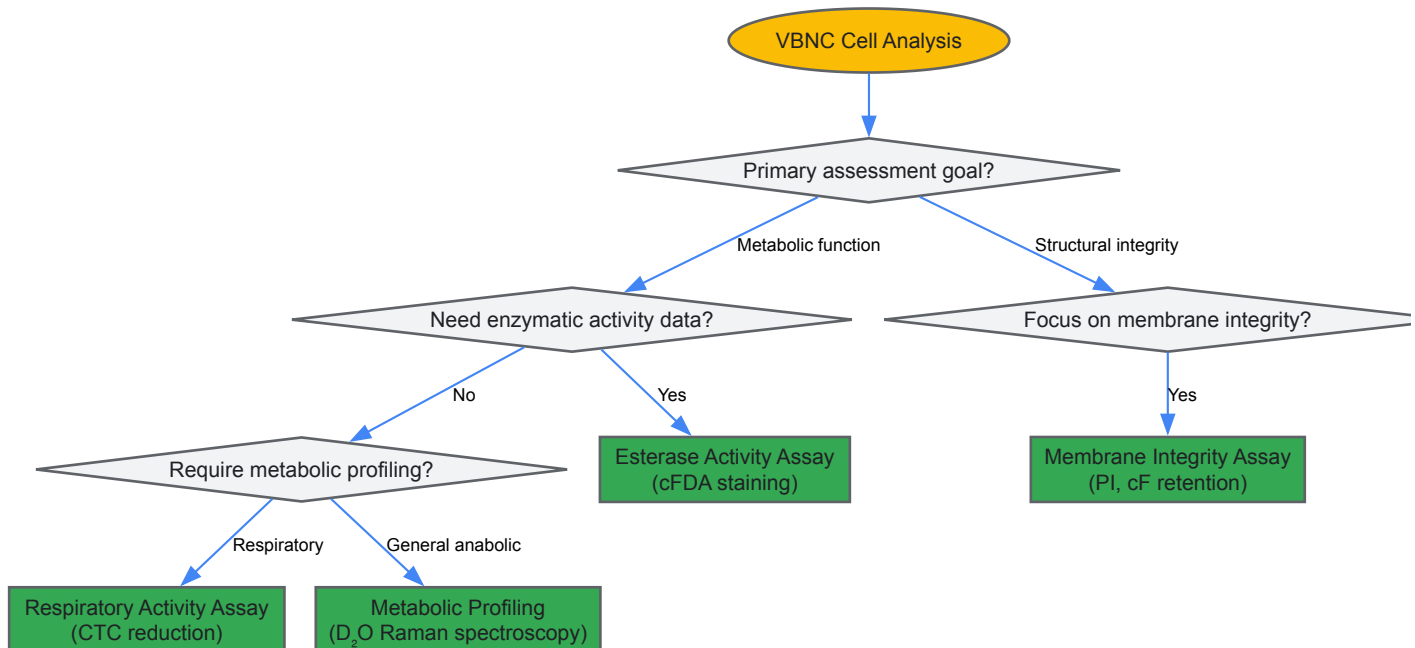
Membrane Integrity Assessment

The intact cytoplasmic membrane of viable cells prevents the penetration of certain fluorescent dyes, forming the basis for membrane integrity assays. VBNC cells maintain membrane integrity, unlike dead cells with compromised membranes [17]. Propidium iodide (PI) serves as a common indicator of membrane damage, as it only penetrates cells with compromised membranes and fluoresces upon binding to nucleic acids [51]. Conversely, carboxyfluorescein (cF) retention indicates membrane integrity, as this dye is retained only within cells with intact membranes [51].

Metabolic Activity Evaluation

Metabolic function in VBNC cells, though reduced, remains detectable through specific substrates. The reduction of tetrazolium salts like 5-cyano-2,3-ditolyl tetrazolium chloride (CTC) to fluorescent formazan indicates respiratory activity [52]. Esterase activity can be measured using fluorogenic substrates such as carboxyfluorescein diacetate (cFDA), which non-fluorescent cFDA is converted to fluorescent carboxyfluorescein (cF) by intracellular esterases [51]. Additionally, deuterium oxide (D₂O)-labeled Raman spectroscopy detects general metabolic activity by measuring microbial incorporation of deuterium into biomass, reflecting the overall anabolic activity in VBNC cells [52].

The following diagram illustrates the decision workflow for selecting appropriate viability assays in VBNC research:



Flow Cytometry Methodologies for VBNC Detection

Sample Preparation Protocol

Proper sample preparation is critical for accurate flow cytometric analysis of VBNC cells. The following protocol outlines essential steps for processing bacterial samples:

- **Harvesting and Washing:** Harvest bacterial cultures by gentle centrifugation at $\sim 200 \times g$ for 5 minutes at 4°C. Wash cells twice with ice-cold phosphate-buffered saline (PBS) containing 5-10% fetal calf serum to maintain cell viability [53].
- **Cell Concentration Adjustment:** Resuspend the final pellet in suspension buffer to a concentration of $0.5\text{--}1 \times 10^6$ cells/mL. Higher concentrations may clog the flow cytometer and affect resolution [53].
- **Viability Staining:** Incubate cells with appropriate viability dyes according to manufacturer protocols, typically in the dark at 4°C. Select dyes with emission spectra that do not overlap with fluorophores used for subsequent immunostaining [53].
- **Fixation and Permeabilization (for intracellular targets):** For intracellular staining, fix cells with 1-4% paraformaldehyde for 15-20 minutes on ice, followed by permeabilization with detergents like Triton X-100 (0.1-1%) or saponin (0.2-0.5%) for 10-15 minutes at room temperature [53].
- **Fc Receptor Blocking:** To prevent non-specific antibody binding, incubate cells with blocking buffers such as 2-10% goat serum, human IgG, or mouse anti-CD16/CD32 for 30-60 minutes in the dark at 4°C [53].

Membrane Integrity Assessment Using Multiparametric Flow Cytometry

Multiparameter flow cytometry enables simultaneous assessment of multiple physiological parameters, allowing discrimination between different subpopulations in bacterial cultures. A representative protocol for membrane integrity assessment in *Oenococcus oeni* can be adapted for VBNC research:

- **Cell Loading with cF:** Deenergize cells with 2-deoxyglucose (2 mM final concentration) for 30 minutes at room temperature. Load cells with 5(6)-carboxyfluorescein diacetate (cFDA) stock solution (2.3 mg/mL in acetone) to assess esterase activity and membrane integrity [51].

- **Propidium Iodide Staining:** Combine cF-loaded cells with propidium iodide (PI) to distinguish membrane-compromised cells. Final concentration should be optimized for specific bacterial species [51].
- **Flow Cytometric Analysis:** Analyze samples using appropriate flow cytometer configurations. Intact cells display cF fluorescence only, permeable cells show both cF and PI fluorescence, and damaged cells exhibit PI fluorescence only [51].

This approach revealed that ethanol-adapted *O. oeni* cells maintained membrane integrity three times better than non-adapted cells when exposed to ethanol stress, demonstrating the utility of this method for assessing physiological adaptations in bacteria [51].

Metabolic Activity Measurement via CTC Reduction

The CTC-FCM method assesses respiratory activity in VBNC cells through the following protocol:

- **CTC Staining:** Incubate bacterial samples with CTC at optimal concentration (typically 5-10 mM) in the dark at room temperature for 60-90 minutes [52].
- **Flow Cytometric Analysis:** Analyze samples using flow cytometry with appropriate excitation (typically 488 nm) and emission detection (typically 630 nm) settings for CTC-formazan.
- **Data Interpretation:** Actively respiring cells reduce CTC to fluorescent CTC-formazan, detectable at the single-cell level. This method has proven effective for detecting essential viability in UV-induced VBNC cells of *Aeromonas* sp., *Pseudomonas* sp., *E. coli*, and *S. aureus* [52].

Comparative Analysis of Viability Assessment Methods

Method Performance in Complex Matrices

The accurate detection of VBNC cells varies significantly across methodologies, particularly in complex matrices like process wash water (PWW) from food processing environments. The following table summarizes the performance characteristics of different VBNC detection methods:

Table 1: Performance Comparison of VBNC Detection Methods

Method	Principle	Detection Target	Advantages	Limitations	Suitable for Complex Matrices
Flow Cytometry with viability dyes [17]	Membrane integrity & metabolic activity	Dye retention/exclusion	Rapid, single-cell resolution, high throughput	Potential overestimation of dead cells in complex matrices, requires specialized equipment	Limited in complex water matrices with high organic content
v-qPCR with PMAx/EMA [17]	DNA amplification inhibition in membrane-compromised cells	DNA from cells with intact membranes	Specific, sensitive, culture-independent	May not detect all VBNC cells, requires optimization of dye concentration	Yes (validated for PWW with 10 μM EMA + 75 μM PMAx)
CTC-FCM [52]	Respiratory activity	CTC-formazan in respiring cells	Detects essential viability, single-cell resolution	Does not detect cells with inactive electron transport chain	Yes for drinking water
D2O-Raman Spectroscopy [52]	Deuterium incorporation	Anabolic activity	Labels overall metabolic activity, single-cell resolution	Requires specialized equipment, complex data analysis	Yes for drinking water

Quantitative Assessment of Bacterial Responses to Stressors

Flow cytometry multiplexing enables quantitative analysis of how different bacterial species respond to pharmaceutical compounds and environmental stressors. The following table compiles IC50 values from studies investigating compound toxicity under varying conditions:

Table 2: Quantitative Assessment of Compound Toxicity via Flow Cytometry Multiplexing (IC50 Values in μM) [59]

Compound	Cell Type	O ₂ Level	PrestoBlue (Metabolic Activity)	LIVE/DEAD Aqua (Membrane Integrity)	Key Findings
Amsacrine	Jurkat	19% O ₂	~0.57	4.96	8.7-fold difference between metabolic and membrane integrity readouts
	Jurkat	1% O ₂	0.26	6.60	25.4-fold difference between readouts
	Ramos	19% O ₂	~23.30	~71.57	Different potency between cell types
	Ramos	1% O ₂	~20.17	~64.95	Consistent differential response
Clofazimine	Jurkat	19% O ₂	10.69	NA	Oxygen-dependent toxicity
	Jurkat	1% O ₂	~30.86	NA	~3-fold reduced potency at low oxygen
	Ramos	19% O ₂	4.77	NA	Cell-type specific sensitivity
	Ramos	1% O ₂	~9.21	NA	~2-fold reduced potency at low oxygen
Benzyl isothiocyanate	Jurkat	19% O ₂	1.48	NA	Similar metabolic response in both cell types at 19% O ₂
	Jurkat	1% O ₂	1.06	NA	Slightly increased potency at low oxygen
	Ramos	19% O ₂	1.50	NA	Similar metabolic response in both cell types at 19% O ₂
	Ramos	1% O ₂	~2.63	NA	~2.5-fold reduced potency at low oxygen

Essential Reagents and Research Solutions

The following table compiles key reagents and their applications in VBNC research:

Table 3: Essential Research Reagents for VBNC Analysis

Reagent Category	Specific Examples	Function & Mechanism	Application Notes
Membrane Integrity Dyes	Propidium iodide (PI) [51]	Penetrates compromised membranes, binds nucleic acids	Impermeant to VBNC cells; indicates dead population
	Carboxyfluorescein diacetate (cFDA) [51]	Converted to membrane-impermeant cF by esterases	Retention indicates intact membrane & enzyme activity
	LIVE/DEAD Fixable Aqua [50]	Reacts with free amines in compromised cells	Distinguishes live/dead populations in multiplex assays
Metabolic Activity Indicators	5-cyano-2,3-ditolyl tetrazolium chloride (CTC) [52]	Reduced to fluorescent formazan by respiratory chain	Detects respiratory activity in VBNC cells
	PrestoBlue Cell Viability Reagent [50]	Resazurin reduction by reducing power of living cells	Measures metabolic activity; used in HTS applications
	D ₂ O (Deuterium Oxide) [52]	Incorporation into biomass during synthesis	Raman-detectable marker for overall anabolic activity
Viability qPCR Dyes	PMAxx [17]	Photoactivatable dye binding DNA in membrane-compromised cells	Inhibits PCR amplification from dead cells in v-qPCR
	EMA [17]	Ethidium monoazide; penetrates compromised membranes	Used in combination with PMAxx for enhanced discrimination
Fixation & Permeabilization Reagents	Paraformaldehyde (1-4%) [53]	Cross-linking fixative preserving cellular structure	Standard fixation for intracellular staining
	Methanol (90%) [53]	Precipitating fixative	Alternative to PFA; may denature some epitopes

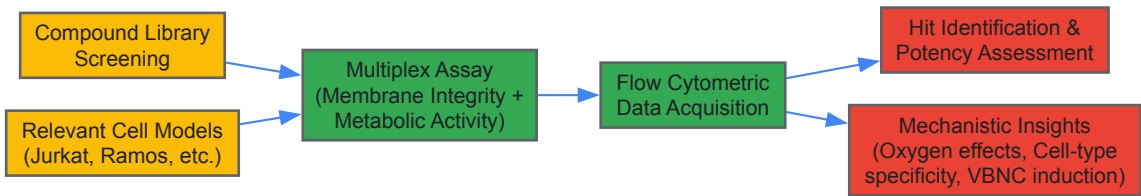
Reagent Category	Specific Examples	Function & Mechanism	Application Notes
	Triton X-100, Saponin [53]	Detergents for membrane permeabilization	Enables antibody access to intracellular targets

Applications in Pharmaceutical Development and VBNC Research

Flow cytometry-based metabolic and membrane integrity assays provide critical insights for pharmaceutical development, particularly in compound screening and mechanistic toxicology:

- **High-Throughput Screening (HTS):** Acoustic flow cytometry platforms like the Attune NxT enable rapid screening of compound libraries, with Z' factors of ~0.8 for membrane integrity assays and ~0.7 for metabolic activity assays, indicating excellent suitability for HTS [50].
- **Mechanistic Toxicity Assessment:** Multiparametric analysis reveals differential compound effects on metabolic pathways versus structural integrity. For instance, amsacrine showed significantly different IC50 values when comparing metabolic activity (PrestoBlue) and membrane integrity (LIVE/DEAD Aqua) readouts [50].
- **Oxygen-Dependent Drug Potency:** Flow cytometry multiplexing demonstrated that oxygen levels significantly impact compound potency. Clofazimine was 3-fold more potent against Jurkat cells at 19% O2 compared to 1% O2 when assessing metabolic activity [50].
- **Detection of VBNC Pathogens in Biofilms:** The technique identifies VBNC cells within oral biofilms, including pathogens like *Porphyromonas gingivalis* and *Enterococcus faecalis*, which contribute to chronic infections and treatment resistance [14].

The following diagram illustrates the integration of these assays in pharmaceutical screening workflows:



Metabolic and membrane integrity assays using flow cytometry and staining techniques provide indispensable tools for investigating the VBNC state in bacteria. These methodologies enable researchers to overcome the limitations of culture-based approaches and gain critical insights into bacterial dormancy, resuscitation potential, and antimicrobial resistance. The comprehensive protocols, reagent specifications, and quantitative data presented in this technical guide offer researchers a robust foundation for advancing VBNC research, with significant implications for pharmaceutical development, food safety, and clinical microbiology. As understanding of the VBNC state continues to evolve, these analytical approaches will play an increasingly vital role in developing effective interventions against persistent bacterial infections and contamination.

Gene Expression and Proteomic Profiling of VBNC Cells

The **viable but non-culturable (VBNC) state** represents a sophisticated survival strategy adopted by numerous bacterial species when confronted with adverse environmental conditions. In this dormant state, bacteria **maintain metabolic activity** and membrane integrity but **lose the ability to form colonies** on conventional laboratory media that typically support their growth [12]. This phenomenon has profound implications across clinical, environmental, and food safety sectors, as VBNC cells can evade standard detection methods while retaining virulence potential and the capacity to resuscitate under favorable conditions [12] [14]. The advent of advanced omics technologies has revolutionized our understanding of this enigmatic physiological state, enabling researchers to decipher the complex molecular reprogramming that occurs during VBNC entry, maintenance, and resuscitation.

Gene expression and proteomic profiling have emerged as powerful tools for elucidating the fundamental mechanisms governing the VBNC state. These technologies provide comprehensive snapshots of the transcriptional and translational adaptations that enable bacterial cells to enter a dormant yet viable condition. Through transcriptomics, scientists can identify key genes and pathways that are upregulated or downregulated in response to stressors such as nutrient limitation, temperature shifts, oxidative stress, and exposure to heavy metals or antibiotics [54] [55]. Similarly, proteomic analyses reveal how protein expression patterns shift to reconfigure cellular metabolism, enhance stress resistance, and maintain viability in the absence of replication [56] [57]. This technical guide synthesizes current methodologies, findings, and applications in VBNC research, providing researchers with a comprehensive framework for investigating this remarkable bacterial survival strategy.

Transcriptomic Landscapes of VBNC Cells

Global Gene Expression Changes

Transcriptomic analyses using **RNA sequencing (RNA-Seq)** have revealed profound reprogramming of gene expression in bacterial cells entering the VBNC state. In *Bacillus subtilis* induced into the VBNC state through osmotic stress and kanamycin treatment, significant alterations were observed across the transcriptome, with **334 genes upregulated** and **514 genes downregulated** compared to untreated control cells [55]. Notably, the transcriptional differences between VBNC cells and kanamycin-sensitive cells were relatively moderate, suggesting common elements in the stress response pathways, with 73 genes upregulated and 185 downregulated in VBNC cells specifically [55].

The induction of the **ICEBs1 conjugative element** represents one of the most prominent transcriptional changes observed in VBNC *B. subtilis* cells. This genetic element, typically activated by quorum sensing or DNA damage stress, showed significant upregulation, potentially indicating a response to antibiotic-induced oxidative stress and genomic damage [55]. Additionally, VBNC cells strongly **upregulate genes involved in proline uptake and catabolism**, suggesting a putative role for proline as a critical nutrient source during dormancy [55]. This transcriptional reprogramming reflects the complex metabolic adjustments necessary for maintaining viability under non-growing conditions.

Table 1: Key Transcriptional Changes in VBNC Cells Across Bacterial Species

Bacterial Species	Induction Method	Upregulated Genes/Pathways	Downregulated Genes/Pathways
<i>Bacillus subtilis</i>	Osmotic stress + kanamycin	ICEBs1 conjugative element, queosine biosynthesis, proline catabolism	General metabolic processes, ribosomal proteins
<i>Pseudomonas syringae</i> pv. <i>syringae</i>	Acetosyringone oxidation	Oxidative stress response, detoxification	Energy metabolism, transport systems
<i>Vibrio cholerae</i>	Low temperature incubation	Regulatory functions, energy metabolism, transport	Cell division, biosynthesis pathways

Stress-Specific Transcriptional Responses

The transcriptional profile of VBNC cells varies significantly depending on the specific environmental stressor inducing this state. In *Pseudomonas syringae* pv. *syringae*, exposure to oxidized acetosyringone—a phenolic compound associated with plant defense responses—triggered a distinct transcriptomic signature characterized by **upregulation of oxidative stress response genes** and **detoxification pathways** [54]. This response enables the pathogen to withstand the redox pulse encountered during plant invasion, potentially contributing to its ability to establish latent infections.

The **queosine biosynthesis pathway** represents another prominently upregulated system in VBNC *B. subtilis* cells, with significant induction of the *queC-queD-queE-queF* operon [55]. Queosine is a modified nucleoside found in tRNA that influences translational accuracy and efficiency. Under antibiotic stress, increased queosine production may represent an adaptive mechanism to minimize translation errors caused by kanamycin, which binds to the 30S ribosomal subunit and promotes protein misfolding. Supporting this hypothesis, mutants deficient in *queG* (encoding the final enzyme in queosine biosynthesis) showed **enhanced susceptibility to kanamycin** after prolonged exposure, underscoring the importance of this pathway in antibiotic tolerance [55].

Proteomic Adaptations in the VBNC State

Time-Resolved Proteomic Dynamics

Advanced **quantitative liquid chromatography tandem mass spectrometry (LC-MS/MS)** enables detailed, time-resolved profiling of proteomic changes throughout the VBNC lifecycle. In *Cupriavidus metallidurans* CH34 exposed to sublethal copper concentrations, proteomic analysis at multiple time points revealed a highly dynamic response characterized by **widespread oxidative stress response proteins** and **downregulated pyruvate metabolism** during VBNC entry [56]. Interestingly, the spontaneous resuscitation phase after 7 days of incubation correlated with **strong upregulation of periplasmic copper detoxification proteins**, suggesting that delayed induction of specific metal resistance determinants facilitates recovery of culturability [56].

The integration of proteomic data with metabolic measurements has provided insights into the energetic basis of VBNC survival and resuscitation. In copper-stressed *C. metallidurans*, the gradual reconstitution of energy reserves through **metabolization of intracellular polyhydroxybutyrate (PHB)** appears to support the resuscitation process [56]. This coordinated metabolic and proteomic reprogramming highlights the sophisticated nature of the VBNC state as an active survival strategy rather than a passive deterioration process.

Table 2: Proteomic Changes During VBNC State Formation and Resuscitation in *Cupriavidus metallidurans* CH34

Phase	Key Proteomic Features	Metabolic Correlations	Functional Significance
Early Entry	Upregulation of oxidative stress response proteins; Downregulation of pyruvate metabolism	Reduced energy metabolism	Protection against copper-induced oxidative damage; Energy conservation

Phase	Key Proteomic Features	Metabolic Correlations	Functional Significance
Maintenance	Persistent stress response; Reduced ribosomal proteins	Low but detectable metabolic activity	Cellular integrity maintenance; Minimal protein synthesis
Resuscitation	Strong upregulation of periplasmic copper resistance proteins; Increased motility/chemotaxis proteins	PHB utilization for energy reconstitution	Metal detoxification; Seeking favorable environments

Temperature-Induced Proteomic Remodeling

Temperature stress represents a common inducer of the VBNC state across bacterial species. In *Vibrio parahaemolyticus* LF1113, proteomic analysis at three different temperatures (4°C, 25°C, and 37°C) revealed **2,899 differentially expressed proteins** and **396 altered metabolites** [58]. The integrated proteomic and metabolomic approach demonstrated that cold stress (4°C) triggered significant reprogramming of pathways involved in **lysine degradation, ABC transporters, and diverse metabolic processes** [58]. These adaptations enable bacterial cells to maintain membrane fluidity and essential metabolic functions under low-temperature conditions that would otherwise inhibit growth.

The proteomic profile of VBNC cells consistently shows **alterations in outer membrane protein composition** across diverse bacterial species. In *Escherichia coli*, the VBNC state is associated with a **marked increase in OmpW** levels [12], while *Helicobacter pylori* transitions to the coccoid VBNC form show changed outer membrane protein patterns [12]. These structural modifications likely contribute to enhanced stress resistance and altered permeability during dormancy. Additionally, VBNC cells often display **reduced levels of ribosomal proteins** and translation factors, consistent with their low metabolic state and limited protein synthesis capacity [12].

Detection and Quantification Methods for VBNC Cells

Molecular Detection Approaches

Conventional culture-based methods fail to detect VBNC cells, necessitating specialized approaches that distinguish them from dead cells. **Viability quantitative PCR (v-qPCR)** combined with photoreactive dyes such as **propidium monoazide (PMA)** or **ethidium monoazide (EMA)** has emerged as a powerful technique for VBNC detection and quantification [17]. These dyes penetrate membrane-compromised dead cells and bind covalently to DNA upon photoactivation, preventing PCR amplification and thus selectively targeting intact cells [17]. For complex matrices like process wash water from food industries, optimal results were achieved using **10 µM EMA and 75 µM PMAxx** (an improved PMA version) incubated at 40°C for 40 minutes followed by 15 minutes of light exposure [17].

More recently, **droplet digital PCR (ddPCR)** has been applied to VBNC research, offering absolute quantification without external standard curves. In high alcohol-producing *Klebsiella pneumoniae* (HiAlc Kpn), PMA-ddPCR targeting single-copy genes (*KP*, *rpoB*, and *adhE*) enabled precise enumeration of VBNC cells [9] [10]. This method demonstrated that ciprofloxacin could inhibit the resuscitation of VBNC-state cells while maintaining their ethanol production capacity after antibiotic removal [10]. The ddPCR approach provides enhanced sensitivity and reliability for quantifying VBNC populations in complex samples like fecal matter, with applications in clinical diagnostics and environmental monitoring.

Complementary Viability Assessment

While molecular methods offer specificity and sensitivity, **flow cytometry** combined with fluorescent staining provides complementary information about cell viability and physiological status. Using kits such as the **LIVE/DEAD BacLight** bacterial viability assay, which contains SYBR Green and propidium iodide (PI), researchers can distinguish cells with intact membranes (viable) from those with compromised membranes (dead) [54]. However, this technique may overestimate dead cells in complex matrices like process wash water due to interference from particulate matter [17].

Additional approaches for VBNC characterization include **metabolomic profiling** to assess metabolic activity, **transmission electron microscopy (TEM)** to visualize morphological changes, and **fluorescence microscopy** to confirm membrane integrity [58] [10]. The combination of multiple independent methods provides the most comprehensive assessment of the VBNC state, overcoming limitations inherent in any single technique.

Experimental Protocols for VBNC Research

Induction of the VBNC State

Protocols for inducing the VBNC state vary depending on the bacterial species and research objectives. For *Cupriavidus metallidurans* CH34, the VBNC state can be induced by washing cells twice with filter-sterilized bottled mineral water and diluting to a starting inoculum of **10^8 cells/mL** in the same mineral water supplemented with **10 µM CuSO4**, followed by incubation in a shaking incubator at 30°C [56]. Samples are typically drawn at multiple time points (0, 1, 2, 3, 4, 5, 24, 48, 72, 96, and 120 h) to monitor the progression into the VBNC state through culturability assessments and molecular analyses [56].

For *Bacillus subtilis*, an effective induction protocol involves pre-adaptation to **0.6 M NaCl osmotic stress** followed by treatment with a lethal dose of **kanamycin** [55]. This combination of stresses locks cells in a hyperpolarized state with metabolic activity but no proliferative capacity. Similarly, *Vibrio parahaemolyticus* can be induced into the VBNC state by incubation in **artificial seawater at 4°C** for extended periods, with culturability monitored regularly until no colonies appear on conventional media [10].

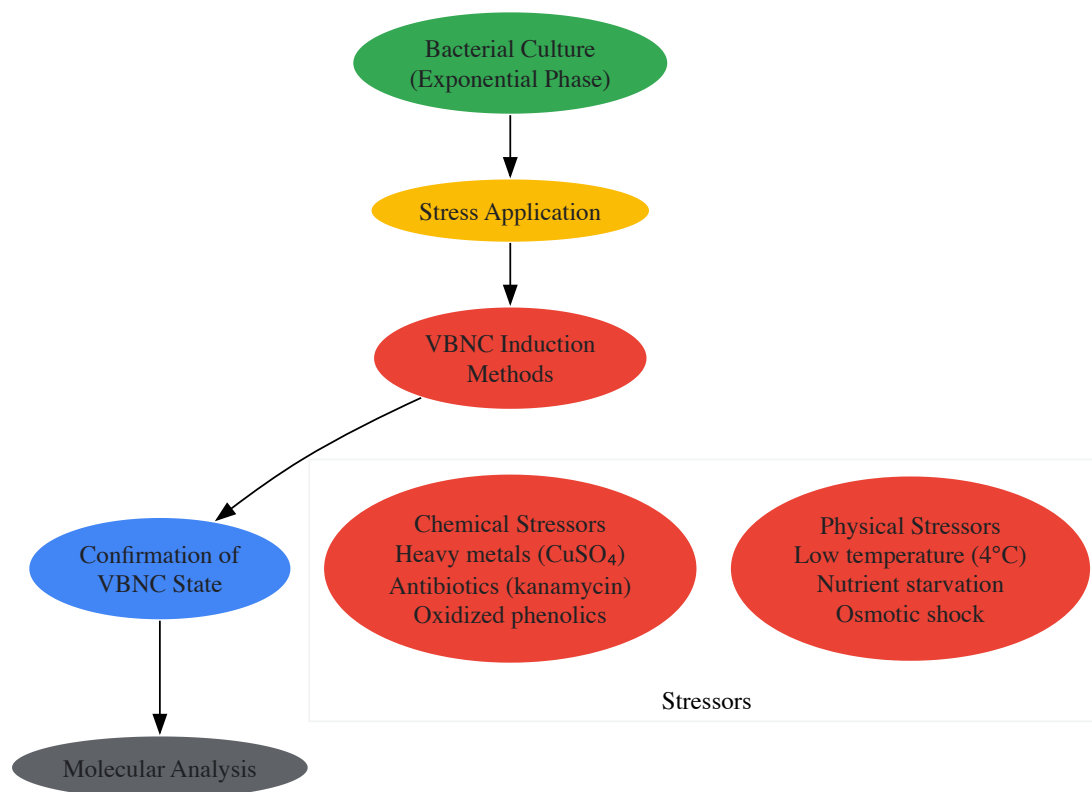


Diagram 1: VBNC State Induction Workflow. This diagram illustrates the general workflow for inducing the VBNC state in bacterial cells, highlighting common chemical and physical stressors used in research protocols.

Proteomic Sample Preparation and Analysis

Comprehensive proteomic analysis of VBNC cells requires careful sample preparation to preserve protein integrity and representativeness. For protein extraction from *C. metallidurans*, samples are pelleted and washed twice with ice-cold PBS, then resuspended in lysis buffer containing **2% sodium dodecyl sulfate (SDS)** in **50 mM ammonium bicarbonate** aqueous solution [56]. After vortexing and incubation at 95°C for 5 minutes, complete lysis is achieved through sonication. The lysate is centrifuged at $14,000 \times g$ at 4°C for 20 minutes, and the supernatant containing soluble proteins is stored at -80°C for further processing [56].

For **tandem mass tag (TMT)-based quantitative proteomics**, extracted proteins are quantified using the bicinchoninic acid (BCA) assay, followed by processing using the **suspension trapping (S-trap) method** [56]. Proteins are digested with trypsin, and the resulting peptides are labeled with TMT reagents according to the manufacturer's protocol. Labeled peptides are fractionated by reverse-phase chromatography before LC-MS/MS analysis on instruments such as the **Q Extractive Plus mass spectrometer** [58]. MS/MS spectra are typically searched against appropriate protein databases using software such as **MASCOT engine** embedded in **Proteome Discoverer** [58].

Transcriptomic Analysis Workflow

RNA sequencing provides comprehensive insights into transcriptional changes during VBNC state formation. For *B. subtilis* transcriptome analysis, RNA is extracted from non-stressed culturable cells, kanamycin-sensitive cells, and VBNC cells, followed by preparation of sequencing libraries [55]. After **high-throughput RNA sequencing**, reads are mapped to reference genomes, and expression values are calculated as **reads per kilobase per million mapped reads (RPKM)**. **Differential gene expression analysis** identifies significantly upregulated and downregulated genes, which are then categorized through **Gene Set Enrichment Analysis (GSEA)** and **Gene Ontology (GO) term analysis** to determine functional enrichment [55].

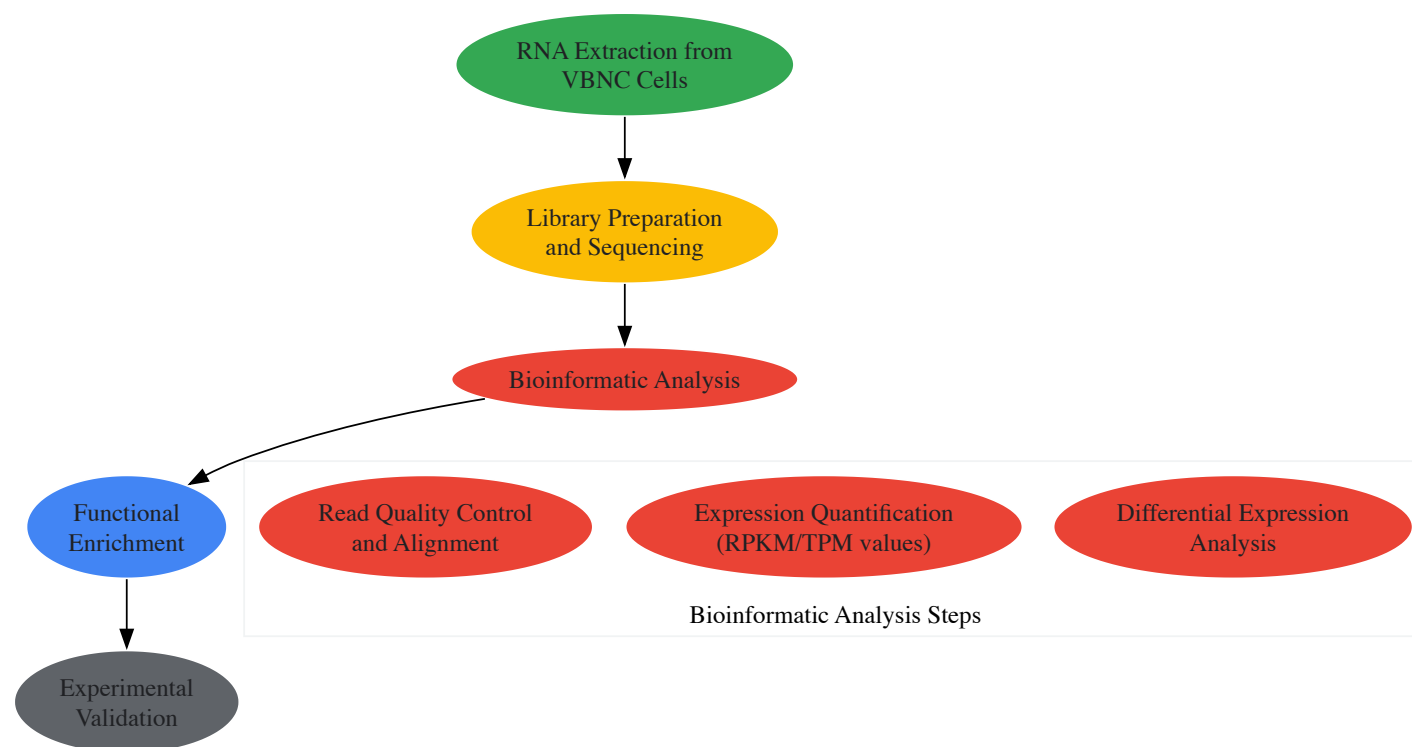


Diagram 2: Transcriptomic Analysis of VBNC Cells. This workflow outlines the key steps in RNA sequencing and analysis to identify gene expression changes characteristic of the VBNC state.

The Scientist's Toolkit: Essential Research Reagents

Table 3: Essential Research Reagents for VBNC Cell Studies

Reagent/Category	Specific Examples	Function in VBNC Research
VBNC Inducers	CuSO4, kanamycin, acetosyringone + H2O2, low temperature incubation	Trigger transition to VBNC state through various stress mechanisms
Viability Stains	LIVE/DEAD BacLight (SYBR Green/PI), Nile Red, PMA, EMA, PMAxx	Differentiate viable and dead cells based on membrane integrity; inhibit DNA amplification from dead cells
Protein Analysis	SDS, ammonium bicarbonate, BCA protein assay kit, TMT reagents, trypsin	Protein extraction, quantification, digestion, and labeling for proteomic studies
Nucleic Acid Analysis	RNA extraction kits, reverse transcription reagents, PCR/qPCR/ddPCR reagents, primers for target genes	Gene expression analysis and viable cell quantification
Chromatography/Mass Spectrometry	LC-MS/MS systems, C18 columns, formic acid, acetonitrile	Separation and identification of proteins, peptides, and metabolites
Bioinformatic Tools	MASCOT engine, Proteome Discoverer, FUNAGE-Pro, GSEA software, KEGG database	Protein identification, functional annotation, pathway analysis

Gene expression and proteomic profiling have transformed our understanding of the VBNC state, revealing it as a **actively regulated survival strategy** with characteristic molecular signatures rather than a passive deterioration process. The integration of transcriptomic, proteomic, and metabolomic data provides unprecedented insights into the complex regulatory networks that govern bacterial dormancy and resuscitation. As detection methods continue to advance, particularly with digital PCR and enhanced viability staining approaches, researchers are better equipped to investigate the VBNC state in complex environments and clinical settings.

The molecular profiling approaches detailed in this technical guide illuminate the **dynamic adaptations** that enable VBNC cells to persist under adverse conditions, including **reconfigured metabolic pathways**, **enhanced stress response systems**, and **structural modifications**. These insights not only advance fundamental knowledge of bacterial physiology but also inform practical applications in clinical diagnostics, food safety, and environmental monitoring. Future research leveraging single-cell omics technologies and temporal resolution of the transition phases will further elucidate the heterogeneous nature of VBNC populations and their role in disease recurrence and environmental persistence.

The **viable but nonculturable (VBNC) state** represents a unique survival strategy adopted by many bacterial pathogens when exposed to adverse environmental conditions. In this state, cells undergo a distinct physiological transition, maintaining metabolic activity and virulence potential while losing the ability to form colonies on routine laboratory media that normally support their growth [37] [59]. This phenomenon poses a substantial challenge for clinical microbiology and public health, as standard culture-based methods—the historical gold standard for pathogen detection—fail to detect these living, potentially infectious cells [60] [37]. Consequently, the reliance on culturability for pathogen tracking results in a significant underestimation of viable pathogen load, misleading assessments of infection sources, and potential diagnostic inaccuracies.

The molecular mechanisms underlying the VBNC state involve significant physiological and genetic reprogramming. Proteome analyses reveal that VBNC cells exhibit protein expression patterns distinct from both exponentially growing and starved cells, confirming the VBNC state as a unique stress response [61]. These changes include morphological alterations such as cell dwarfing and rounding, increased peptidoglycan cross-linking in the cell wall, reduced metabolic rates, and enhanced resistance to physical and chemical stresses, including antibiotics [37]. Understanding this state is therefore critical for developing accurate pathogen tracking methodologies from the gut microbiome to clinical bloodstream samples, enabling more precise interventions for infection control.

Advanced Detection Methodologies for VBNC Cells

Accurate detection of VBNC cells requires moving beyond traditional culture plates. Modern viability assessments are categorized into three main strategies based on accepted viable criteria: culturability, metabolic activity, and membrane integrity [60]. The following sections detail the current methodologies, highlighting standardized protocols and their applications in complex samples.

Viability qPCR (v-qPCR) with Photoreactive Dyes

Viability quantitative PCR (v-qPCR) combined with photoreactive dyes like propidium monoazide (PMA) or ethidium monoazide (EMA) has emerged as a powerful tool for discriminating between viable and dead cells in complex matrices. This method is based on the principle that these dyes can penetrate cells with compromised membranes (dead cells), bind covalently to DNA upon photoactivation, and thereby inhibit PCR amplification. In contrast, viable cells (including VBNC cells) with intact membranes exclude the dye, allowing their DNA to be amplified and detected [17].

Table 1: Key Reagents for v-qPCR-based Detection of VBNC Cells

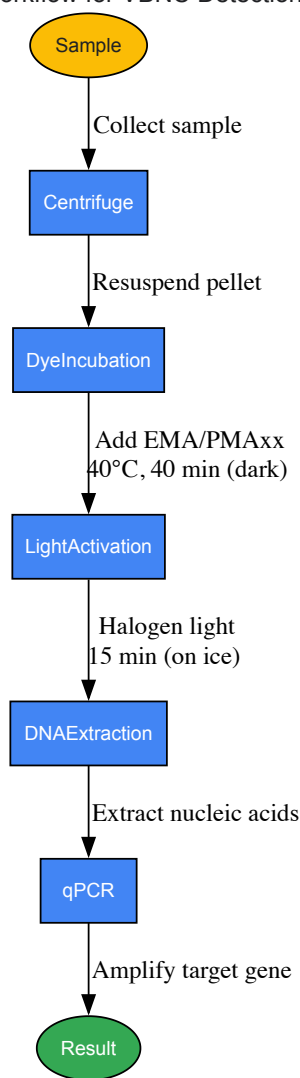
Reagent / Solution	Function	Considerations for Use
PMAxx Dye	Improved version of PMA; penetrates only dead cells with damaged membranes and inhibits DNA amplification.	Less prone to penetration into viable cells compared to EMA.
Ethidium Monoazide (EMA)	Photosensitive dye that crosses compromised membranes of dead cells.	Can sometimes penetrate viable cells, potentially leading to false negatives.
PMA/EMA Solution	Ready-to-use mixture of dyes in distilled water.	Must be stored at -20°C protected from light.
Sodium Thiosulfate Pentahydrate	Neutralizes residual chlorine in water samples to prevent continued antimicrobial action.	Critical for accurate assessment in sanitizer-treated samples.
Phosphate-Buffered Saline (PBS)	Used for washing cell pellets and diluting samples.	Provides a non-reactive, isotonic buffer.

A validated protocol for detecting *Listeria monocytogenes* VBNC cells in process wash water (PWW) involves a combination of EMA and PMAxx [17]:

- **Sample Preparation:** Centrifuge water samples (e.g., 1 mL) at 2,500 × g for 5 minutes. Resuspend the pellet in PBS.
- **Dye Treatment:** Add EMA and PMAxx to the sample at final concentrations of **10 µM and 75 µM**, respectively.
- **Incubation and Photoactivation:** Incubate the sample in the dark at **40°C for 40 minutes**, followed by exposure to a **650-W halogen light source for 15 minutes** on ice.
- **DNA Extraction and qPCR:** Proceed with standard DNA extraction and quantitative PCR targeting the pathogen of interest.

This protocol has been successfully validated in industrial settings, demonstrating its robustness for detecting VBNC cells in challenging environments like chlorine-treated wash water [17]. The workflow for this method is outlined in the diagram below.

Figure 1. Workflow for VBNC Detection via v-qPCR



Metabolic Activity Assays

Detection based on metabolic activity leverages the fact that VBNC cells maintain a basal level of metabolism. A common approach uses fluorescein diacetate (FDA), a non-fluorescent, lipophilic compound that passively diffuses across the cell membrane. Once inside a metabolically active cell, non-specific intracellular esterases hydrolyze FDA into fluorescein, a fluorescent product that accumulates inside cells with intact membranes, generating a detectable signal [60].

Another strategy involves monitoring the uptake and utilization of specific substrates like glucose. The fluorescent glucose analog **2-NBDG** can be consumed by some viable bacteria and subsequently degraded, leading to a loss of fluorescence. Conversely, dead cells cannot metabolize the compound and retain fluorescence. Alternatively, enzymatic assays can measure the depletion of native glucose from a sample, indirectly indicating metabolic activity [60]. A significant limitation of these methods is that deeply dormant VBNC cells may have silenced their metabolism to undetectable levels, and not all bacterial species can uptake artificial substrates like 2-NBDG [60].

Flow Cytometry with Vital Stains

Flow cytometry, when combined with fluorescent viability stains, offers a powerful tool for rapidly quantifying different cell populations in a sample. Stains like SYTO9 (which labels all cells) and propidium iodide (PI, which penetrates only dead cells) are commonly used to distinguish intact (viable/VBNC) from compromised (dead) cells based on membrane integrity [60] [17].

However, this method's effectiveness can be compromised in complex sample matrices like process wash water from the food industry. High organic matter content can cause interference and autofluorescence, leading to an overestimation of dead cells and making it less suitable for certain environmental or clinical samples without extensive purification [17].

Table 2: Comparison of Key VBNC Detection Methods

Method	Principle	Advantages	Limitations
Viability qPCR (v-qPCR)	Membrane integrity; DNA amplification inhibition in dead cells.	Specific, sensitive, applicable to complex samples, can be quantitative.	Requires optimization of dye concentration; may not detect cells with intact but impermeable membranes.
Metabolic Assays (e.g., FDA, 2-NBDG)	Metabolic activity via substrate conversion.	Can detect active VBNC cells; can be simple.	Dormant VBNC cells may be missed; not universal (e.g., 2-NBDG not consumed by all species); pH-sensitive.
Flow Cytometry	Membrane integrity using fluorescent dyes.	Rapid, high-throughput, provides cell count.	Prone to interference in complex matrices; requires expensive equipment.
Whole Genome Sequencing (WGS)	Genomic relatedness between strains from different sources.	High-resolution, strain-level tracking; identifies transmission pathways.	Does not directly prove viability; requires bioinformatics expertise.

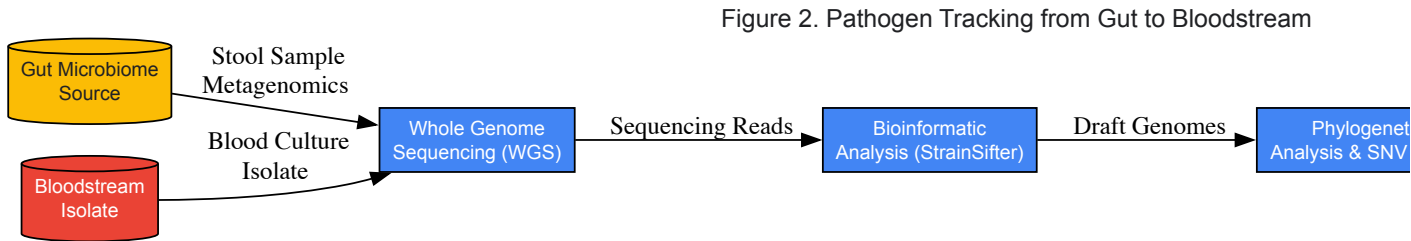
Genomic Workflows for High-Resolution Pathogen Tracking

For establishing definitive links between pathogens found in different locales—such as matching a bloodstream isolate to a gut microbiome reservoir—strain-level genomic tracking is essential. Whole genome sequencing (WGS) provides the highest resolution for this purpose.

Strain-Level Tracking with Bioinformatics Pipelines

Bioinformatic tools like **StrainSifter** enable precise matching of pathogens from an infection site to potential sources by comparing whole genome sequences [62]. This approach was pivotal in a study of hematopoietic cell transplantation (HCT) recipients, where it confirmed that bloodstream infections (BSI) with typically enteric organisms like *Escherichia coli* and *Klebsiella pneumoniae* often originated from identical strains colonizing the patient's own gut, demonstrating gut translocation as the route of infection [62]. In some cases, zero single-nucleotide variants (SNVs) were found between BSI and gut strains, confirming they were the same clone [62].

This methodology challenges conventional assumptions about infection sources. For instance, StrainSifter analysis also identified cases where classically non-enteric pathogens like *Pseudomonas aeruginosa* and *Staphylococcus epidermidis* were present in the gut microbiome, suggesting a possible gut origin for infections that were previously assumed to come from skin or environmental contamination [62]. The process of tracking pathogens from the gut to the bloodstream is illustrated below.



Tracking Horizontal Gene Transfer in Complex Microbiomes

Beyond tracking entire bacterial strains, understanding the flow of mobile genetic elements via **horizontal gene transfer (HGT)** is crucial for monitoring the spread of antibiotic resistance and virulence genes. Longitudinal metagenomic analysis of gut samples can reveal these dynamics. A recent study analyzing 676 fecal samples collected four years apart identified over 5,600 high-confidence HGT events, demonstrating that an individual's mobile gene pool is highly personalized and stable over time [63]. Furthermore, HGT was found to be influenced by host factors, such as proton pump inhibitor usage being linked to increased transfer of multidrug transporter genes [63]. Tools like HDMI are specifically designed to detect recent HGT events directly from metagenomic data, providing a deeper understanding of how adaptive functions disseminate within microbial communities, including those involving VBNC pathogens [63].

Significance and Applications in Clinical and Industrial Settings

The inability to detect VBNC cells has direct and serious implications for public health and industrial safety. In clinical settings, this can lead to misdiagnosis, ineffective treatment, and an incomplete understanding of infection reservoirs. The persistence of VBNC pathogens in the gut microbiome, for example, represents a potential hidden reservoir for recurrent infections, particularly in immunocompromised patients like HCT recipients [62] [37].

In the food industry, sanitizers like chlorine, while effective at killing culturable cells, can induce the VBNC state in pathogens like *Listeria monocytogenes* and *Salmonella enterica* [17]. If only culture-based methods are used to monitor wash water, these viable pathogens will be missed, leading to a false sense of security and potential cross-contamination of food products. The optimized v-qPCR protocol provides a more reliable method for monitoring water quality and preventing outbreaks [17].

Perhaps most critically, many pathogens retain virulence in the VBNC state or can rapidly resuscitate under favorable conditions, regaining full culturability and pathogenicity. For instance, *Vibrio vulnificus* in the VBNC state can resuscitate in a human host and cause fatal infections [37]. This underscores the importance of detecting these cells not merely as a scientific curiosity, but as a crucial component of comprehensive infectious disease management and prevention.

Navigating Challenges: Resistance, Resuscitation, and Therapeutic Gaps

The **viable but non-culturable (VBNC) state** is a dormant survival strategy adopted by numerous bacterial species when confronted with unfavorable environmental conditions [14] [19]. In this state, bacteria lose the ability to form colonies on routine culture media—the gold standard of microbiological detection—yet remain alive with reduced metabolic activity and retain pathogenic potential [14] [48]. The **resuscitation** of VBNC cells back to a metabolically active, culturable state represents a critical crossroads in bacterial survival dynamics and poses a significant dilemma for public health and clinical intervention [19]. When VBNC cells resuscitate, they not **resume cell division** but can also **regain full virulence**, leading to recurrent infections and disease outbreaks from sources previously considered safe [64]. This whitepaper synthesizes current research on the conditions and molecular mechanisms driving this regrowth phenomenon, providing a technical guide for researchers and drug development professionals navigating the challenges of bacterial dormancy and resurgence.

Resuscitation from the VBNC state is not merely a passive reversal of entry conditions but an **active physiological process** requiring specific signaling and biosynthetic events [19]. While the simple removal of the initial stress factor can sometimes trigger resuscitation, this is not universally sufficient, indicating the involvement of more complex regulatory pathways [19]. The process involves the **re-synthesis of cytoplasmic proteins and cell wall peptidoglycan**; studies demonstrate that inhibiting protein synthesis with chloramphenicol or peptidoglycan synthesis with penicillin effectively blocks resuscitation in species like *Vibrio vulnificus* and *Enterococcus faecalis* [19]. The ability to resuscitate is time-dependent, constrained within a "**resuscitation window**" beyond which cells may lose revival capacity, a phenomenon influenced by the duration of the VBNC state and the intensity of the inducing stress [19].

Table 1: Key Confirmed Factors Inducing Resuscitation from the VBNC State

Resuscitation Factor	Bacterial Species Example(s)	Specific Resuscitation Condition
Temperature Upshift	<i>Vibrio vulnificus</i> , <i>Escherichia coli</i> , <i>Staphylococcus aureus</i>	Transfer from low temperature (e.g., 4°C) to optimal growth temperature [19]
Nutrient Replenishment	<i>Salmonella bovis</i> morbificans, <i>Enterococcus faecalis</i> , <i>Listeria monocytogenes</i>	Addition of rich media (e.g., LB broth) or specific nutrients to starved cells [19] [64]
Host-Derived Factors	<i>Listeria monocytogenes</i> , <i>Erwinia amylovora</i>	Inoculation into chicken embryo models or plant seedlings [65] [64]
Chelation of Stressors	<i>Acidovorax citrulli</i> , <i>Escherichia coli</i>	Addition of EDTA to chelate toxic copper ions [19] [64]
Resuscitation Promoting Factors (Rpfs)	<i>Mycobacterium tuberculosis</i> (and other Actinobacteria)	Supplementation with Rpfs, which are bacterial cytokines involved in peptidoglycan remodeling [19]
Quorum Sensing Signals	<i>Vibrio</i> spp.	Presence of autoinducer molecules facilitating cell-cell communication [19]
Oxidative Stress Alleviation	<i>E. coli</i> O157:H7, <i>Salmonella</i> sp.	Addition of H ₂ O ₂ scavengers like sodium pyruvate or catalase to the medium [19]

Molecular Mechanisms and Energetics of Regrowth

Recent research on *E. coli* O157:H7 has illuminated the central role of **cellular energy management** in resuscitation. A study found that mutation of the *rfaL* gene, encoding O-antigen ligase, significantly shortened the lag phase preceding resuscitation [66]. Further investigation revealed that these mutant VBNC cells contained **higher levels of ATP** than their wild-type counterparts. During the resuscitating lag phase, this stored ATP was consumed to activate the **Handler and salvage pathways for NAD+ synthesis** [66]. This critical cofactor is essential for balancing redox reactions and recovering overall metabolic activity. This finding suggests a fundamental resuscitation strategy: VBNC cells **utilize residual ATP reserves to reboot central metabolism**, thereby driving the exit from dormancy [66].

Toxin-Antitoxin Systems and Gene Regulation

At the genetic level, **type II toxin-antitoxin (TAS) systems** are implicated in the control of dormancy entry and exit [48]. Under normal conditions, a stable toxin and an unstable antitoxin form a non-toxic complex. Under stress, the antitoxin is degraded, freeing the toxin to act on cellular targets, leading to a sharp decrease in translation, replication, and growth—hallmarks of the VBNC state [48]. Resuscitation is hypothesized to involve the **re-synthesis of the antitoxin**, neutralizing the toxin and allowing the resumption of metabolic processes. Other global regulators, such as **rpoS** (the stationary phase sigma factor) and **oxyR** (involved in oxidative stress response), also contribute to this complex regulatory network governing the switch between dormancy and active growth [48].

This protocol is adapted from a study on a plant pathogen and exemplifies the use of chemical stressors [64].

- **VBNC Induction:**

- **Culture Preparation:** Grow *A. citrulli* strain AAC00-1 in LB broth to the exponential growth phase (12 h at 28°C with shaking at 120 rpm).
- **Stress Application:** Harvest cells by centrifugation ($10,610 \times g$ for 5 min) and resuspend in a solution containing 5-50 μM copper sulfate (concentration determines time to full VBNC state).
- **Monitor Culturability:** Plate aliquots onto LB agar every 24-48 hours. The VBNC state is achieved when **colony-forming units (CFU) drop to zero** while viability assays (e.g., LIVE/DEAD staining) confirm cells are still alive.

- **Resuscitation Methods:**

- **Chelation:** Treat VBNC cells with the chelating agent EDTA to remove inhibitory Cu^{2+} ions.
- **Nutrient Replenishment:** Transfer VBNC cells to fresh LB broth or oligotrophic media amended with casein hydrolysate or host plant (watermelon) seedling juice.
- **Virulence Assay:** Confirm successful resuscitation by demonstrating that resuscitated cells regain the ability to colonize and infect host seedlings, restoring virulence equivalent to untreated cultures [64].

This modern molecular protocol allows for the absolute quantification of viable cells without cultivation, crucial for tracking resuscitation [16].

- **VBNC Induction:** Induce the VBNC state in *Klebsiella pneumoniae* via starvation in Artificial Seawater (ASW) at 4°C. Monitor CFUs until plates show no growth.
- **Sample Treatment:**
 - **Optimize PMA:** Test PMA (propidium monoazide) concentrations from 5-200 μM and incubation times (5-30 min in the dark) to establish optimal conditions for selective penetration into dead cells.
 - **Cross-linking:** Expose PMA-treated samples to a halogen light source to cross-link the dye with DNA from membrane-compromised (dead) cells, preventing its PCR amplification.
- **Absolute Quantification:**
 - **DNA Isolation:** Extract genomic DNA from treated samples.
 - **Droplet Digital PCR (ddPCR):** Partition the DNA sample into thousands of nanodroplets. Perform PCR amplification targeting three single-copy genes (*KP*, *rpoB*, *adhE*) for robust counting.
 - **Analysis:** Use Poisson statistics to absolutely quantify the copy numbers of the target genes from the fraction of positive droplets. This number corresponds directly to the number of **viable (membrane-intact) cells** in the original VBNC suspension, providing a baseline for resuscitation studies [16].
- **Resuscitation Inhibition:** To study resuscitation dynamics, the antibiotic ciprofloxacin can be added to the resuscitation medium to inhibit the recovery of VBNC cells, allowing researchers to isolate and quantify the resuscitation process specifically [16].

VBNC Cell Quantification via PMA-ddPCR

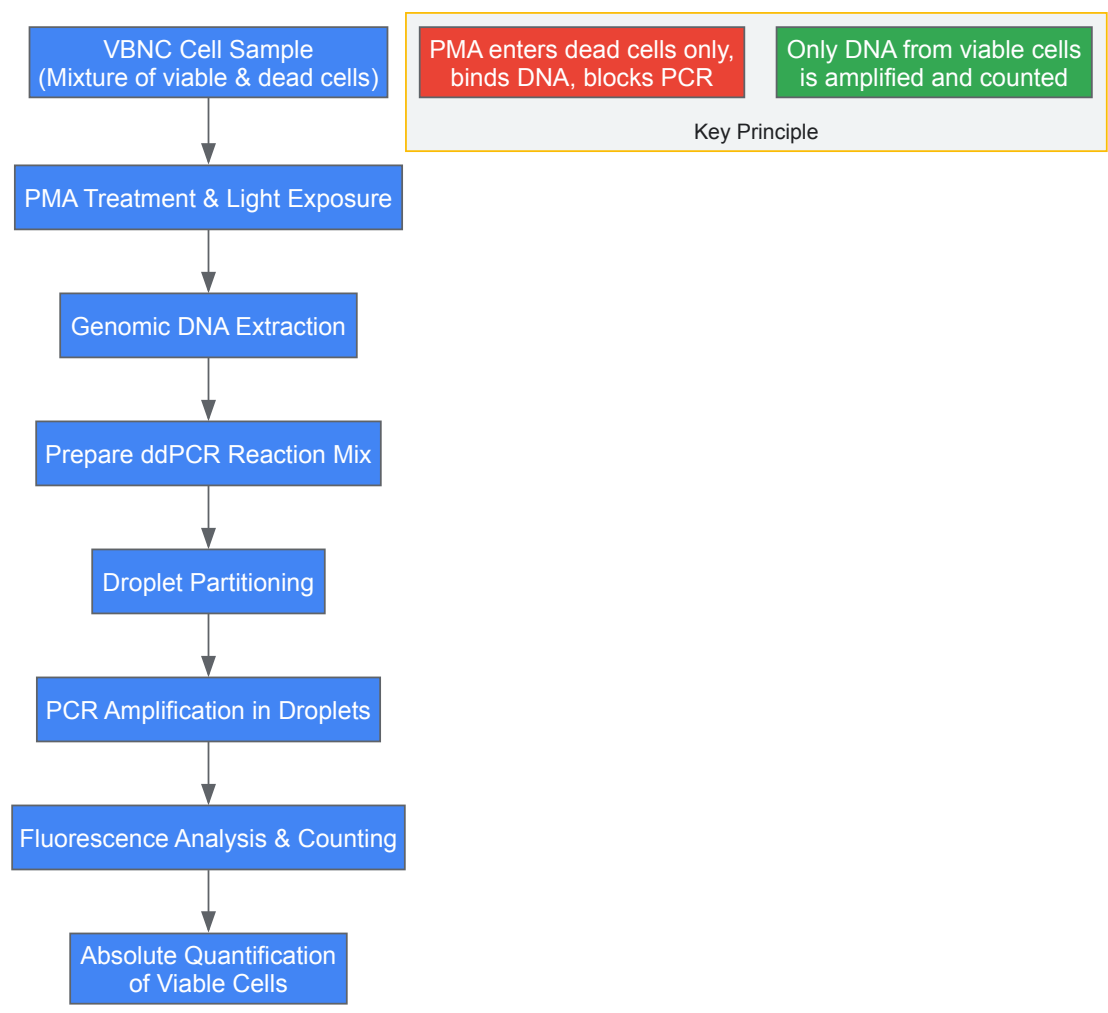


Diagram 1: Experimental workflow for the absolute quantification of VBNC cells using PMA-ddPCR.

The Scientist's Toolkit: Essential Reagents for VBNC Research

Table 2: Key Research Reagent Solutions for VBNC and Resuscitation Studies

Reagent / Material	Function in VBNC Research	Example Application
Propidium Monoazide (PMA)	DNA-binding dye that penetrates only membrane-compromised cells; used with qPCR/ddPCR to differentiate viable from dead cells.	Absolute quantification of VBNC <i>Klebsiella pneumoniae</i> in fecal samples [16].
LIVE/DEAD BacLight Kit	Fluorescent staining (SYTO9/PI) for microscopy/flow cytometry to visualize and enumerate cells with intact vs. damaged membranes.	Determining the ratio of viable to dead <i>Acidovorax citrulli</i> during copper-induced VBNC state [64].
Resuscitation Promoting Factors (Rpfs)	Bacterial cytokines (lysozyme-like enzymes) that stimulate peptidoglycan turnover and break dormancy.	Resuscitation of <i>Mycobacterium tuberculosis</i> and other Actinobacteria from the VBNC state [19].
Sodium Pyruvate / Catalase	Scavengers of hydrogen peroxide (H2O2) in culture media, mitigating oxidative stress that can prevent colony formation.	Recovery of "VBNC" <i>Vibrio vulnificus</i> cells by neutralizing media-derived H2O2 [19] [48].
Ethylenediaminetetraacetic Acid (EDTA)	Chelating agent that binds and neutralizes heavy metal ions (e.g., Cu2+), reversing metal-induced VBNC states.	Resuscitation of <i>Acidovorax citrulli</i> from a copper-induced VBNC state [64].
In Vivo Models (e.g., Chicken Embryo)	Complex host environment providing physiological signals (nutrients, immune factors) that can trigger resuscitation.	Revival of VBNC <i>Listeria monocytogenes</i> that could not be resuscitated by in vitro methods [65].

Table 3: Advanced Molecular Techniques for Mechanism Elucidation

Technique	Parameter Measured	Insight into Resuscitation Mechanism
Time-Lapse Microscopy	Single-cell growth and division kinetics.	Revealed that <i>rfaL</i> mutation shortens the resuscitating lag phase in <i>E. coli</i> O157:H7 [66].
Metabolomics	Global profile of small molecule metabolites.	Identified ATP consumption and activation of NAD+ synthesis pathways as critical for resuscitation [66].
Droplet Digital PCR (ddPCR)	Absolute quantification of gene copies without a standard curve.	Enabled precise counting of viable <i>K. pneumoniae</i> cells in complex samples like mouse feces [16].
Transmission Electron Microscopy (TEM)	Ultra-structural cell morphology.	Visualized cell wall and cytoplasmic changes in VBNC <i>K. pneumoniae</i> and upon resuscitation [16].

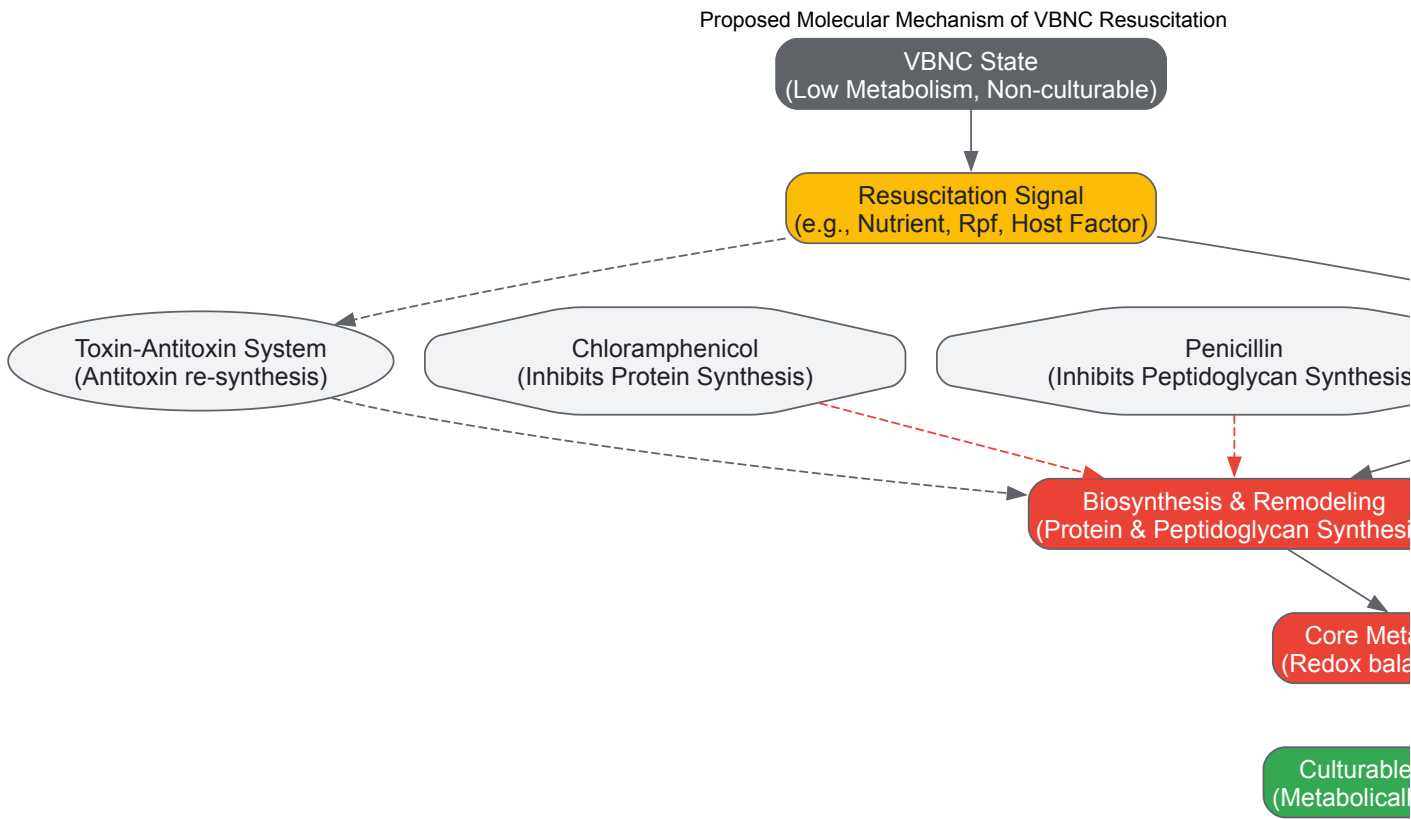


Diagram 2: A proposed molecular mechanism for resuscitation from the VBNC state, integrating energy mobilization, biosynthetic processes, and regulatory systems.

The resuscitation of VBNC bacteria from a dormant to a proliferative state is a multifaceted process governed by specific environmental conditions, energetic constraints, and molecular signaling pathways. Overcoming the dilemma posed by this phenomenon requires a deep understanding of these mechanisms, which is now becoming possible through advanced techniques like PMA-ddPCR, metabolomics, and sophisticated *in vivo* models. The experimental protocols and research tools detailed in this whitepaper provide a framework for systematically investigating VBNC resuscitation across different bacterial species. Future research must focus on identifying universal molecular targets within the resuscitation pathway. The continued elucidation of these mechanisms is paramount for developing novel therapeutic and public health strategies to permanently silence the "wake-up call" of dormant bacterial pathogens.

Enhanced Resistance of VBNC Cells to Antibiotics and Disinfectants

The **viable but non-culturable (VBNC) state** is a dormant survival strategy adopted by numerous bacterial species when faced with environmental stress. In this state, cells undergo a dramatic reduction in metabolic activity and lose the ability to form colonies on routine culture media, yet they remain alive with intact membranes and can resuscitate when conditions become favorable [14] [48]. This phenomenon presents a formidable challenge in clinical, food safety, and environmental settings, as VBNC cells **evade detection by standard culture-based methods** while exhibiting **markedly enhanced tolerance** to antimicrobial agents, including antibiotics and disinfectants [67] [68] [69].

The induction of the VBNC state is triggered by a variety of sub-lethal stresses common in disinfection processes. Physical methods like **UV irradiation** and chemical agents such as **chlorine, chloramines, and oxidizing disinfectants** have been widely reported to induce this dormant state in pathogens like *Escherichia coli*, *Pseudomonas aeruginosa*, and *Legionella pneumophila* [67] [68] [70]. Understanding the underlying molecular mechanisms governing this enhanced resistance is crucial for developing effective strategies to combat persistent and recalcitrant infections.

Mechanisms of Enhanced Resistance in VBNC Cells

Bacteria in the VBNC state deploy a multi-faceted arsenal of physiological and molecular adaptations to survive exposure to concentrations of antimicrobials that would readily kill their culturable counterparts. The core mechanisms include reduced metabolic activity, oxidative damage repair, and structural and genetic adaptations.

Physiological and Metabolic Adaptations

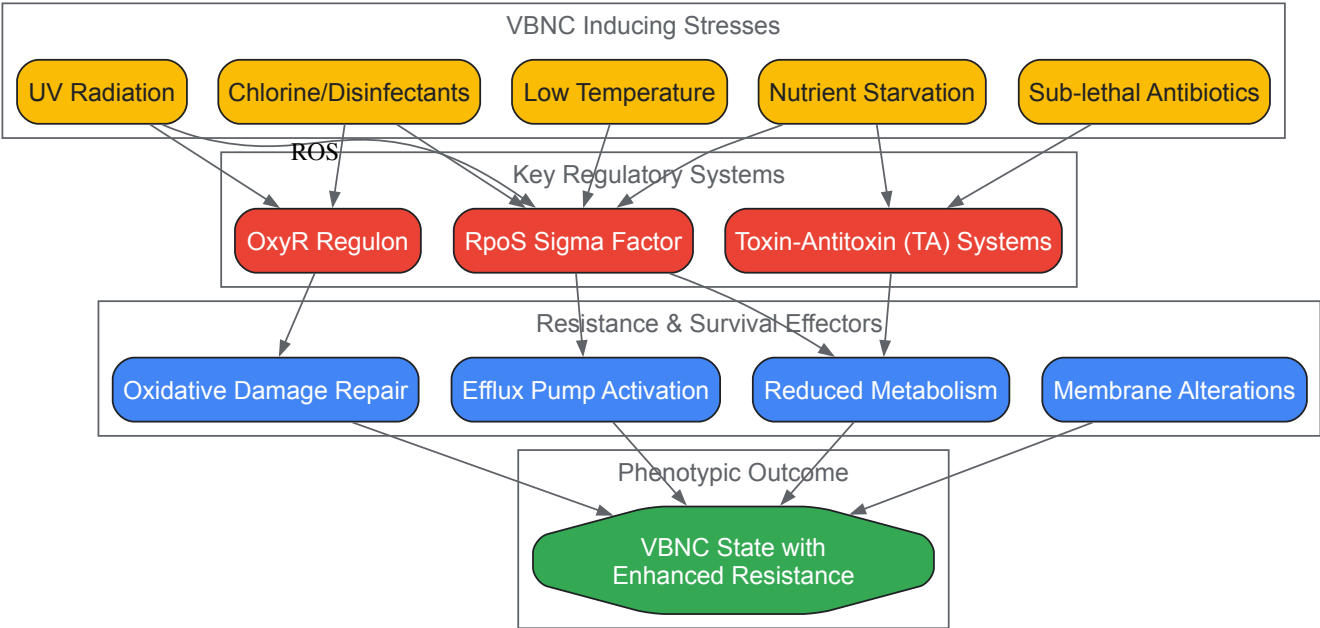
- **Drastically Reduced Metabolism:** Upon entering the VBNC state, bacteria sharply downregulate their metabolic activity and cell division processes [14] [48]. Since most conventional antibiotics target active cellular processes like cell wall synthesis, protein production, and DNA replication, this metabolic dormancy inherently reduces the number of functional targets for these drugs, leading to heightened tolerance [69] [48].
- **Activation of Stress Responses:** The entry into the VBNC state is orchestrated by a complex regulatory network that senses stress. Key players include the **RpoS stationary phase sigma factor** and the **OxyR regulator** of the oxidative stress response [67] [68]. These systems coordinate a global shift in gene expression to bolster the cell's defenses.
- **Membrane Permeability Alterations:** Studies on VBNC *E. coli* induced by low-level chlorination have shown that these cells undergo changes in their cell walls and membranes, leading to **increased membrane permeability** [69]. Despite this increase, which would typically facilitate antimicrobial intake, the dormant cells survive, suggesting that other resistance mechanisms effectively neutralize this vulnerability [69].

Molecular Mechanisms and Genetic Regulation

At the molecular level, the resilience of VBNC cells is supported by the expression of specific genetic determinants and protective cellular processes.

- **Upregulation of Efflux Pumps:** VBNC cells can actively expel toxic compounds. Research has documented the **increased expression of efflux pump genes** in VBNC bacteria, which helps to reduce the intracellular concentration of antimicrobials [71]. This mechanism is a significant contributor to both disinfectant and antibiotic cross-resistance.
- **Toxin-Antitoxin (TA) Systems:** Modules such as the **Type II Toxin-Antitoxin systems** are crucial genetic regulators inducing and maintaining dormancy [48]. Under stress, the unstable antitoxin degrades, freeing the stable toxin to inhibit essential cellular processes like translation and replication, thereby inducing dormancy and a concomitant increase in antimicrobial resistance [48].
- **Oxidative Damage Repair:** Sub-lethal disinfection often generates reactive oxygen species (ROS). VBNC cells counteract this by **repairing oxidative damage** to cellular components. The **OxyR regulon** plays a pivotal role in this defense, triggering the expression of detoxifying enzymes like catalase to manage oxidative stress [67] [68].

The following diagram synthesizes the primary signaling pathways and regulatory networks involved in the formation of and resistance in the VBNC state, integrating key inducters, regulatory nodes, and effector mechanisms.



Quantitative Data on VBNC Resistance

The enhanced resistance of VBNC cells is not merely a qualitative observation but is substantiated by robust quantitative data. The following tables consolidate key findings from recent research, illustrating the degree of resistance conferred by the VBNC state against various disinfectants and antibiotics.

Table 1: VBNC Induction by Disinfectants and Resulting Resistance

Disinfectant/Stress	Bacterial Species	Key Findings on VBNC Induction & Resistance
Sub-lethal Photocatalysis	<i>E. coli</i> (Antibiotic-Resistant & Susceptible)	Induced VBNC state in 6h; Resuscitated cells showed increased antibiotic resistance and repaired oxidative damage [67].
Low-Level Chlorine (0.5 mg/L)	<i>E. coli</i>	Induced VBNC state; Cells exhibited enhanced persistence to 9 typical antibiotics; Transcriptomics revealed upregulation of adhesins, stress, and antibiotic resistance genes [69].
Chlorine	<i>Enterococcus faecalis</i>	VBNC state cells were resistant to antibiotics at >500 times the Minimal Inhibitory Concentration (MIC) [68].
UV254 Radiation	<i>Pseudomonas aeruginosa</i>	Induced VBNC state; Reactivation observed after disinfection, posing a contamination risk [70].
Multiple (e.g., disinfectants, cold, starvation)	<i>Listeria monocytogenes</i>	VBNC state induced by various food production stresses; Cells retain virulence and can resuscitate, evading standard detection methods [72].

Table 2: Documented Antibiotic Tolerance of VBNC Cells

Bacterial Species	Antibiotic/Disinfectant Challenge	Level of Tolerance/Resistance Documented
<i>E. coli</i>	Ampicillin, Ofloxacin	VBNC cells persisted exposure to 128x MIC of Ampicillin and 64x MIC of Ofloxacin [69].
<i>Vibrio vulnificus</i>	Multiple challenges (heat, antibiotics, heavy metals)	Entry into VBNC state protected cells against a variety of potentially lethal environmental challenges [69].
<i>E. coli</i> (VBNC from chlorination)	Nine typical antibiotics	Metabolic activity reduced but persistence to antibiotics enhanced post-disinfection [68] [69].
VBNC State Bacteria (General)	Wide spectrum of antimicrobials	Exhibit " extraordinary tolerance " to adverse conditions, including antibiotics and biocides, due to dormancy and reduced metabolic activity [48].

Advanced Detection and Methodological Protocols

A significant challenge in VBNC research is the inability to detect these cells via standard plate counts (Heterotrophic Plate Count - HPC). This necessitates the use of vital stains and molecular methods that distinguish viability from culturability. The following workflow and reagent toolkit are essential for accurate research in this field.

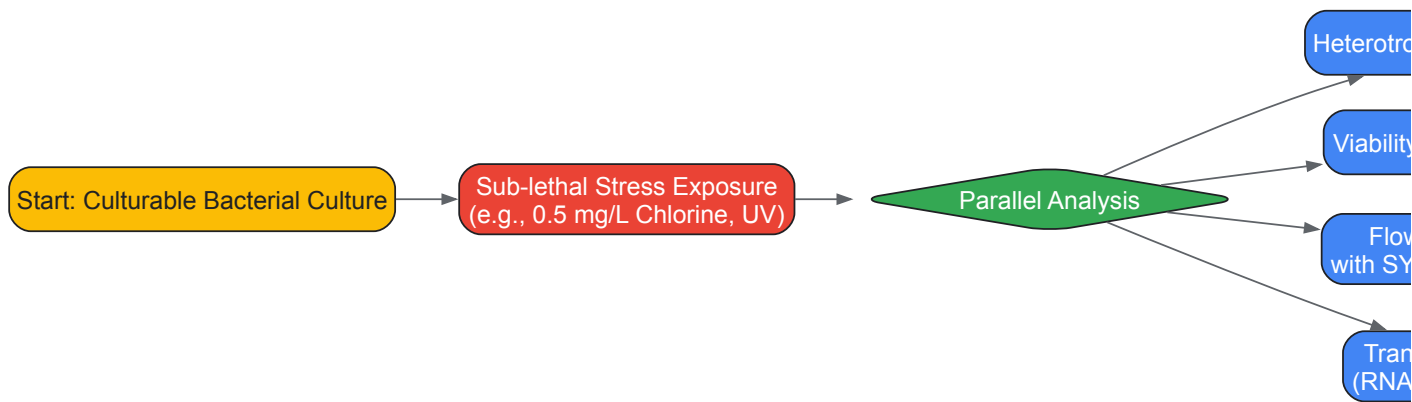
Key Research Reagent Solutions

Table 3: Essential Reagents and Kits for VBNC Research

Reagent / Kit	Primary Function in VBNC Research
Propidium Monoazide (PMA)	A DNA intercalating dye that penetrates only cells with compromised membranes. Used in PMA-qPCR to selectively inhibit amplification of DNA from dead cells, allowing quantification of viable (membrane-intact) cells [70].
SYTO 9 / Green Fluorescent Nucleic Acid Stains	Membrane-permeant stains that label all cells, both live and dead. When used with a membrane-impermeant counterstain like PI, it enables viability counting via Flow Cytometry (FCM) [69] [73].
Flow Cytometry (FCM) - Label-Free	Rapid, single-cell analysis based on light scatter (FSC/SSC) to assess cell integrity and count without staining. Effective for detecting disinfectant-induced injury and VBNC states within hours [73].
RNA Sequencing (RNA-seq)	High-throughput transcriptomic analysis to uncover global gene expression changes, including upregulation of stress response, efflux pump, and antibiotic resistance genes during the VBNC state [67] [69].
Catalase & Pyruvate	Added to culture media to scavenge hydrogen peroxide (H2O2). Can resuscitate some "VBNC" cells (e.g., <i>V. vulnificus</i>) whose non-culturability is due specifically to a loss of H2O2 detoxification capacity [48].
DNeasy PowerSoil Pro Kit	Used for robust microbial DNA extraction from complex samples, a critical step prior to downstream qPCR or sequencing applications [70].
Quantitative PCR (qPCR)	Molecular method to quantify gene copies of a target organism. When coupled with PMA (PMA-qPCR), it quantifies viable cells (both culturable and VBNC), with VBNC count = PMA-qPCR count - HPC count [70].

Core Experimental Workflow for Inducing and Analyzing VBNC Cells

The following diagram outlines a generalized protocol for inducing the VBNC state and characterizing its resistance profile, integrating the key reagents and methods listed above.



Detailed Protocol Steps:

- **VBNC State Induction:**
 - **Culture Preparation:** Grow the target bacterium (e.g., *E. coli*, *P. aeruginosa*) to mid-logarithmic phase in an appropriate liquid medium like LB broth. Harvest cells by centrifugation, wash, and resuspend in a sterile buffer like Phosphate Buffered Saline (PBS) to remove residual nutrients [70].
 - **Stress Application:** Expose the bacterial suspension to a sub-lethal concentration of the chosen stressor. For disinfectants, this could be **0.5 mg/L chlorine for 6 hours** [69] or a defined UV254 dose [70]. The sub-lethal dose must be determined empirically as the concentration that reduces the culturable count by several logs without eliminating the entire population.

- **Monitoring and Detection:**

- **Culturability Assessment (HPC):** Serially dilute the stress-exposed and control cultures and plate on non-selective rich agar media. Incubate under optimal conditions for the bacterium. The **absence of colony formation** confirms loss of culturability [68] [72].
- **Viability Staining and Flow Cytometry (FCM):** Stain the bacterial suspension with a viability dye combination like **SYTO 9 and Propidium Iodide (PI)**. SYTO 9 stains all cells, while PI stains only those with damaged membranes. Analyze by FCM to distinguish and quantify sub-populations: SYTO 9+/PI- (viable), SYTO 9+/PI+ (injured), and SYTO 9-/PI+ (dead) [69] [73]. Label-free FCM, which analyzes forward and side scatter, can also detect morphological changes indicative of the VBNC state [73].
- **Molecular Viability Detection (PMA-qPCR):** Treat the sample with **PMA** to cross-link DNA in membrane-compromised (dead) cells. Protect from light, then photo-lyse the dye. Extract total DNA and perform **qPCR** targeting a species-specific gene. PMA-treated samples will only amplify DNA from viable (membrane-intact) cells. The VBNC population is calculated by subtracting the HPC count from the PMA-qPCR count [70].

- **Resistance Profiling:**

- **Antibiotic Tolerance Assay:** Expose VBNC cells, their resuscitated counterparts, and normal culturable cells to a range of antibiotics. Use **MIC assays** and **time-kill curves** to compare tolerance levels. VBNC cells are expected to survive concentrations **multiples of the MIC** (e.g., 128x MIC) for significantly longer durations [69] [48].
- **Transcriptomic Analysis (RNA-seq):** To understand the genetic basis of resistance, extract high-quality RNA from VBNC and control cells. Perform **RNA-seq** for global transcriptome profiling. This analysis typically reveals **upregulation of genes** involved in efflux pumps, oxidative stress response (e.g., *oxyR*), toxin-antitoxin systems, and general stress response (e.g., *rpoS*) [67] [69].

Implications and Future Perspectives

The formation of VBNC cells in response to disinfection represents a significant "Achilles' heel" in modern microbial control strategies. The induction of this dormant state by sub-lethal treatments, coupled with their exceptional resilience, means that our current disinfection protocols may inadvertently be selecting for and protecting the most dangerous pathogens [68]. This has direct consequences for public health, contributing to **persistent infections, recurrent outbreaks**, and the **spread of antibiotic resistance** [67] [74].

Future research must focus on overcoming this challenge. Promising directions include the development and implementation of **combination disinfection strategies**, such as **UV/NaClO or UV/PAA**, which have been shown to more effectively inactivate VBNC *P. aeruginosa* and prevent reactivation by causing irreparable cellular damage [70]. Furthermore, there is an urgent need to move beyond culture-dependent diagnostics. Integrating **PMA-qPCR and FCM** into routine environmental and clinical monitoring can provide a more accurate assessment of microbial threats, uncovering the "hidden" VBNC populations that plate counts miss [73] [72]. Finally, a deeper exploration of the molecular mechanisms of VBNC, particularly the interplay between TA systems, oxidative stress regulators, and efflux pumps, will unveil new targets for therapeutic agents designed to eradicate dormant cells or prevent their resuscitation.

Retention and Expression of Virulence Factors in the Dormant State

The viable but non-culturable (VBNC) state is a dormant survival strategy adopted by numerous bacterial pathogens when faced with environmental stress. While cells in this state are metabolically active but cannot form colonies on conventional culture media, a critical public health concern lies in their potential to retain and reactivate virulence factors, enabling them to cause disease upon resuscitation. This whitepaper synthesizes current research on the preservation and expression of virulence determinants in VBNC pathogens. It provides a detailed guide on advanced methodologies for their detection and quantification, focusing on molecular and single-cell approaches that circumvent the limitations of traditional culture techniques. The findings underscore the significant threat VBNC cells pose to food safety, clinical microbiology, and drug development, necessitating revised detection frameworks and therapeutic countermeasures.

The viable but non-culturable (VBNC) state is a unique survival strategy adopted by a wide range of bacteria in response to adverse environmental conditions such as starvation, extreme temperatures, osmotic pressure, and exposure to antibiotics or disinfectants [18] [12]. In this state, bacteria fail to grow on standard culture media upon which they would normally proliferate, rendering them undetectable by conventional microbiological plating methods. However, they maintain viability, including metabolic activity, and possess the capacity to resuscitate and regain culturability when favorable conditions are restored [18]. To date, over 100 bacterial species, including major human pathogens like *Escherichia coli*, *Vibrio cholerae*, *Listeria monocytogenes*, and *Klebsiella pneumoniae*, have been documented to enter the VBNC state [18] [72].

A paramount concern within public health and clinical microbiology is whether pathogens in the VBNC state retain their virulence potential. These cells can evade routine detection protocols, leading to an underestimation of microbial risk in water, food, and clinical settings. Furthermore, if VBNC cells remain virulent or can fully regain their pathogenicity upon resuscitation, they represent a significant hidden reservoir for disease outbreaks. This technical guide explores the current evidence on the retention and expression of virulence factors in the VBNC dormant state, framing it within the broader context of bacterial pathogenesis and persistence research. It also details the sophisticated experimental protocols required to study these elusive bacterial populations.

Virulence Factor Retention in VBNC Pathogens

The retention of virulence factors in VBNC cells is not a uniform phenomenon; it varies by bacterial species, specific strain, and the inducing conditions of the VBNC state. The table below summarizes key findings from recent research on specific pathogens.

Table 1: Documented Evidence of Virulence Factor Retention in VBNC Pathogens

Bacterial Pathogen	Virulence Factor / Pathogenic Trait	Evidence of Retention in VBNC State	Experimental Model / Assay	Citation
<i>Shigella dysenteriae</i>	Shiga toxin (Stx)	Gene (<i>stx</i>) remains functional; biologically active toxin produced.	Genetic and cytotoxicity assays.	[12]
<i>Klebsiella pneumoniae</i> (HiAlc Kpn)	Ethanol production (via <i>adhE</i>); Resuscitation potential	Maintains ethanol production and ability to resuscitate after antibiotic removal.	PMA-ddPCR, ethanol assay kit, resuscitation cultures.	[16]
<i>Salmonella</i> Enteritidis	Ulcerative colitis exacerbation	VBNC cells exacerbated colitis severity, compromised intestinal barrier, and altered gut microbiome in a mouse model.	DSS-induced murine colitis model, histopathology, cytokine measurement, 16S rRNA sequencing.	[74]
<i>Listeria monocytogenes</i>	Resuscitation and Virulence	VBNC cells regained growth and virulence after appropriate resuscitation.	Resuscitation models, cell culture assays, animal models.	[72]
<i>Vibrio cholerae</i>	Colonization ability	VBNC cells maintained membrane integrity and colonizing ability.	Iron-dextran-treated mouse model.	[12]

In certain cases, virulence may be attenuated in the VBNC state but recovered upon resuscitation. For instance, *Vibrio alginolyticus* and *Vibrio parahaemolyticus* VBNC cells were shown to lose virulence after resuscitation in a mouse model, but this virulence was restored after two consecutive passages in a rat ileal loop [12]. Similarly, *Aeromonas hydrophila* lost its ability to lyse erythrocytes and adhere to cells in the VBNC state but regained these properties after temperature-induced resuscitation [12]. This demonstrates that the pathogenic potential can be latent and fully recoverable.

Molecular Mechanisms and Physiological Adaptations

The transition to the VBNC state is accompanied by profound changes in the bacterial transcriptome and proteome, reprogramming the cell for survival. While overall metabolic activity is reduced, specific pathways related to stress tolerance and virulence regulation are actively modulated.

- Stress Response and Virulence Regulation:** Molecular studies indicate that the entry into the VBNC state involves the upregulation of proteins associated with stress defense, such as antioxidants (e.g., peroxiredoxins, AhpC/Tsa family), chaperones, and DNA repair enzymes [12]. In *Vibrio vulnificus*, VBNC cells highly express glutathione S-transferase, which helps mitigate oxidative damage [12]. The expression of virulence factors is often integrated with these stress responses. For example, the LuxS/AI-2 quorum sensing system in *Lactiplantibacillus plantarum* was shown to govern the biofilm-to-VBNC transition by modulating metabolic resilience and stress adaptation, directly regulating genes involved in carbohydrate metabolism and oxidative defense [75].
- Membrane Integrity and Toxin Production:** A key feature of VBNC cells is an intact cell membrane, which is crucial for distinguishing them from dead cells. Pathogens like *Shigella dysenteriae* not only maintain membrane integrity but also continue to produce potent toxins like Shiga toxin in the VBNC state [12]. This indicates that the genetic machinery for toxin production remains operational, even in the absence of active replication.
- Metabolic Reprogramming:** Proteomic analyses of VBNC cells often show altered levels of proteins involved in central metabolism. In *E. coli* O157, VBNC food isolates exhibited changes in oxidation-responsive factors and outer-membrane proteins [12]. This metabolic reprogramming is thought to support energy homeostasis and maintain a basal level of physiological function necessary for survival and potential resuscitation.

The following diagram illustrates the core regulatory network and physiological changes driving the VBNC state and its association with virulence.

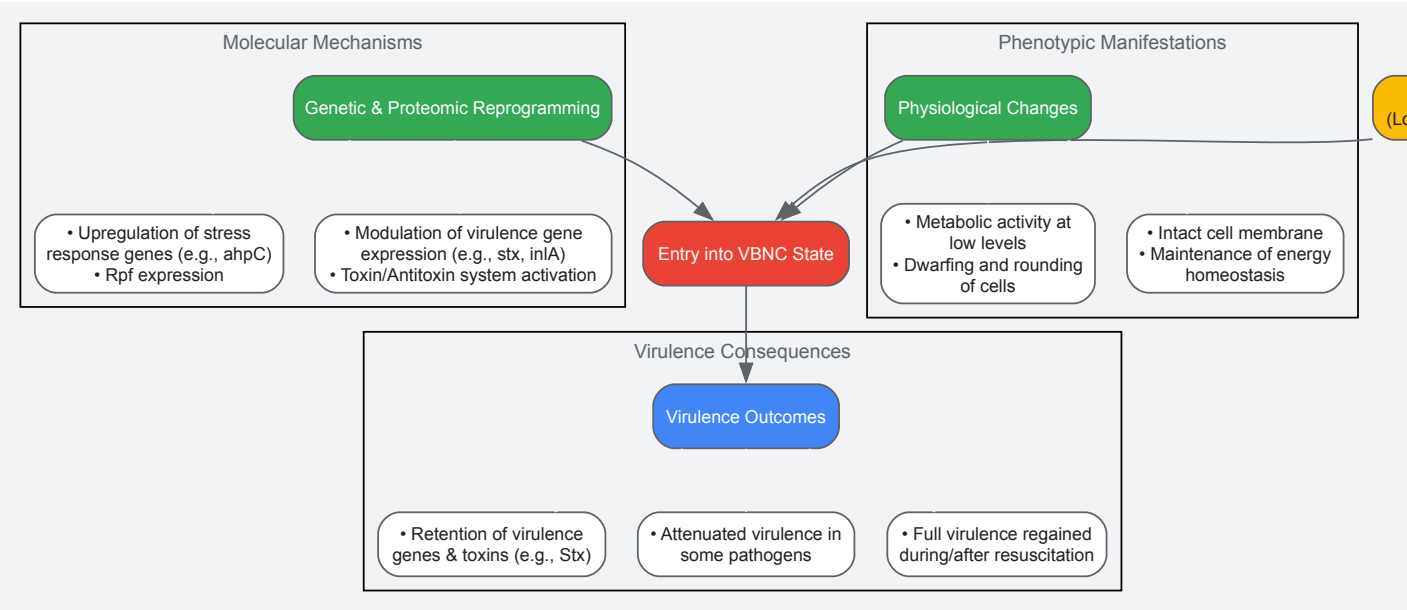


Figure 1: Regulatory Network Linking VBNC State and Virulence

Advanced Detection and Quantification Methodologies

Studying VBNC cells requires moving beyond traditional culture methods. The following sections detail key protocols for inducing, detecting, and quantifying VBNC cells, with a focus on assessing their viability and virulence.

Propidium Monoazide (PMA) coupled with Droplet Digital PCR (ddPCR)

This method allows for the direct quantification of viable (membrane-intact) cells by selectively inhibiting the amplification of DNA from dead cells with compromised membranes.

Table 2: Key Research Reagents for VBNC Detection (PMA-ddPCR)

Reagent / Tool	Function / Description	Application in VBNC Research
Propidium Monoazide (PMA)	DNA-intercalating dye that penetrates only membrane-damaged cells. Photoactivated to bind DNA covalently.	Distinguishes viable from non-viable cells; PMA-bound DNA from dead cells is not amplified in PCR.
Droplet Digital PCR (ddPCR)	Microfluidic-based PCR that partitions a sample into thousands of nanodroplets for absolute DNA quantification without a standard curve.	Absolutely quantifies gene copy numbers from single viable cells, offering high sensitivity and precision.
Single-Copy Gene Primers	PCR primers targeting genes present only once per bacterial chromosome (e.g., <i>rpoB</i> , <i>adhE</i>).	Enumerates bacterial cells directly, as one cell equals one (or few) copy number of the target gene.
Halogen Light Source	High-intensity light source for activating PMA after incubation.	Crucial for cross-linking PMA to DNA from dead cells; samples are typically kept on ice to prevent heating.

Detailed Experimental Protocol:

- VBNC Induction:** Resuspend the bacterial culture (e.g., *Klebsiella pneumoniae* or *Vibrio cholerae*) in a starvation medium such as Artificial Seawater (ASW) at a defined concentration (e.g., 1×10^8 CFU/mL). Incubate at a low temperature (e.g., 4°C) for several weeks. Monitor culturability by regular plating on non-selective media. The VBNC state is confirmed when no colonies form on plates after 48 hours of incubation [16] [47].
- PMA Treatment Optimization:**
 - Add PMA to the VBNC sample to a final concentration typically between 5-200 μM. Optimal concentration should be determined empirically (e.g., 50 μM was used for *Vibrio cholerae*) [16] [47].
 - Incubate the sample in the dark for 5-30 minutes to allow PMA penetration into dead cells.
 - Photoactivate the sample by exposing it to a 650W halogen light source for 15 minutes at a distance of 20 cm. Place the sample tube on ice during this process to avoid overheating [16].

- **DNA Preparation via Oil-Enveloped Bacterial Method (Direct without extraction):**

- Mix the PMA-treated bacterial suspension directly with the ddPCR supermix (e.g., EvaGreen Supermix) [47].
- The subsequent heating steps during PCR cycling will lyse the cells and release genomic DNA within the droplets, eliminating the need for a separate DNA extraction step and reducing processing time.

- **Droplet Digital PCR (ddPCR)**

- Generate droplets from the sample-pcr-mix emulsion using a droplet generator.
- Perform PCR amplification on a thermal cycler using primers for a single-copy gene (e.g., *rpoB* for *K. pneumoniae* or a chromosomal gene for *V. cholerae*).
- After amplification, read the droplets on a droplet reader. The system uses Poisson statistics to provide an absolute count of the target DNA molecules (copies/μL), which corresponds directly to the number of viable VBNC cells in the original sample [16] [47].

The workflow for this combined methodology is outlined below.



Figure 2: Workflow for PMA-ddPCR VBNC Quantification

Single-Cell Microfluidics and Live/Dead Staining

This approach allows for the direct observation and tracking of individual VBNC cells, providing insights into their physiology and heterogeneity.

Detailed Experimental Protocol:

- **Device and Cell Loading:** Use a microfluidic device like the "mother machine," which contains thousands of tiny channels designed to trap individual bacterial cells. Load a stationary-phase bacterial culture into the device, confining cells in the lateral channels [76].
- **Antibiotic Treatment and Staining:**
 - Perfuse the device with a high dose of an antibiotic (e.g., 25x MIC of ampicillin) for a set period (e.g., 3 hours) to kill susceptible cells and induce a non-culturable state in others.
 - Replace the antibiotic medium with a rich growth medium (e.g., Lysogeny Broth) to allow for the resuscitation of persister cells.

- Finally, perfuse the device with a LIVE/DEAD stain (e.g., BacLight). This stain typically contains SYTO 9 (labels all cells green) and propidium iodide (PI), which only penetrates cells with damaged membranes (labeling them red) [76].

• Time-Lapse Microscopy and Analysis:

- Image the entire device throughout the experiment using time-lapse fluorescence microscopy.
- Track individual cells over time to classify their fate:
 - **Susceptible Lysed:** Cells that lyse during/after antibiotic treatment.
 - **Persister:** Cells that survive antibiotic treatment and resume growth in the rich medium.
 - **VBNC:** Cells that remain intact and stain green (live) after antibiotic treatment but do not initiate growth upon medium exchange. These cells are confirmed viable by membrane integrity but are non-culturable under the experimental conditions [76].

Implications for Drug Development and Therapeutic Strategies

The existence of virulent VBNC cells has profound implications for therapeutic development. Standard antibiotics, which typically target active cellular processes like cell wall synthesis or replication, are often ineffective against the dormant, metabolically altered VBNC populations. This contributes to chronic and recurrent infections [76] [12]. Therefore, novel therapeutic strategies are needed.

- **Targeting Resuscitation Pathways:** A promising approach is the identification of compounds that inhibit resuscitation. For example, research on *Klebsiella pneumoniae* showed that the antibiotic ciprofloxacin could inhibit the resuscitation of VBNC cells [16]. Understanding and targeting bacterial cytokines like Resuscitation-Promoting Factor (Rpf) could provide a means to lock pathogens in the dormant state, facilitating their clearance by the immune system [12].
- **Anti-Virulence Therapies:** Given that VBNC cells can retain virulence factors, drugs that disarm these factors (e.g., toxin neutralizers) could mitigate disease severity even if the bacteria are not killed, potentially allowing the host immune system to manage the persistent infection.
- **Advanced Diagnostics for Clinical Trials:** Incorporating VBNC detection methods (like PMA-ddPCR) into clinical trials for antibacterial drugs is crucial. This would provide a more accurate assessment of a drug's efficacy in eradicating all viable bacterial subpopulations, not just the culturable ones, potentially reducing relapse rates.

The VBNC state represents a critical survival mechanism for numerous bacterial pathogens, allowing them to persist in a dormant, hidden form. Evidence confirms that many of these cells retain their virulence factors or the genetic capacity to reactivate them fully upon resuscitation, posing a significant and often undetected threat to public health. Effectively studying this phenomenon requires a move beyond traditional microbiology to embrace sophisticated molecular and single-cell techniques like PMA-ddPCR and microfluidics. For the drug development community, addressing the challenge of VBNC cells is imperative. Future efforts must focus on developing therapeutics that specifically target dormant and resuscitating bacteria, and diagnostic pipelines must be updated to include viability testing to truly gauge the sterility of clinical and industrial samples.

Overcoming Detection Failures in Diagnostic and Environmental Monitoring

The **viable but non-culturable (VBNC) state** is a dormant survival strategy adopted by many bacteria in response to environmental stress. In this state, bacteria maintain metabolic activity and viability but cannot form colonies on conventional culture media, the gold standard in many clinical and environmental laboratories [77]. This phenomenon poses a significant challenge for public health, as VBNC pathogens remain potentially virulent and capable of causing infections yet evade detection by routine culture-based methods [74] [77]. Consequently, the presence of VBNC cells can lead to underestimation of pathogen prevalence, compromised outbreak investigations, and persistent threats in food, water, and healthcare settings [77] [14]. This technical guide examines the core mechanisms, detection failures, and advanced solutions for identifying VBNC pathogens, providing researchers and drug development professionals with the methodologies needed to overcome these critical detection gaps.

The VBNC State: Mechanisms and Public Health Implications

Physiological Basis and Induction Factors

Bacteria enter the VBNC state as a survival mechanism under stressful conditions. This transition involves a coordinated cellular shutdown of replication while preserving essential life functions. Key inducers include:

- **Temperature fluctuations** (particularly cold stress) [77] [6]
- **Nutrient deprivation** [77] [78]
- **Exposure to biocides** such as chlorine and disinfectants [77] [79]
- **Osmotic challenges** and changes in pH [6] [78]
- **Oxidative stress** [9]

Upon entering the VBNC state, cells undergo significant physiological changes: reduction in cell size, alteration in membrane fatty acid composition, and downregulation of metabolic activity to a maintenance level [14]. Critically, **pathogenicity is often retained**; VBNC cells can continue to

express virulence factors and maintain infectivity. For example, VBNC *Campylobacter jejuni* maintains expression of virulence-associated genes (*flaA*, *flaB*, *cadF*, *ciaB*, *cdtA*, *cdtB*, *cdtC*) and can still invade human intestinal epithelial cells [6].

A defining characteristic of VBNC cells is their capacity to **resuscitate** when favorable conditions return. Recent research on *Escherichia coli* O157:H7 has revealed that resuscitation involves utilizing residual ATP to activate Handler and salvage pathways for NAD+ synthesis, which helps balance redox reactions and recover metabolic activity [66]. The molecular mechanism involves **ATP consumption during the resuscitating lag phase** being highly correlated with resuscitation efficiency [66]. This resuscitation potential creates a hidden reservoir of pathogens that can lead to disease outbreaks after perceived control measures have been implemented.

Table 1: Pathogens Confirmed to Enter the VBNC State and Their Associated Risks

Pathogen	Primary Sources	Health Risks	Documented Evidence
<i>Escherichia coli</i> O157:H7	Contaminated food, water	Hemorrhagic colitis, hemolytic uremic syndrome	[78] [66]
<i>Campylobacter jejuni</i>	Poultry products	Gastroenteritis, Guillain-Barré syndrome	[6]
<i>Salmonella</i> Enteritidis	Food products	Ulcerative colitis, systemic infection	[74]
<i>Klebsiella pneumoniae</i> (HiA1c)	Clinical, gut microbiome	Non-alcoholic fatty liver disease	[9]
<i>Cronobacter sakazakii</i>	Food processing environments	Neonatal infections	[79]
<i>Vibrio cholerae</i>	Water	Cholera	[77] [14]
<i>Legionella pneumophila</i>	Water systems	Legionnaires' disease	[6]
<i>Porphyromonas gingivalis</i>	Oral cavity	Periodontal disease, systemic infections	[14]

Limitations of Conventional Detection Methods

Culture-Based Failures

Conventional culture methods rely on the ability of microorganisms to proliferate on or in nutrient media forming visible colonies [77]. This approach fails completely with VBNC cells because, by definition, these cells have lost this cultivability while maintaining viability [77] [78]. This fundamental limitation has profound implications:

- **False negatives** in clinical diagnostics and environmental monitoring [77]
- **Underestimation** of pathogen prevalence and contamination levels [77] [14]
- **Delayed outbreak responses** due to failure to identify causative agents [77]
- **Inaccurate assessment** of sterilization and disfficacy efficacy [6] [79]

Economic and Public Health Impact

The inability to detect VBNC pathogens translates into significant economic and public health burdens. The World Health Organization estimates that pathogens like *E. coli*, primarily linked to food and water contamination, are associated with 485,000 deaths from diarrheal diseases annually, with a worldwide economic loss of nearly **12 billion USD per annum** [80]. Waterborne diseases alone affect over seven million people in the United States annually, with an economic burden of **\$2.2–3.7 billion USD** [80].

Advanced Detection Technologies and Methodologies

Molecular Detection Approaches

Molecular methods that detect genetic material rather than depend on growth have revolutionized VBNC pathogen detection.

Table 2: Advanced Molecular Detection Methods for VBNC Pathogens

Method	Principle	Target Pathogens	Detection Limit	Advantages
PMA-qPCR	Propidium monoazide dye exclusion + quantitative PCR	<i>E. coli</i> O157:H7, <i>Cronobacter sakazakii</i>	10 ⁴ CFU/mL [78]	Distinguishes viable cells, quantitative, relatively fast
PMA-ddPCR	Propidium monoazide + droplet digital PCR	<i>Cronobacter sakazakii</i> , <i>K. pneumoniae</i>	1 copy/mL [79] [9]	Absolute quantification without standard curve, high sensitivity
CRISPR-based Detection	CRISPR-Cas system for nucleic acid detection	Multiple foodborne pathogens [81]	High sensitivity and specificity	Rapid, potential for point-of-care use

Method	Principle	Target Pathogens	Detection Limit	Advantages
RT-qPCR	Reverse transcriptase quantitative PCR	Various pathogens [77]	Varies by target	Detects viable cells via mRNA detection
LAMP	Loop-mediated isothermal amplification	Waterborne pathogens [80]	Comparable to PCR	Isothermal, suitable for field use

Experimental Protocol: PMA-ddPCR for VBNC Cell Quantification

For detection of VBNC *Cronobacter sakazakii* on stainless steel surfaces after desiccation and disinfectant exposure [79]:

Sample Preparation:

- Suspend bacterial cells in appropriate buffer or growth medium.
- Induce VBNC state through:
 - Nutrient deprivation (varying LB medium concentrations: 0%, 25%, 50%, 100%)
 - Environmental stress (NaCl concentrations: 0.9%, 5%, 10%, 15%)
 - Acid stress (acetic acid concentrations: 0%, 0.3%, 0.7%, 1.0%)
 - Temperature stress (incubation at 4°C or -20°C) [78]

Viability Staining:

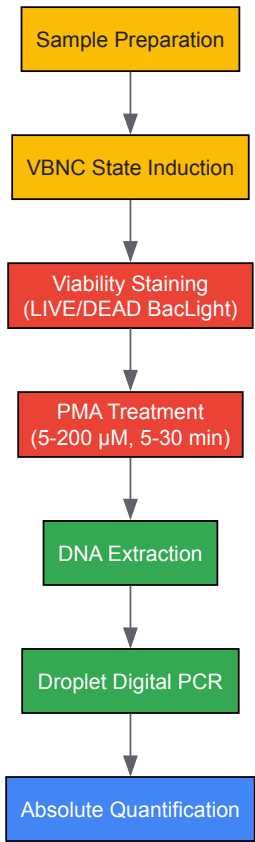
- Use LIVE/DEAD BacLight bacterial viability kit.
- Centrifuge 500 µL bacterial sample at 5000 × g for 15 minutes.
- Wash pellet with saline twice and resuspend in staining solution.
- Analyze under fluorescence microscopy: viable cells stain green, dead cells red [78].

PMA Treatment:

- Optimize PMA concentration (5 µM to 200 µM) and incubation time (5-30 minutes) [9].
- Add PMAx dye to sample to final concentration of 50 µM.
- Incubate in dark for 10 minutes with occasional mixing.
- Expose to LED light source for 15 minutes for photo-induced crosslinking.

DNA Extraction and Digital PCR:

- Extract genomic DNA using commercial kits.
- Partition sample into ~20,000 nanoliter-sized droplets.
- Perform PCR amplification with target-specific primers and probes.
- Count positive and negative droplets for absolute quantification without standard curve [9].



VBNC Detection Workflow

Research Reagent Solutions

Table 3: Essential Research Reagents for VBNC Studies

Reagent/Kit	Application	Function	Example Use
LIVE/DEAD BacLight	Viability staining	Differential staining of live (green) and dead (red) cells based on membrane integrity	Confirming VBNC state formation [78]
PMA/PMAxx	Viability PCR	Selective DNA modification in dead cells (membrane-compromised)	Differentiating viable cells in PCR detection [79] [9]
Carvacrol	Antimicrobial studies	Plant-based antimicrobial for combination studies	Testing efficacy against VBNC C. jejuni [6]
Al2O3 Nanoparticles	Antimicrobial studies	Nanoparticles with antimicrobial properties	Synergistic combinations against VBNC pathogens [6]
Droplet Digital PCR Reagents	Absolute quantification	Partitioning samples for digital counting	Quantifying VBNC cells without standard curve [9]

Innovative Approaches and Future Directions

Mathematical Modeling for Antimicrobial Studies

Conventional synergism testing methods (disk diffusion, checkboard assays) that rely on growth inhibition are unsuitable for VBNC bacteria [6]. A novel mathematical model applying the **combination index** derived from the Loewe additivity model enables quantitative evaluation of synergy against VBNC cells [6]. This approach considers the dose of each drug required to achieve a comparable quantitative effect, valuable for drugs with nonlinear dose-effect curves [6].

Protocol for Studying Antimicrobial Interactions Against VBNC Campylobacter jejuni [6]:

- **Prepare Simulated Poultry Processing Environment:**
 - Create suspension with chicken cecal contents (10% w/v) in sterile water

- Centrifuge at $5,000 \times g$ for 20 minutes, filter through 0.45- μm membrane

- **Induce VBNC State:**

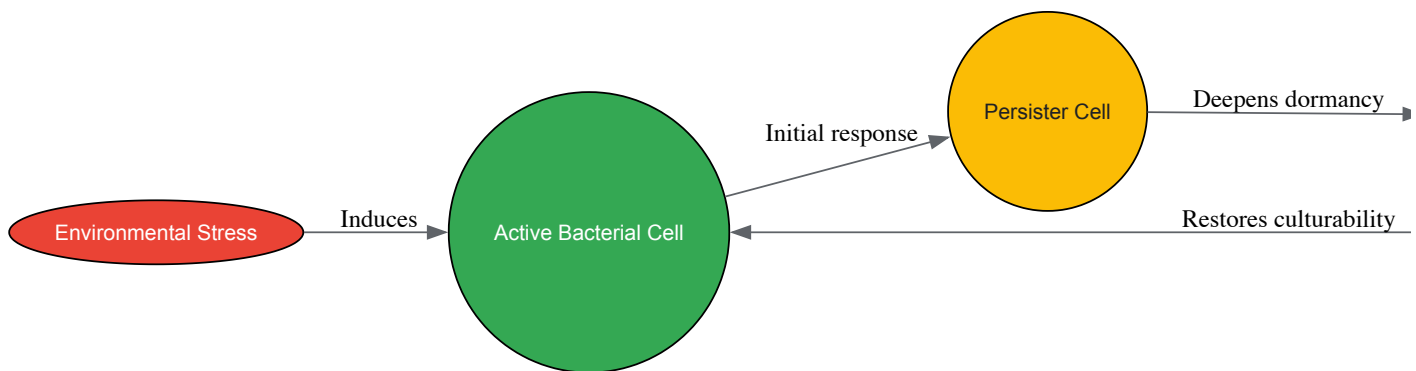
- Suspend *C. jejuni* in microcosm water at 4°C for 15 days [6]
- Confirm VBNC state by plate count (<1 CFU/mL) and viability staining

- **Time-Kill Assay:**

- Treat VBNC cells with individual antimicrobials (carvacrol, diallyl sulfide, Al_2O_3 NPs)
- Use concentrations from $0.5\times$ to $4\times$ minimum inhibitory concentration (MIC)
- Incubate at 25°C for 24 hours with agitation
- Sample at 0, 1, 2, 4, 8, 12, 24 hours for viability assessment

- **Mathematical Analysis:**

- Calculate combination index (CI) using Chou-Talalay method
- $\text{CI} < 1$ indicates synergy, $\text{CI} = 1$ additive effect, $\text{CI} > 1$ antagonism



VBNC State Transition Pathway

Integrated Systems and Artificial Intelligence

Emerging technologies integrate molecular detection with portable devices for field-deployable pathogen detection. The KRAKEN system exemplifies this approach, using **quantitative real-time PCR** to detect genetic material without requiring bacterial culturability [77]. Such systems enable continuous, autonomous pathogen testing in water supplies, food processing facilities, and clinical environments.

Artificial intelligence and machine learning are increasingly applied to analyze complex datasets generated by these advanced detection systems, providing:

- **Predictive models** for VBNC formation and resuscitation risks
- **Pattern recognition** in outbreak scenarios involving VBNC pathogens
- **Optimization** of detection protocols for specific matrices and pathogens [80]

The challenge of detecting viable but non-culturable pathogens requires a fundamental shift from culture-dependent to molecular and integrated approaches. Methods such as PMA-ddPCR, CRISPR-based detection, and mathematical modeling for antimicrobial testing provide powerful tools to overcome the limitations of conventional diagnostics. As research continues to unravel the molecular mechanisms of VBNC formation and resuscitation, the development of targeted detection and control strategies will be crucial for mitigating the public health threats posed by these elusive pathogens. Future efforts should focus on validating these advanced methods in real-world settings, reducing their cost and complexity, and integrating them into routine surveillance systems to better protect global health security.

Current Gaps in Anti-VBNC Therapeutic Strategies

The **viable but non-culturable (VBNC) state** represents a sophisticated survival strategy adopted by numerous bacterial pathogens when confronted with unfavorable environmental conditions. Bacteria in the VBNC state exhibit a state of **dormancy** with **markedly reduced metabolic activity**, enabling them to withstand otherwise lethal stresses, including antibiotic exposure, nutrient starvation, and extreme temperatures [48] [14]. Critically, while these cells lose the ability to form colonies on standard laboratory media—the cornerstone of conventional microbiological detection—they maintain **viability and pathogenicity**, and can resuscitate when conditions improve [48] [6]. This phenomenon poses a formidable challenge to public health, clinical treatment, and food safety, as VBNC cells **evade routine culture-based detection** and exhibit **drastically increased tolerance** to antimicrobials and antibiotics [48] [33]. The World Health Organization has identified antimicrobial resistance as a critical public health challenge, and the VBNC state represents a significant, yet underexplored, facet of this problem [48]. This

whitepaper delineates the principal gaps impeding the development of effective therapeutic strategies against VBNC pathogens and outlines the essential methodologies and reagents required to bridge these gaps.

Physiological and Diagnostic Gaps in VBNC Research

The Fundamental Challenge of Detection and Quantification

The core obstacle in VBNC research is their inherent **non-culturability** on routine media. This single characteristic invalidates the majority of standard antimicrobial efficacy tests, which rely on growth inhibition or cell death as measurable endpoints [48] [6]. Consequently, the true prevalence of VBNC cells in clinical, environmental, and industrial settings remains largely unknown, creating a significant blind spot in our understanding of persistent and recurring infections.

This diagnostic gap manifests in clinical scenarios such as **cystic fibrosis (CF) lung infections**, where VBNC cells of pathogens like *Pseudomonas aeruginosa* are suspected to contribute to chronicity despite antibiotic therapy [82]. Similarly, in the oral cavity, pathogens including *Porphyromonas gingivalis*, *Enterococcus faecalis*, and *Helicobacter pylori* can enter the VBNC state, potentially driving chronic periodontitis and systemic infections that are difficult to eradicate [33] [14]. The inability to reliably detect and quantify these cells in patient samples is a major barrier to accurate diagnosis and effective treatment.

Molecular Mechanisms and Physiological Barriers

The extreme tolerance of VBNC cells to antimicrobials is not fully understood, but several key mechanisms have been identified. Entry into the VBNC state involves profound physiological remodeling, including:

- **Metabolic Downregulation:** A severe reduction or alteration of metabolic activity is a hallmark of the VBNC state, reducing the efficacy of antibiotics that target active cellular processes [48] [14].
- **Membrane Integrity:** VBNC cells maintain an intact cell membrane, a key parameter of viability that also serves as a barrier to some compounds [48].
- **Genetic Regulation: Type II toxin-antitoxin (TAS) systems** are implicated in the induction of dormancy. Under stress, unstable antitoxins are degraded, allowing stable toxins to inhibit processes like translation and replication, thereby inducing a dormant, highly resistant state [48].
- **Structural Changes:** Alterations in the levels and composition of **outer membrane proteins (OMPs)**, **peptidoglycan cross-linking**, and **fatty acid profiles** have been observed, which may further contribute to reduced permeability and increased resilience [14].

Table 1: Documented Physiological Changes in VBNC Cells and Their Proposed Impact on Antimicrobial Tolerance

Physiological Change	Example Pathogen	Proposed Impact on Antimicrobial Tolerance
Upregulation of OmpW	<i>Escherichia coli</i> [14]	Potential barrier function, reducing compound uptake
Increased Peptidoglycan Cross-linking	<i>Enterococcus faecalis</i> [14]	Enhanced structural integrity and resilience
Altered Fatty Acid Profiles	<i>Vibrio vulnificus</i> [14]	Modification of membrane fluidity and permeability
Activation of Toxin-Antitoxin Systems	<i>Escherichia coli</i> [48]	Arrest of growth, rendering cells refractory to cidal drugs
Global Downregulation of Metabolism	Universal [48] [14]	Reduced activity of metabolic pathways targeted by antibiotics

Methodological Gaps in Anti-VBNC Compound Screening

The current paradigm for evaluating antimicrobial efficacy is fundamentally ill-suited for VBNC cells. Standard methods like disk diffusion, checkboard assays, and broth microdilution depend on bacterial growth, which is absent in dormant VBNC populations [6]. This creates a critical methodological gap, leaving researchers without robust, standardized tools to screen for compounds that can either kill VBNC cells or prevent their resuscitation.

Advanced Viability Testing Methodologies

To overcome the limitations of culture-based methods, several culture-independent techniques have been adapted for VBNC research. These methods typically rely on indicators of viability such as membrane integrity, metabolic activity, and detection of RNA or protein synthesis.

Protocol 1: PMA-ddPCR for Absolute Quantification of Viable Cells

This protocol uses **propidium monoazide (PMA)**, a DNA-intercalating dye that penetrates only cells with compromised membranes (dead cells). Upon light exposure, PMA covalently cross-links to the DNA, preventing its amplification by PCR. Thus, only DNA from viable cells with intact membranes can be amplified and quantified.

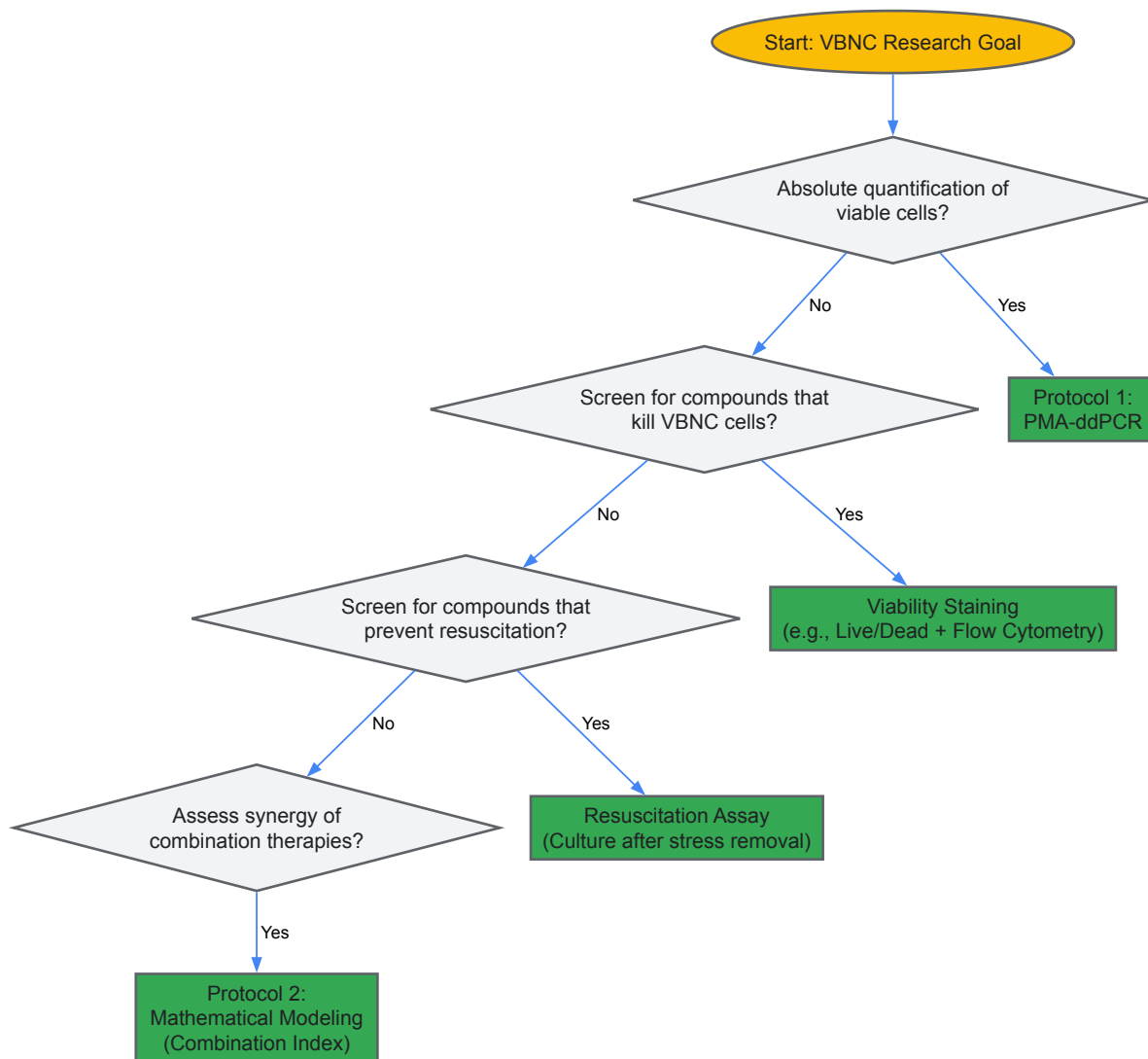
- **Sample Preparation:** Induce the VBNC state in a bacterial suspension (e.g., *Klebsiella pneumoniae* in artificial seawater at 4°C). Confirm non-culturability by plating on appropriate media with no colony formation after 48 hours [16].
- **PMA Treatment Optimization:**
 - Treat sample with PMA at a range of final concentrations (e.g., 5-200 µM).
 - Incubate in the dark for 5-30 minutes to allow PMA penetration into dead cells.
 - Place samples on ice and expose to a 650W halogen light source for 15 minutes at a 20cm distance to photo-activate the dye [16].
- **DNA Extraction:** Isolate genomic DNA from the PMA-treated cell suspensions using a commercial kit (e.g., Wizard Genomic DNA Purification Kit) [16].
- **Droplet Digital PCR (ddPCR):**
 - Prepare ddPCR reaction mix targeting single-copy genes (e.g., *rpoB*, *KP*, *adhE* for *K. pneumoniae*).
 - Generate droplets and perform PCR amplification.
 - Quantify the target DNA concentration (copies/µL) by counting positive and negative droplets using Poisson statistics. This provides an absolute count of viable (PMA-negative) cells without requiring a standard curve [16].

Protocol 2: Mathematical Modeling for Synergy Screening

For evaluating combination therapies against VBNC cells, growth-independent models are required. A novel application of a mathematical model based on the Loewe additivity concept can be used.

- **Time-Kill Assay:** Expose VBNC cells (e.g., *Campylobacter jejuni*) to single and combined antimicrobials (e.g., carvacrol, diallyl sulfide, Al₂O₃ nanoparticles) under relevant environmental conditions (e.g., simulated poultry processing). Sample at intervals (e.g., 0, 1, 2, 4 hours) [6].
- **Viable Cell Quantification:** Use a viability stain (e.g., PMA) coupled with qPCR or ddPCR to quantify the number of remaining viable cells at each time point, expressed as log CFU/mL.
- **Calculate Combination Index (CI):**
 - The CI is calculated using the equation: $CI = (D_1)/(D_x)_1 + (D_2)/(D_x)_2$ for two-drug combinations.
 - Where (D_x)₁ and (D_x)₂ are the doses of drug 1 and drug 2 alone required to produce x% kill (e.g., a 1-log reduction).
 - D₁ and D₂ are the doses of drug 1 and drug 2 in combination that also produce an x% kill.
 - **CI < 1 indicates synergy**, CI = 1 indicates additivity, and CI > 1 indicates antagonism [6].
- **Interpretation:** This model allows for a quantitative, mechanism-independent assessment of antimicrobial interactions against non-growing VBNC cells, identifying promising synergistic combinations for further development.

The workflow below illustrates the decision-making process for selecting the appropriate VBNC research methodology based on the experimental goal.



The Scientist's Toolkit: Essential Research Reagent Solutions

Bridging the gaps in VBNC research requires a specific set of reagents and tools designed to handle non-culturable cells. The following table details key solutions for a functional VBNC research laboratory.

Table 2: Key Research Reagent Solutions for VBNC Investigation

Research Reagent / Tool	Function & Rationale in VBNC Research
Propidium Monoazide (PMA)	Viability dye; selectively penetrates dead cells with compromised membranes, allowing PCR-based quantification of only viable (VBNC) cells [16].
Droplet Digital PCR (ddPCR)	Enables absolute quantification of target genes without a standard curve; superior for low-concentration samples and provides robust data with single-copy genes [16].
Viability qPCR (v-PCR)	Standard quantitative PCR coupled with PMA/EMA treatment; a common method to distinguish and quantify viable cells, though requires a standard curve [16] [33].
Flow Cytometry with Vital Stains	Allows high-throughput, multi-parameter analysis of cell viability and physiology using stains for membrane integrity (e.g., propidium iodide) and metabolic activity [16].
Transmission Electron Microscopy (TEM)	Visualizes ultrastructural morphological changes in VBNC cells, such as cell wall and membrane condensation, providing visual evidence of the dormant state [16].
Type II Toxin-Antitoxin System Mutants	Genetic tools to elucidate molecular mechanisms of VBNC induction and maintenance, e.g., via deletion or overexpression of toxin genes [48].
Simulated Environmental Systems	Custom microcosms (e.g., Artificial Seawater, simulated food processing fluids) to induce the VBNC state under controlled, relevant conditions [16] [6].

The pervasive threat of VBNC bacteria is exacerbated by a critical lack of targeted therapeutic strategies. The principal gaps—**diagnostic invisibility, physiological intractability, and methodological inadequacy**—are interlinked and must be addressed concurrently. Future research must prioritize the development of standardized, culture-independent protocols for drug screening, such as the PMA-ddPCR and mathematical modeling approaches outlined herein. A deeper, mechanistic understanding of the genetic triggers (e.g., TAS systems) and physiological adaptations that define the VBNC state is paramount to identifying vulnerable pathways that can be exploited therapeutically. The ultimate goal is to move beyond merely detecting VBNC cells and toward developing a new class of "anti-dormancy" agents capable of eradicating these persistent cells or securely locking them in a non-resuscitable state, thereby eliminating a major source of recurrent infections and treatment failures in human medicine.

Assessing the Proven Threat: VBNC Pathogens in Model Systems and Outbreaks

The **viable but non-culturable (VBNC) state** is a unique survival strategy adopted by many bacteria in response to adverse environmental conditions. In this state, bacteria **cannot form colonies** on routine culture media—the gold standard in clinical, food, and water safety testing—yet they remain **metabolically active**, maintain membrane integrity, and can retain **pathogenicity** [5] [2]. This phenomenon presents a formidable challenge to public health, as these pathogens evade detection by conventional methods, leading to a significant underestimation of viable bacterial counts and potentially contributing to unexplained disease outbreaks [5] [83]. The VBNC state was first identified in 1982 in *Escherichia coli* and *Vibrio cholerae* [5] [37], and it is now recognized that over 100 bacterial species, including major human pathogens, can enter this dormant condition [83] [4]. Induction factors are diverse, spanning starvation, extreme temperatures, osmotic pressure, and critically, food processing and water disinfection techniques such as chlorination, pasteurization, and treatment with preservatives [5] [2] [3]. This technical guide synthesizes current evidence linking VBNC pathogens to foodborne and waterborne outbreaks, providing detailed experimental protocols for their study and detection, framed within the broader context of VBNC research.

Documented Case Studies of VBNC-Associated Outbreaks

While direct causation is often challenging to prove conclusively, several epidemiological investigations and experimental studies provide compelling evidence that VBNC pathogens have played a role in past foodborne and waterborne outbreaks. The primary challenge in attribution is that the initial outbreak investigation, relying on standard culture methods, typically fails to detect the VBNC pathogen, leaving the outbreak without a confirmed etiological agent. Post-hoc laboratory studies on the implicated food or water matrix then demonstrate that the suspected pathogen can enter and resuscitate from the VBNC state under the relevant conditions.

The table below summarizes key documented cases where VBNC pathogens have been strongly implicated in outbreaks.

Table 1: Documented Case Studies of VBNC Pathogens in Foodborne and Waterborne Outbreaks

Implicated Pathogen	Vehicle/Matrix	Country	Year/Period	Key Evidence	Reference
<i>Salmonella</i> Oranienburg	Dried processed squid	Japan	2002 (Reported)	Resuscitation experiments confirmed the pathogen could enter the VBNC state in response to NaCl stress present in the dried squid.	[5]
Enterohemorrhagic <i>E. coli</i> O157	Salted salmon roe	Japan	Not Specified	Laboratory studies proposed <i>E. coli</i> O157 likely entered the VBNC state within the high-salt environment of the salmon roe.	[5]
<i>Vibrio cholerae</i> O1	Environmental Water Sources	Multiple	Ongoing Risk	VBNC cells remain viable in water, resuscitate in human/animal intestines, and have been shown to cause fluid accumulation in rabbit ileal loop assays.	[2]
<i>E. coli</i> (Uropathogenic)	Potable Water Systems	N/A (Recurrent Infections)	N/A	Studies link recurrent urinary tract infections to VBNC <i>E. coli</i> in water, which resuscitate in the host and demonstrate antibiotic resistance.	[2]
<i>E. coli</i> & <i>Pseudomonas aeruginosa</i>	Drinking Water System	Eastern China	2024 (Study)	PMA-qPCR detected VBNC forms in tap and potable water despite compliance with culturalbe bacteria standards, indicating a persistent health risk.	[83]

Beyond specific outbreaks, the pervasive presence of VBNC pathogens in treated water systems represents a continuous public health risk. A 2024 study comprehensively assessed the occurrence of VBNC *E. coli* and *P. aeruginosa* in a drinking water system in eastern China. The findings are summarized below.

Table 2: Occurrence of VBNC Pathogens in a Drinking Water System (2024 Study)

Water Type	VBNC <i>E. coli</i> Concentration (CFU/100 mL)	VBNC <i>P. aeruginosa</i> Concentration (CFU/100 mL)	Key Contributing Factors
Source Water	Up to 10 ²	Up to 10 ²	Initial high microbial load from environment.
Tap Water	10 ⁰ - 10 ²	10 ⁰ - 10 ²	Residual disinfectants (e.g., chlorine) induce VBNC state.
Potable Water	10 ⁰ - 10 ²	10 ⁰ - 10 ²	Loss of disinfectant residual, allowing for potential resuscitation.

This study also developed predictive models using artificial neural networks (ANN) and multiple linear regression (MLR), finding a strong **nonlinear relationship** between VBNC pathogen levels and conventional water quality parameters like turbidity and residual chlorine. A quantitative microbial risk assessment (QMRA) based on this data indicated that the health risks posed by VBNC pathogens in potable water could exceed acceptable benchmarks, highlighting a significant and often overlooked public health threat [83].

Detection Methodologies for VBNC Pathogens

Confirming the presence of VBNC cells requires a combination of methods that demonstrate both **viability** and **non-culturability**. A standard experimental workflow involves inducing the VBNC state, confirming the loss of culturability, and then applying viability assays.

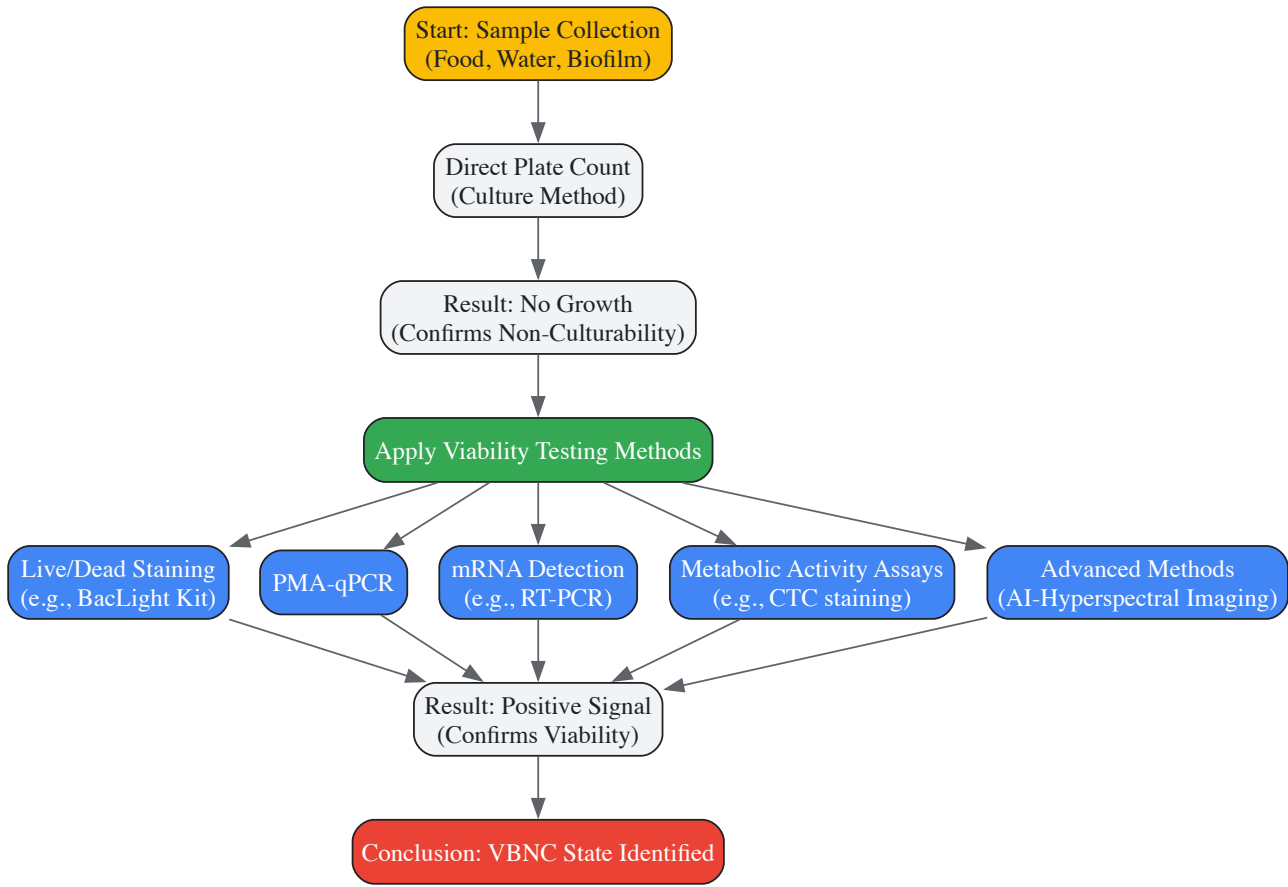


Diagram 1: Experimental Workflow for VBNC State Identification

Key Direct Detection Methods

- PMA-qPCR (Propidium Monoazide quantitative Polymerase Chain Reaction):** This is currently the **optimal method** for detecting specific VBNC pathogens in complex samples like food and water [83]. PMA dye penetrates only cells with compromised membranes (dead cells) and covalently cross-links to their DNA, inhibiting its amplification in PCR. In contrast, viable cells (including VBNC) with intact membranes exclude the dye, allowing their DNA to be amplified and quantified. This method specifically targets and quantifies viable cells of a specific pathogen.

- **Typical Protocol:** A water sample is centrifuged to pellet cells. The pellet is resuspended and treated with PMA dye. The sample is then exposed to bright light (photolysis) to activate the PMA cross-linking. DNA is extracted, and qPCR is performed targeting a pathogen-specific gene (e.g., *rfbE* for *E. coli* O157). The difference between qPCR signals from non-PMA-treated and PMA-treated samples indicates the quantity of viable (VBNC) cells [83].
- **Live/Dead Staining with Epifluorescence Microscopy:** This method assesses **cell membrane integrity**. Kits like the LIVE/DEAD BacLight use a mixture of two nucleic acid stains: SYTO 9 (green fluorescence, penetrates all cells) and propidium iodide (red fluorescence, penetrates only damaged membranes). Under an epifluorescence microscope, viable (including VBNC) cells appear green, while dead cells appear red [2] [83]. This provides a total viable count but cannot distinguish between species.
- **Metabolic Activity Detection:** Methods like CTC (5-cyano-2,3-ditolyl tetrazolium chloride) staining are used. Viable cells reduce the colorless CTC to red, fluorescent formazan crystals, which can be enumerated using fluorescence microscopy or flow cytometry [83]. This confirms respiratory activity in VBNC cells.
- **mRNA Detection:** Reverse Transcriptase-PCR (RT-PCR) can detect messenger RNA, which has a short half-life. The presence of pathogen-specific mRNA (e.g., for virulence genes or housekeeping genes) is a strong indicator of viability and metabolic activity, as it confirms ongoing gene expression in VBNC cells [12] [37]. For instance, VBNC *E. coli* O157 has been shown to express mRNA for the *rfbE* and *fliC* genes after 10 months in water [12].
- **AI-Enabled Hyperspectral Microscopy (HMI):** A novel, rapid detection framework. This method induces the VBNC state (e.g., in *E. coli* with low-level H₂O₂), collects hyperspectral images capturing physiological changes in cells, and uses a deep learning model (e.g., EfficientNetV2) to classify cells as "normal" or "VBNC" based on pseudo-RGB images derived from characteristic spectral wavelengths. This approach has achieved **97.1% accuracy** and holds promise for automated, culture-free detection [84].

Detailed Experimental Protocol: Studying VBNC Pathogens In Vitro

The following section provides a detailed methodology for inducing, detecting, and resuscitating VBNC pathogens, based on protocols cited in the literature. This serves as a template for researchers to investigate the VBNC state in specific pathogen-food/water matrix pairs.

Induction of the VBNC State

The goal is to subject a culturable bacterial population to sub-lethal stress until plate counts drop to zero while viability is maintained.

- **Pathogen Strain Selection:** Use a well-characterized strain of the target pathogen (e.g., *E. coli* O157:H7, *Salmonella* Enteritidis, *Listeria monocytogenes*).
- **Induction Stressors:**
 - **Chlorine Stress:** Resuspend cells in a water matrix (e.g., sterilized surface water) or PBS containing a low concentration of sodium hypochlorite (e.g., 0.1-0.5 mg/L) [83]. Incubate in the dark at a relevant temperature (e.g., 4°C or 25°C).
 - **Food Matrix Stress:** Inoculate the pathogen into a specific food, such as refrigerated pasteurized juice, milk, or a high-salt product like dried squid. The natural physicochemical properties (low pH, preservatives, low *a_w*) of the food itself can induce the VBNC state [5] [2].
 - **Oxidative and Acid Stress:** For laboratory models, exposure to 0.01% hydrogen peroxide or 0.001% peracetic acid for several days can efficiently induce the VBNC state in *E. coli* [84].
- **Monitoring Induction:** At regular intervals (e.g., daily), sample the suspension/food homogenate.
 - Perform **serial dilution and plate counting** on appropriate agar. The VBNC state is considered induced when the plate count reaches zero (CFU/mL = 0).
 - Simultaneously, perform **live/dead staining** to confirm that a significant population of cells remains viable (green fluorescence).

Confirmation of the VBNC State

Once culturability is lost, apply the detection methods from Section 3 to confirm viability.

- **PMA-qPCR:** This is the confirmatory method of choice. Compare qPCR results from PMA-treated and untreated samples of the induced culture. A high qPCR signal in the PMA-treated sample confirms the presence of intact, viable cells that are non-culturable [83].
- **Virulence Gene Expression:** Use RT-PCR to detect mRNA of key virulence genes. For example, the detection of Shiga toxin (*stx*) mRNA in VBNC *E. coli* O157 confirms the pathogen retains its pathogenic potential even in the dormant state [12] [2].

Resuscitation is the process of reversing the VBNC state, causing cells to regain culturability. This is crucial for fulfilling Koch's postulates and proving potential pathogenicity.

- **In Vitro Methods:**
 - **Temperature Upshift:** A common method is to incubate the VBNC cell suspension at a higher, permissive temperature (e.g., shifting from 4°C to 25-37°C) [5].

- **Nutrient Supplementation:** Adding fresh, rich media (e.g., Tryptic Soy Broth) or specific compounds like sodium pyruvate can facilitate resuscitation [5].
- **Resuscitation-Promoting Factors (Rpfs):** The addition of purified Rpf, a bacterial cytokine, has been shown to stimulate the resuscitation of VBNC cells in both Gram-positive and Gram-negative bacteria [12].
- **In Vivo Methods:**
 - **Animal Passage:** The most reliable method for resuscitation. VBNC cells are introduced into an animal model (e.g., oral administration to mice, rabbit ileal loop, or injection into embryonated eggs) [5] [12]. The cells resuscitate in the host environment, and the pathogen can be re-cultured from host tissues or fluids.
 - **Co-culture with Amoebae:** For pathogens like *Legionella pneumophila*, co-culturing VBNC cells with amoebae (e.g., *Acanthamoeba castellanii*) can provide the necessary signals for resuscitation [83].

The Scientist's Toolkit: Essential Research Reagents and Materials

Research into the VBNC state requires a combination of classic microbiological tools and modern molecular biological reagents. The following table lists key solutions and materials essential for experiments in this field.

Table 3: Key Research Reagent Solutions for VBNC Studies

Reagent/Material	Function/Application	Example Use Case
PMA (Propidium Monoazide)	Chemical dye that selectively binds DNA in membrane-compromised (dead) cells, preventing its PCR amplification.	Used in PMA-qPCR to differentiate between viable (VBNC) and dead cells in a sample, enabling specific quantification.
LIVE/DEAD BacLight Kit	Fluorescent staining kit for assessing bacterial membrane integrity via epifluorescence microscopy or flow cytometry.	Provides a direct visual count of viable (green) vs. dead (red) cells in a population, confirming viability after loss of culturability.
CTC (5-cyano-2,3-ditolyl tetrazolium chloride)	Tetrazolium dye that is reduced to fluorescent formazan by metabolically active bacteria.	Detects respiratory activity in VBNC cells, serving as a marker for metabolic viability.
RNA Protect Reagents / RNA Extraction Kits	Stabilize and purify high-quality RNA from bacterial samples for downstream molecular analysis.	Essential for RT-PCR experiments aimed at detecting gene expression (mRNA) in VBNC cells, proving metabolic activity.
Resuscitation-Promoting Factor (Rpf)	Bacterial cytokine protein that stimulates the resuscitation of VBNC cells.	Used in in vitro experiments to trigger the recovery of culturability in a population of VBNC cells.
Specific Primers & Probes	Oligonucleotides designed to target species-specific genes or virulence genes for PCR and qPCR.	Allows for the precise identification and quantification of a specific VBNC pathogen (e.g., detecting <i>E. coli</i> O157 via <i>rfbE</i> gene).
Axenic Amoeba Cultures	(e.g., <i>Acanthamoeba castellanii</i>) Provide a host environment for the resuscitation of certain VBNC pathogens.	Used in co-culture experiments to resuscitate VBNC <i>Legionella pneumophila</i> and other amoeba-resistant bacteria.

The viable but non-culturable state represents a critical frontier in microbial food and water safety. As documented in the case studies and data presented, VBNC pathogens are not a mere laboratory curiosity but a **demonstrated public health threat** implicated in outbreaks and contributing to the persistence of pathogens in treated water systems. The failure of conventional, culture-based methods to detect these cells leads to a dangerous false sense of security. Mitigating the risks posed by VBNC pathogens requires a paradigm shift in regulatory and monitoring practices. The future of outbreak investigation and routine safety testing must integrate **molecular and advanced detection methods**, such as PMA-qPCR and AI-enabled diagnostics, which can differentiate between truly safe samples and those harboring dormant but dangerous pathogens. Further research is urgently needed to develop more efficient resuscitation techniques, to fully understand the environmental triggers and genetic regulation of this state, and to design novel interventions that either prevent entry into the VBNC state or effectively eliminate these persistent cells from our food and water supplies.

The viable but non-culturable (VBNC) state represents a unique survival strategy adopted by numerous bacterial pathogens when confronted with unfavorable environmental conditions. First discovered in *Escherichia coli* and *Vibrio cholerae* in 1982, this physiological state is now recognized in at least 67 species of pathogenic bacteria, including the high-concern pathogens *Klebsiella pneumoniae*, *Campylobacter jejuni*, and *Escherichia coli* O157:H7 [37] [85]. In the VBNC state, bacteria undergo a dramatic physiological transformation, losing the ability to form colonies on routine culture media—the very foundation of conventional microbiological detection—while maintaining metabolic activity and viability [37]. This fundamental characteristic poses a substantial threat to public health and food safety, as standard plating methods routinely employed in clinical and quality control laboratories fail to detect these dormant pathogens, leading to potentially dangerous underestimation of contamination levels [37] [8] [86].

The transition to the VBNC state is typically triggered by various environmental stresses commonly encountered in food processing, clinical settings, and natural environments. These inducing factors include nutrient starvation, temperature extremes (particularly refrigeration at 4°C or freezing at -20°C), exposure to sanitizers like chlorine, osmotic stress, and antibiotic pressure [37] [85] [10]. From a physiological perspective, VBNC cells undergo significant morphological and compositional changes, including reduction in cell size (dwarfing), cell rounding, increased peptidoglycan cross-linking, and alterations in membrane fatty acid composition [37]. Perhaps most notably, VBNC cells exhibit markedly enhanced resistance to physical and chemical challenges, including antibiotics, extreme temperatures, pH extremes, and chlorine-based sanitizers, making them considerably harder to eradicate than their culturable counterparts [37] [8].

The critical public health concern surrounding VBNC pathogens stems from their potential to resuscitate under favorable conditions and regain culturability and virulence. This resuscitation can occur in appropriate hosts or environments, potentially leading to disease outbreaks that appear to originate from "sterile" sources that passed conventional microbiological testing [37] [85] [86]. For researchers, drug development professionals, and food safety specialists, understanding the validated risks posed by specific pathogens in the VBNC state is paramount for developing effective detection methods, intervention strategies, and risk assessment models.

VBNC State Formation and Risks by Pathogen

Klebsiella pneumoniae

Klebsiella pneumoniae, particularly the high alcohol-producing (HiAlc Kpn) strain, has been investigated for its ability to enter the VBNC state and its associated health implications. HiAlc Kpn has been linked to non-alcoholic fatty liver disease (NAFLD) through its persistent colonization of the gut microbiome [9] [10]. Research has demonstrated that HiAlc Kpn can be induced into the VBNC state when stored in artificial seawater (ASW) at 4°C, with entry into this state confirmed when no colony formation occurs on Luria-Bertani (LB) agar plates after 48 hours of incubation [10]. The metabolic activity of these VBNC cells, a key indicator of viability, has been successfully quantified using advanced molecular methods combining propidium monoazide (PMA) with droplet digital PCR (ddPCR) targeting single-copy genes (*KP*, *rpoB*, and *adhE*) [9] [10].

A significant finding regarding the risk potential of VBNC *K. pneumoniae* concerns its response to antibiotic treatment. Studies have revealed that ciprofloxacin, a broad-spectrum fluoroquinolone antibiotic, inhibits the resuscitation of VBNC-state HiAlc Kpn cells. However, upon antibiotic removal, these cells maintain their capacity for resuscitation and continued ethanol production [9] [10]. This phenomenon underscores the persistent threat posed by VBNC *K. pneumoniae*, as standard antibiotic treatments may suppress detection and recovery without eliminating the underlying bacterial reservoir, potentially leading to recurrent infections or chronic conditions like NAFLD.

Campylobacter jejuni

Campylobacter jejuni, a leading cause of foodborne gastroenteritis worldwide, can enter the VBNC state under various stress conditions, including temperature shifts and nutrient limitation. When induced at 4°C under aerobic conditions, *C. jejuni* undergoes significant morphological changes, transitioning from its characteristic spiral shape in the exponential phase to a coccoid form in the VBNC state [37] [87]. Physiological characterization of VBNC *C. jejuni* has revealed a notable increase in cell volume (from 1.73 µL/mg protein for culturable cells to 10.96 µL/mg protein after 30 days in microcosm water), along with significantly reduced internal potassium content and membrane potential [87]. Furthermore, the adenylate energy charge (AEC) decreases dramatically (from 0.66 to 0.26), with AMP becoming the only detectable nucleotide after extended incubation, indicating a profound downregulation of metabolic activity [87].

The pathogenic potential and colonization capacity of VBNC *C. jejuni* have been recently evaluated using a mouse model. When administered to mice, culturable cells of *C. jejuni* strain NCTC11168 successfully colonized the intestinal tract, as evidenced by increased bacterial numbers in stool detected through both colony counting and quantitative PCR. In stark contrast, VBNC-state cells did not yield any culturable cells in infected mice and failed to establish colonization [88]. Importantly, no clinical signs, weight loss, or significant increases in inflammatory biomarkers (lipocalin-2 and myeloperoxidase) were observed in mice administered VBNC cells [88]. These findings suggest that while *C. jejuni* in the VBNC state may not pose an immediate risk for causing food poisoning through direct infection, the potential risk from possible resuscitation in the human gut requires further investigation.

Escherichia coli O157:H7

Escherichia coli O157:H7, a pathogen of significant public health concern due to its low infectious dose and severe complications, has been extensively studied for its ability to enter, persist in, and resuscitate from the VBNC state. Research has systematically documented the induction of the VBNC state in *E. coli* O157:H7 under various low-temperature conditions. In one comprehensive study, an initial inoculum of 2.1×10^7 CFU/mL of *E. coli* O157:H7 cells was induced into the VBNC state in different media: normal saline at -20°C after 176 days, distilled water at -20°C after 160 days, and LB broth at -20°C after just 80 days [85]. These findings indicate that both temperature and nutrient availability significantly influence the kinetics of VBNC state induction, with rich media like LB broth accelerating the process under freezing conditions.

Morphological examination of VBNC *E. coli* O157:H7 cells reveals a shift from the typical rod shape to a shorter rod form, accompanied by a decrease in cell size [85]. The resuscitation of VBNC *E. coli* O157:H7 has been successfully achieved using various approaches, with studies identifying that the addition of 5% Tween 80 to LB medium and 3% Tween 80 to saline most effectively promoted resuscitation of cells induced by low temperature [85].

The ability of *E. coli* O157:H7 to enter the VBNC state and subsequently resuscitate poses a substantial food safety risk, as evidenced by outbreaks linked to VBNC cells in salted salmon roe [86]. This pathogen's capacity to maintain virulence potential while in the VBNC state further compounds the public health threat, necessitating specialized detection methods that can accurately identify these dormant cells in food products and environmental samples.

Comparative Analysis of VBNC Pathogens

Table 1: VBNC State Formation and Characteristics Across Pathogens

Parameter	<i>Klebsiella pneumoniae</i>	<i>Campylobacter jejuni</i>	<i>Escherichia coli</i> O157:H7
Primary Induction Conditions	Storage in ASW at 4°C [10]	Microcosm water at 4°C; aerobic conditions [87] [88]	LB broth at -20°C (80 days); saline at -20°C (176 days) [85]
Morphological Changes	Not specified in results	Spiral to coccoid shape; cell volume increase from 1.73 to 10.96 µL/mg protein [37] [87]	Rod to short rod; decreased cell size [85]
Key Metabolic Features	Maintains metabolic activity detectable by PMA-ddPCR [9] [10]	Reduced AEC (0.66 to 0.26); decreased internal K+ and membrane potential [87]	Maintains metabolic activity detectable by DVC and AODC methods [85]
Resuscitation Triggers	Removal of ciprofloxacin pressure; fresh YPD medium [10]	Not specified in results	5% Tween 80 in LB; 3% Tween 80 in saline [85]
Virulence Retention	Maintains ethanol production post-resuscitation [10]	Lacks colonization ability in mouse model [88]	Retains virulence potential; linked to foodborne outbreaks [86]
Detection Challenges	Escapes culture on LB agar [10]	Not detectable by standard plating [87]	Conventional methods fail detection [85] [86]

Advanced Detection Methodologies for VBNC Cells

Molecular Detection Technologies

The accurate detection and quantification of VBNC pathogens require sophisticated approaches that bypass the limitations of conventional culture methods. Several molecular technologies have been developed and validated for this purpose, with viability PCR (v-PCR) emerging as a particularly powerful tool. This methodology utilizes nucleic acid intercalating dyes such as propidium monoazide (PMA) or ethidium monoazide (EMA), which penetrate cells with compromised membranes—characteristic of dead cells—and form covalent bonds with DNA upon photoactivation, thereby inhibiting PCR amplification [8] [10] [86]. This selective amplification allows v-PCR to specifically target DNA from viable cells (including VBNC cells) with intact membranes, providing a more accurate assessment of viable pathogen load than conventional PCR.

Recent advancements in v-PCR methodologies have further refined detection capabilities. Research on *Listeria monocytogenes* detection in process wash water (PWW) from the fruit and vegetable industry demonstrated that a combination of EMA (10 µM) and PMAx (an improved version of PMA; 75 µM), incubated at 40°C for 40 minutes followed by a 15-minute light exposure, effectively inhibited qPCR amplification from dead cells while allowing detection of VBNC cells [8]. This optimized viability qPCR (v-qPCR) protocol proved particularly effective for complex water matrices like PWW, where flow cytometry methods were unsuitable due to interference from organic matter that led to overestimation of dead cells [8].

Droplet digital PCR (ddPCR) has emerged as another highly promising technology for VBNC cell detection, offering absolute quantification without the need for external standard curves. A landmark study on HiAlec Kpn established a PMA-ddPCR method targeting three single-copy genes (*KP*, *rpoB*, and *adhE*) to enumerate viable cells [9] [10]. The researchers optimized PMA concentration (testing 5-200 µM) and incubation time (5-30 minutes), establishing that this approach could directly quantify viable cells without reference standards while maintaining technical robustness comparable to qPCR [10]. In mouse fecal samples, the method detected activity reductions ranging from 0.64 to 1.13 log₁₀ DNA copies/mL, demonstrating its sensitivity even in complex biological matrices [9] [10].

Loop-mediated isothermal amplification (LAMP) has also been adapted for VBNC pathogen detection, offering a rapid, sensitive alternative that doesn't require sophisticated thermal cycling equipment. PMA-LAMP assays developed for VBNC *E. coli* O157:H7 (targeting the *wzy* gene) and *Salmonella enterica* (targeting the *agfA* gene) achieved detection limits of 9.0 CFU/reaction and 4.6 CFU/reaction in pure culture, respectively [86]. When applied to fresh produce samples, the assays detected VBNC *E. coli* O157:H7 at 5.13 × 10³ to 5.13 × 10⁴ CFU/g and VBNC *S. enterica* at 1.05 × 10⁴ to 1.05 × 10⁵ CFU/g, with the entire assay completed in approximately 30 minutes—significantly faster than traditional qPCR methods [86].

Comparative Method Performance

Table 2: Comparison of VBNC Detection Methods

Method	Principle	Applications in Search Results	Advantages	Limitations
PMA-qPCR	PMA dye inhibits DNA amplification from membrane-compromised (dead) cells; qPCR then detects viable cells [8] [10]	Detection of VBNC <i>Listeria monocytogenes</i> in process wash water; HiA1c Kpn quantification [8] [10]	Quantitative; relatively fast (1-2 hours); specific	Requires optimization for each matrix; may overestimate if dead cells have intact membranes [8]
PMA-ddPCR	PMA treatment followed by partitioning into nanodroplets for absolute digital quantification [9] [10]	Absolute quantification of HiA1c Kpn viable cells in mouse fecal samples [9] [10]	Absolute quantification without standard curves; high precision; resistant to PCR inhibitors	Higher equipment costs; more complex workflow
PMA-LAMP	PMA treatment combined with isothermal DNA amplification [86]	Detection of VBNC <i>E. coli</i> O157:H7 and <i>Salmonella</i> in fresh produce [86]	Rapid (30 minutes); isothermal conditions; equipment-free result interpretation	Semi-quantitative at best; primer design more complex
Flow Cytometry with Viability Stains	Membrane-permeant vs. impermeant fluorescent dyes distinguish viable/dead cells [8]	Attempted detection of VBNC <i>Listeria</i> in process wash water [8]	Rapid; visual confirmation; single-cell resolution	Overestimates dead cells in complex matrices; interference from organic matter [8]
Direct Viable Count (DVC)	Cell elongation in presence of nutrients + DNA gyrase inhibitors indicates viability [85]	Enumeration of VBNC <i>E. coli</i> O157:H7 [85]	Direct visualization of metabolic activity; no specialized equipment	Labor-intensive; subjective; not suitable for all bacteria

Experimental Protocols for VBNC Research

VBNC State Induction Protocol for Escherichia coli O157:H7

The induction of *E. coli* O157:H7 into the VBNC state following a standardized protocol yields reproducible results essential for systematic research. The process begins with preparing an overnight culture of *E. coli* O157:H7 in LB broth, incubated at 37°C with shaking at 190 rpm. This culture is then subcultured into fresh LB broth and grown to mid-logarithmic phase, which is determined by monitoring optical density at 600 nm (OD₆₀₀) [85]. Exponential phase cells are harvested by centrifugation at 8,000 × g for 5 minutes, followed by three washes with 0.85% (w/v) sterile saline solution to remove residual nutrients. The washed cells are resuspended in sterile saline solution at a final density of approximately 10⁷ CFU/mL [85].

For VBNC state induction, aliquots of the cell suspension are transferred to various induction conditions that reflect different environmental scenarios: physiological saline at -20°C, sterile distilled water at -20°C, LB broth at -20°C, sterile distilled water at 4°C, and LB broth at 4°C [85]. To avoid the confounding effects of repeated freezing and thawing during periodic sampling, researchers should distribute the bacterial suspension in 1 mL aliquots into multiple 1.5 mL tubes before low-temperature incubation. Culturability is assessed every 4 days by plating 10 µL suspensions on LB agar plates and incubating at 37°C for 24-48 hours. Cells are considered to have entered the VBNC state when culturable counts drop below 0.1 CFU/mL for three consecutive days, while total cell counts (determined by acridine orange direct count) remain substantially higher than culturable counts [85].

PMA-ddPCR Quantification Protocol for Klebsiella pneumoniae

The accurate quantification of VBNC *K. pneumoniae* using PMA-ddPCR requires careful optimization and execution. Begin by optimizing PMA concentration through tests across a range of 5-200 µM in ultrapure water to determine the ideal concentration that maximizes suppression of DNA amplification from dead cells while minimizing toxicity to viable cells [10]. Simultaneously, optimize incubation time by testing intervals of 5, 10, 20, and 30 minutes at laboratory temperature, followed by photoactivation using a 650W double-ended halogen light source positioned 20 cm from sample tubes for 15 minutes [10].

For sample processing, mix bacterial suspensions with the optimized PMA concentration and incubate in the dark for the determined optimal time. After photoactivation, extract DNA using a standardized commercial kit. Design primer-probe sets targeting three single-copy genes in *K. pneumoniae* (*KP*, *rpoB*, and *adhE*) to ensure quantification accuracy through averaging of multiple genetic targets [10]. Prepare the ddPCR reaction mixture according to manufacturer specifications, then partition samples into approximately 20,000 nanodroplets using a droplet generator. Perform PCR amplification with optimized thermal cycling conditions, and subsequently read droplets using a droplet reader to determine the fraction of positive reactions [9] [10]. Calculate the absolute copy number concentration of target genes using Poisson distribution analysis, which provides quantification of viable VBNC cells without requiring external standard curves [10].

Research Reagent Solutions Toolkit

Table 3: Essential Research Reagents for VBNC Studies

Reagent/Chemical	Function in VBNC Research	Specific Applications in Search Results
Propidium Monoazide (PMA)	DNA intercalating dye that penetrates membrane-compromised cells; inhibits PCR amplification after photoactivation [8] [10]	Differentiation of viable/dead cells in PMA-qPCR and PMA-ddPCR; used at 5-200 µM for HiA1c Kpn [9] [10]
PMAxx	Enhanced version of PMA with improved penetration of dead cells [8]	Combined with EMA for VBNC <i>Listeria</i> detection in process wash water [8]
Ethidium Monoazide (EMA)	DNA intercalating dye; penetrates cells using efflux pumps [8]	Used with PMAxx (10 µM EMA + 75 µM PMAxx) for optimal detection in complex matrices [8]
Nalidixic Acid	DNA gyrase inhibitor; prevents DNA replication [85]	Used in DVC method at 0.002% concentration to cause cell elongation in viable <i>E. coli</i> O157:H7 [85]
Acridine Orange	Fluorescent nucleic acid stain for total cell counting [85]	Used in AODC method at 0.01% concentration for total <i>E. coli</i> O157:H7 counts [85]
Tween 80	Non-ionic surfactant; may facilitate membrane repair or nutrient uptake [85]	Resuscitation of VBNC <i>E. coli</i> O157:H7 (5% in LB medium; 3% in saline) [85]
Ciprofloxacin	Fluoroquinolone antibiotic; DNA gyrase inhibitor [10]	Inhibits resuscitation of VBNC HiA1c Kpn at 3, 9, or 18 µg/mL concentrations [10]
Chlorine (Sodium Hypochlorite)	Oxidizing sanitizer; induces VBNC state [8]	Induction of VBNC state in <i>Listeria monocytogenes</i> at 10 mg/L in process wash water [8]

Risk Assessment and Pathogenicity Considerations

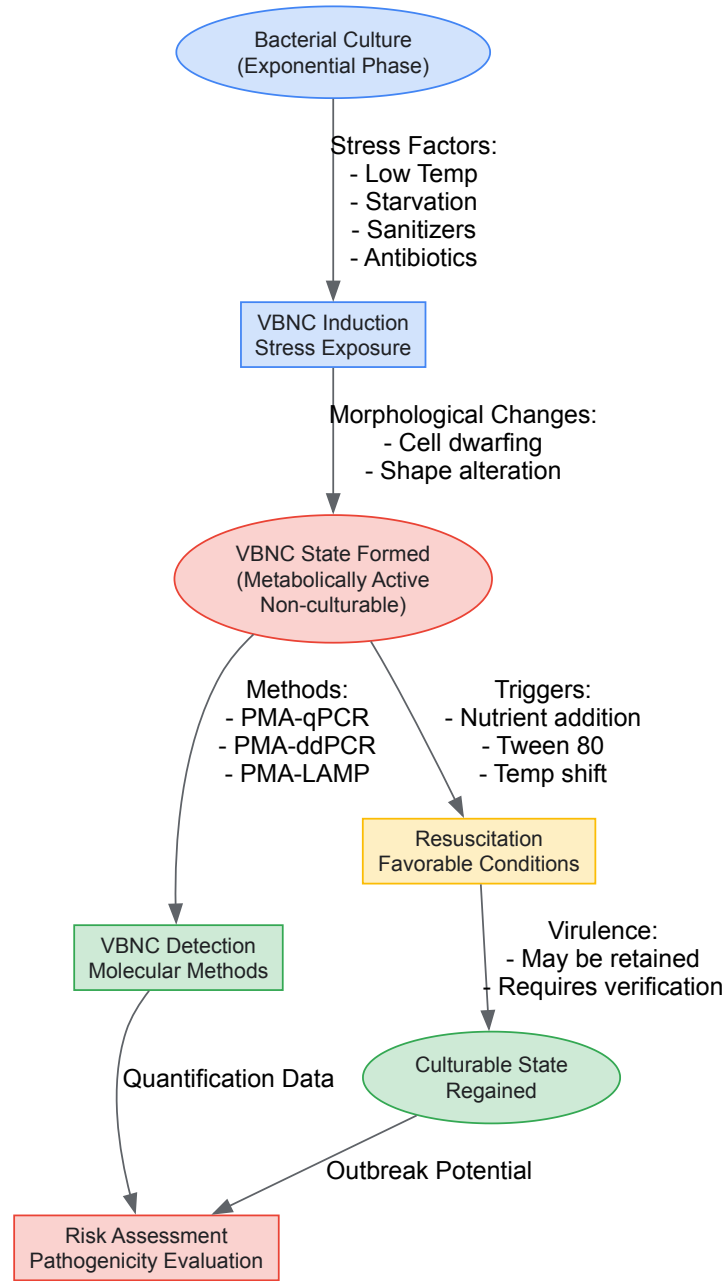
The pathogenicity and associated risks of VBNC bacteria vary significantly across species and environmental contexts, necessitating careful assessment for each pathogen of concern. For *Campylobacter jejuni*, recent evidence suggests potentially reduced immediate risk, as studies using a mouse model demonstrated that VBNC cells lacked the ability to colonize the intestinal tract or induce clinical signs of disease, inflammation, or weight loss [88]. However, these findings should be interpreted with caution, as differences between mouse and human physiology may affect pathogen behavior, and the possibility of resuscitation in more favorable environments cannot be excluded.

In contrast, *Escherichia coli* O157:H7 in the VBNC state presents a validated public health threat, with documented outbreaks linked to the consumption of foods contaminated with VBNC cells, notably salted salmon roe [86]. The ability of VBNC *E. coli* O157:H7 to resuscitate under appropriate conditions and regain full virulence potential underscores the significant risk this pathogen poses throughout the food production chain [85] [86]. Similarly, VBNC *Klebsiella pneumoniae* maintains its metabolic capabilities, particularly in HiA1c Kpn strains, which continue to produce ethanol after resuscitation—a factor implicated in the development and progression of non-alcoholic fatty liver disease [10]. The concerning finding that ciprofloxacin inhibits resuscitation without eliminating VBNC HiA1c Kpn cells highlights the potential for recurrent infections following antibiotic therapy cessation [9] [10].

The resistance profiles of VBNC cells further compound their risk assessment. Bacteria in the VBNC state universally exhibit enhanced tolerance to environmental stresses and antimicrobial agents compared to their culturable counterparts [37]. This heightened resistance stems from their reduced metabolic rate, morphological changes (including reduced cell size), and structural modifications such as increased peptidoglycan cross-linking [37]. This multi-factorial resistance mechanism enables VBNC pathogens to persist in food processing environments despite sanitation protocols, in clinical settings despite antibiotic treatments, and in environmental reservoirs where they may serve as silent contamination sources.

Visualizing VBNC Research Workflows

VBNC Research Workflow: From Induction to Risk Assessment



VBNC Research Workflow Diagram

The diagram above illustrates the comprehensive workflow for VBNC research, encompassing state induction, detection, resuscitation, and risk assessment. This visual representation highlights the critical pathways and decision points in studying VBNC pathogens, emphasizing the transition between physiological states and the methodologies required at each stage.

The validated risks posed by *Klebsiella pneumoniae*, *Campylobacter jejuni*, and *Escherichia coli* O157:H7 in the VBNC state necessitate continued research and methodological refinement. While these pathogens share the common ability to enter this dormant state, they display important differences in their induction conditions, pathogenicity retention, and resuscitation behaviors. The development and optimization of advanced detection methodologies, particularly PMA-based molecular approaches coupled with qPCR, ddPCR, or LAMP platforms, have significantly improved our capacity to accurately identify and quantify these elusive pathogens. Future research directions should focus on elucidating the genetic mechanisms regulating VBNC state entry and exit, developing intervention strategies specifically targeting VBNC cells, and validating detection methods across diverse food matrices and clinical samples. As our understanding of the VBNC state continues to evolve, so too will our ability to mitigate the substantial public health risks posed by these dormant pathogens.

Comparative Analysis of Antibiotic Resistance in VBNC vs. Culturable Cells

The viable but non-culturable (VBNC) state is a dormant survival strategy adopted by many bacteria in response to environmental stressors, such as nutrient starvation, temperature shifts, and exposure to disinfectants or antibiotics [48] [37]. In this state, cells undergo a profound physiological transformation: they fail to grow on routine culture media—the gold standard for microbial viability assessment—but maintain metabolic activity and

membrane integrity, and can resuscitate when favorable conditions return [37] [14]. This phenomenon poses a significant challenge to public health, as standard clinical and environmental detection methods fail to identify these pathogens, leading to an underestimation of microbial risk [89] [37].

A critical and concerning characteristic of VBNC cells is their markedly enhanced tolerance to antimicrobial agents, including antibiotics [48] [69]. This guide provides a comparative analysis of antibiotic resistance in VBNC cells versus their culturable counterparts. It synthesizes current research to elucidate the underlying mechanisms, presents quantitative data, outlines key experimental methodologies, and discusses the implications for drug development and infection control. Understanding this aspect of phenotypic plasticity is essential for researchers and scientists aiming to develop novel strategies to combat persistent and recalcitrant bacterial infections.

Resistance Profiles: VBNC vs. Culturable Cells

VBNC cells exhibit a significantly increased tolerance to a wide spectrum of antibiotics compared to culturable cells. This tolerance is phenotypic and does not necessarily involve the acquisition of new genetic resistance determinants, though existing ones are often retained [89] [69].

Table 1: Documented Antibiotic Tolerance of VBNC Cells in Various Bacterial Species

Bacterial Species	Induction Method	Antibiotic Challenge (vs. Culturable Cells)	Key Findings	Source
<i>Escherichia coli</i>	Low-level chlorination (0.5 mg/L)	Ampicillin (128x MIC), Ofloxacin (64x MIC)	VBNC cells persisted under extreme antibiotic concentrations that kill culturable cells.	[69]
<i>Vibrio vulnificus</i>	Environmental stress	Various antibiotics, heat, pH, ethanol, heavy metals	Exhibited higher resistance to a multitude of physical and chemical challenges.	[37]
Antibiotic-Resistant <i>E. coli</i> (RP4 plasmid)	UV/Chlorine (AOP)	Kanamycin, Tetracycline, Ampicillin	Retained antibiotic resistance determinants (ARGs) in the VBNC state and upon resuscitation.	[89]
<i>E. faecalis</i>	Unknown	Chlorine & Antibiotics	Showed greater tolerance to chlorine and antibiotics in the VBNC state.	[37]

Table 2: Retention of Antibiotic Resistance in Resuscitated VBNC Cells

Strain Type	Resistance Determinant	Status in VBNC State	Status after Resuscitation	Notes	Source
RIF <i>E. coli</i>	Chromosomal mutation (<i>rpoB</i> gene)	Retained	Retained	Resistance mediated by target site mutation remains stable.	[89]
RP4 <i>E. coli</i>	Plasmid-borne genes	Retained	Retained, sometimes enhanced	Horizontal gene transfer potential may be retained.	[89]
<i>E. coli</i> DH5α (MCR/CTX)	Plasmid-borne genes	Retained	Retained, with increased resistance in some cases	Resuscitated cells showed heterogeneous colony size but maintained resistance.	[67]

Underlying Mechanisms of Enhanced Tolerance

The extreme antibiotic tolerance of VBNC cells is not attributed to a single mechanism but is a multifactorial phenotype arising from their dormant physiology.

- **Metabolic Dormancy and Reduced Target Activity:** Most antibiotics target active cellular processes like cell wall synthesis, protein production, and DNA replication. VBNC cells have a drastically reduced metabolic rate and are not actively growing or dividing. This absence of target activity makes these drugs largely ineffective [48] [90]. The state represents a form of non-genetic **phenotypic plasticity** that grants broad tolerance [48].
- **Cell Envelope Modifications:** Bacteria undergo several physical changes to their cell envelope upon entering the VBNC state.
 - **Increased Peptidoglycan Cross-Linking:** Studies on *Enterococcus faecalis* have shown significantly higher levels of peptidoglycan cross-linking in VBNC cells, potentially strengthening the cell wall and reducing permeability [37] [14].
 - **Membrane Fatty Acid Alterations:** *Vibrio vulnificus* remodels its membrane in the VBNC state, increasing the proportion of unsaturated fatty acids. This can change membrane fluidity and integrity, potentially reducing the uptake of toxic compounds [37] [14].
- **Stress Response and Genetic Regulation:** The induction into the VBNC state triggers a robust stress response. Transcriptomic analysis of VBNC *E. coli* induced by low-level chlorination revealed significant upregulation of genes involved in oxidative stress response, transcriptional regulation, and those encoding for toxic proteins [69]. Key global regulators like **RpoS** (the stationary phase sigma factor) are involved in the formation and maintenance of the VBNC state, controlling a large regulon of stress defense genes [89] [37]. The **SOS response** and **stringent**

response pathways, which are also implicated in antibiotic resistance evolution, can trigger VBNC formation by reducing biosynthesis and altering cell physiology [89].

- **Efflux Pumps and Persistence:** Multidrug efflux pumps, which are often implicated in intrinsic and acquired antibiotic resistance, may also contribute to the VBNC state by accelerating intracellular nutrient depletion during stress induction [89]. Furthermore, the line between **persister cells** (a small, transiently tolerant subpopulation) and VBNC cells is blurring, with some researchers proposing they exist on a continuum of dormancy [48] [37]. Persisters are known to be enriched for activity of toxin-antitoxin (TAS) systems, which halt cell growth under stress, and these systems may also play a role in initiating and stabilizing the VBNC state [48].

The following diagram synthesizes these core concepts into a unified framework for understanding VBNC state induction, maintenance, and associated antibiotic tolerance mechanisms.

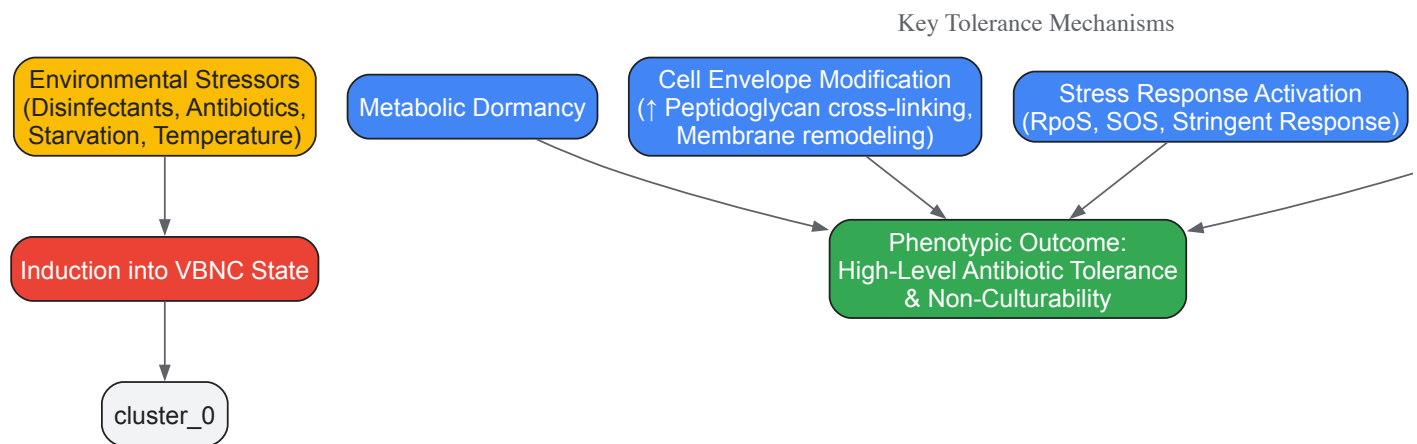


Figure 1: Integrated Framework of VBNC State Induction and Antibiotic Tolerance

Essential Research Protocols

Studying VBNC cells requires specialized methods that move beyond traditional culturing. Below are detailed protocols for key experiments in this field.

Protocol 1: Induction of the VBNC State by UV/Chlorine

This protocol is adapted from studies investigating the effect of advanced oxidation processes (AOPs) on antibiotic-resistant bacteria (ARB) [89].

- **Objective:** To induce the VBNC state in antibiotic-resistant and sensitive *E. coli* using a UV/chlorine disinfection system.
- **Materials & Strains:**
 - **Strains:** Antibiotic-sensitive *E. coli* K12, rifampicin-resistant *E. coli* K12 (RIF, chromosomal mutation), and *E. coli* K12 carrying the RP4 plasmid (conferring resistance to kanamycin, tetracycline, ampicillin).
 - **Culture Conditions:** Grow strains to mid-exponential phase in LB broth with appropriate antibiotics for resistant strains.
 - **Reaction Vessel:** 50-mL quartz flask reactor.
 - **Disinfectant:** Sodium hypochlorite (NaOCl) solution.
 - **UV Source:** 365 nm LED lamp with a calibrated fluence rate (e.g., 100 mW/cm²).
 - **Buffer:** Phosphate-buffered saline (PBS), pH 7.4.
- **Procedure:**
 - Harvest bacterial cells by centrifugation, wash twice, and resuspend in PBS to a density of ~10⁸ CFU/mL.
 - Add NaOCl to the bacterial suspension in the quartz reactor to a final concentration of 1-2 mg/L.
 - Immediately expose the mixture to UV light, stirring continuously to ensure homogeneity. A typical UV fluence used is 36 mJ/cm² (e.g., 3 minutes at a set irradiance).
 - Sample at regular time intervals (e.g., every 1-2 minutes) to monitor culturability and viability.
 - Neutralize any residual chlorine in the sample immediately using sodium thiosulfate.
- **Success Metric:** The VBNC state is confirmed when culturable cells on LB agar become undetectable (<0.1 CFU/mL), while a significant population of viable cells remains, as determined by viability assays.

Protocol 2: Differentiating Viable and Culturable Cell Counts

Accurately quantifying total viable cells (including VBNC) versus culturable cells is fundamental.

- **Objective:** To quantify the total viable cell count and the culturable cell count in a sample containing VBNC cells.
- **Materials:**

- **Viability Stains:** LIVE/DEAD *BacLight* Bacterial Viability Kit (SYTO 9 and Propidium Iodide) or similar.
- **Flow Cytometer** or Epifluorescence Microscope.
- **Culture Media:** Standard LB agar plates.
- **Microcentrifuge Tubes** and Pipettes.
- **Procedure:**
 - **Culturable Count (CFU/mL):**
 - Perform serial decimal dilutions of the bacterial sample in PBS.
 - Spread plate appropriate dilutions onto LB agar plates in duplicate.
 - Incubate plates at 37°C for 24-48 hours and count the resulting colonies.
 - **Total Viable Count (Cells/mL) via Flow Cytometry:**
 - Dilute the bacterial sample to a concentration of ~10⁵-10⁶ cells/mL.
 - Mix 1 mL of the diluted sample with a predefined ratio of SYTO 9 and PI stains as per manufacturer's instructions.
 - Incubate the mixture in the dark for 15-30 minutes.
 - Analyze using flow cytometry. SYTO 9 penetrates all bacteria (green fluorescence for intact membranes), while PI only penetrates cells with damaged membranes (red fluorescence, considered dead).
 - The population staining positive for SYTO 9 and negative for PI is quantified as the total viable cell count.
- **Data Interpretation:** The VBNC cell count is calculated as: **Total Viable Count (Flow Cytometry) - Culturable Count (CFU)**. A significant positive value indicates a VBNC population [89] [69].

Protocol 3: Transcriptomic Analysis of VBNC Cells

This protocol explains how to investigate the molecular mechanisms of antibiotic tolerance in VBNC cells.

- **Objective:** To compare global gene expression profiles of VBNC cells versus culturable cells using RNA sequencing (RNA-seq).
- **Materials:**
 - **Sample:** VBNC state cells and control culturable cells.
 - **RNA Stabilization:** RNAlater or similar reagent.
 - **RNA Extraction Kit:** For bacterial RNA, including a DNase digestion step.
 - **RNA Quality Control:** Bioanalyzer or similar instrument.
 - **Library Prep Kit:** Strand-specific RNA-seq library preparation kit.
 - **Sequencing Platform:** e.g., Illumina NovaSeq.
- **Procedure:**
 - **Sample Preparation:** Concentrate VBNC and control cells by centrifugation. Immediately stabilize the cell pellet in RNAlater to preserve RNA integrity.
 - **RNA Extraction:** Extract total RNA following the kit protocol. Ensure genomic DNA is thoroughly removed.
 - **Quality Control:** Assess RNA integrity (RIN > 8.0 is desirable).
 - **Library Preparation & Sequencing:** Deplete ribosomal RNA, construct cDNA libraries, and sequence on an appropriate platform to generate sufficient depth (e.g., 20 million reads per sample).
 - **Bioinformatic Analysis:**
 - Map reads to a reference genome.
 - Quantify gene expression (e.g., using FPKM or TPM).
 - Perform differential expression analysis (e.g., with DESeq2) to identify significantly up- or down-regulated genes in VBNC cells.
- **Key Analysis Focus:** Examine expression of genes related to:
 - **Antibiotic Resistance Genes (ARGs)**
 - **Stress Response** (e.g., *rpoS*, *oxyR*, *soxS*)
 - **Membrane Proteins & Efflux Pumps** (e.g., *omp* genes, *tolC*)
 - **Cellular Metabolism and Toxin-Antitoxin Systems** [89] [69].

The Scientist's Toolkit: Key Research Reagents

Table 3: Essential Reagents and Materials for VBNC Research

Reagent / Material	Function / Application	Example Use-Case
SYTO 9 / Propidium Iodide (PI)	Viability staining. Differentiates cells with intact (SYTO9+/PI-) vs. damaged membranes.	Flow cytometric quantification of total viable cells in a VBNC population [69].
5-Cyano-2,3-ditolyl-tetrazolium Chloride (CTC)	Detects respiratory activity. CTC is reduced to an insoluble fluorescent formazan in respiring cells.	Confirming metabolic activity in non-culturable cells [37].

Reagent / Material	Function / Application	Example Use-Case
RNA Later Stabilization Solution	Preserves RNA integrity in bacterial cells immediately upon sampling.	Prevents degradation of mRNA for accurate transcriptomic analysis of VBNC cells [69].
RiboZero rRNA Depletion Kit	Removes abundant ribosomal RNA prior to RNA-seq library construction.	Enriches for mRNA sequences, improving transcriptome coverage and data quality [69].
Sodium Thiosulfate	Neutralizes residual chlorine disinfectant in samples.	Stops the disinfectant action immediately after sampling during chlorine-based VBNC induction [89].
Resuscitation Promoting Factors (Rpfs)	Bacterial cytokine-like proteins that stimulate resuscitation from dormancy.	Used in experiments to trigger the recovery of VBNC cells into a culturable state [67].
qPCR Reagents & Probes	Quantifies specific gene expression (e.g., ARGs, stress genes) and 16S rRNA gene copy number.	Validating transcriptomic data and tracking persistence of resistance genes in VBNC cells [89] [67].

The comparative analysis unequivocally demonstrates that VBNC cells represent a high-risk reservoir for antibiotic tolerance in the environment. Their non-culturability leads to a critical diagnostic gap, while their inherent resistance mechanisms allow them to withstand both disinfectants and therapeutic antibiotics. The retention, and sometimes enhancement, of antibiotic resistance upon resuscitation underscores their role in the persistence and potential spread of antibiotic resistance [89] [67].

Future research must pivot towards bridging the knowledge-to-action gap. Key priorities include:

- **Developing Novel Detection Methods:** Moving beyond culture-based diagnostics to incorporate viability-based assays (e.g., flow cytometry, viability-PCR) into environmental and clinical surveillance frameworks [48] [14].
- **Targeting Resuscitation Pathways:** Investigating the molecular signals that trigger resuscitation could lead to compounds that either lock bacteria in the harmless VBNC state or force their resurrection in the presence of antibiotics, making them vulnerable again [67].
- **Exploring Anti-VBNC Therapeutics:** High-throughput screening for agents that can specifically kill VBNC cells by disrupting their unique physiology (e.g., membrane integrity in dormancy) is a promising avenue for drug development [14].

Addressing the challenge posed by VBNC cells requires a paradigm shift in how we perceive bacterial viability, assess microbial risk, and develop antibacterial strategies. A comprehensive "One Health" approach, integrating research from environmental, clinical, and molecular microbiology, is essential to mitigate the hidden threat they pose to public health.

Mathematical Modeling for Studying Antimicrobial Efficacy Against VBNC Cells

The **viable but non-culturable (VBNC) state** is a dormant survival strategy adopted by numerous bacterial pathogens when exposed to environmental stressors. In this state, cells undergo a transition where they **fail to grow on routine culture media** yet maintain metabolic activity and viability, presenting a significant challenge for public health and food safety [11]. Importantly, VBNC cells are not dead; they retain an **intact cell membrane**, continue **gene expression and protein synthesis**, and sustain **metabolic activity** including respiration and ATP production, but enter a state of low metabolic activity that renders them undetectable by conventional plating methods [11]. This phenomenon has been observed in over one hundred bacterial species, including major human pathogens such as *Campylobacter jejuni*, *Listeria monocytogenes*, *Salmonella enterica*, *Escherichia coli*, and *Vibrio cholerae* [72] [11].

The VBNC state poses a substantial threat to public health for several reasons. VBNC cells **cannot be detected by standard culture-dependent methods** mandated by food safety regulations, leading to potential false negatives in microbiological monitoring [72] [91]. These dormant cells exhibit **dramatically increased tolerance** to antimicrobial agents, including antibiotics and disinfectants, making them difficult to eradicate from clinical and food processing environments [91] [11]. Perhaps most concerning is that VBNC cells **maintain virulence potential** and can resuscitate under favorable conditions, potentially causing infections after evading detection [6] [11]. For example, VBNC *Campylobacter jejuni* maintains expression of virulence-associated genes and the ability to invade human intestinal epithelial cells [6]. These characteristics make the VBNC state a critical consideration in the broader context of bacterial pathogenesis and antimicrobial resistance research.

Detection Methodologies for VBNC Cells

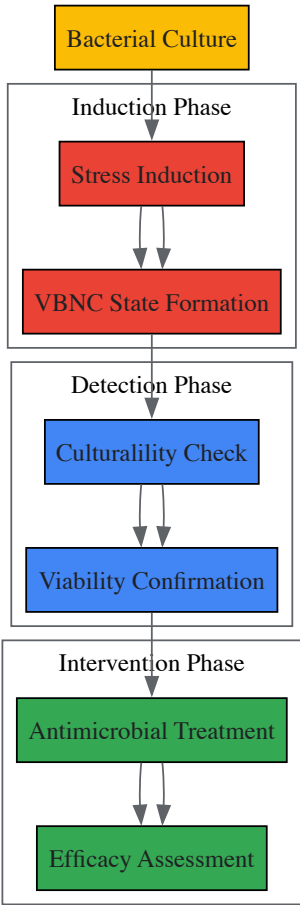
Limitations of Culture-Based Methods

Standard culture-based methods like plate counts are inadequate for detecting VBNC cells because these dormant cells will not form visible colonies on routine media within the standard incubation timeframe [91]. This limitation has significant implications for food safety and clinical diagnostics, as pathogens in the VBNC state may go undetected while still posing infection risks. The inability to culture VBNC cells also presents a substantial challenge for evaluating antimicrobial efficacy, as traditional metrics like Minimum Inhibitory Concentration (MIC) rely on observable growth inhibition [91].

Table 1: Detection Methods for VBNC Bacterial Cells

Method Category	Specific Technique	Key Principle	Applications in VBNC Research
Viability Staining	Live/Dead BacLight, CMA/PI, Fluorescence microscopy	Differential staining based on membrane integrity	Distinguishes viable (green) from dead (red) cells; provides visual confirmation [8] [92]
Molecular Methods with Viability Markers	PMA/EMA-qPCR or ddPCR	Dyes penetrate compromised membranes of dead cells and inhibit PCR amplification	Quantifies viable cells with intact membranes; applicable in complex matrices [8] [7] [92]
Molecular Methods with Viability Markers	PMAxx-EMA combined qPCR	Enhanced dye combination for better discrimination in complex samples	Differentiates dead and VBNC cells in process wash water; reduces false positives [8]
Molecular Methods with Viability Markers	DyeTox13 and DyeTox13+EMA with ddPCR	Novel dyes targeting metabolic activity with digital PCR quantification	Detects viable <i>Salmonella</i> in flour; higher sensitivity at low concentrations [92]
Metabolic Activity Assays	ATP measurement, Flow cytometry with metabolic probes	Detection of ongoing metabolic processes via ATP levels or substrate reduction	Confirms metabolic activity in VBNC cells; measures respiratory activity [7] [11]

The experimental workflow for detecting and studying VBNC cells typically follows a structured process as illustrated below:



Mathematical Modeling Approaches for Antimicrobial Efficacy

The Combination Index Model for Synergy Quantification

Mathematical modeling provides powerful tools for quantifying antimicrobial efficacy against VBNC cells, overcoming limitations of culture-based methods. The **Combination Index (CI) method**, derived from the Loewe additivity model, offers a particularly valuable approach for evaluating

synergistic interactions between antimicrobial agents [6] [93]. This model calculates whether drug combinations produce effects greater than expected from simple additive effects, using the equation:

$$CI = (D_1)/(D_x)_1 + (D_2)/(D_x)_2 + \alpha((D_1)(D_2))/((D_x)_1(D_x)_2)$$

Where (Dx)₁ and (Dx)₂ represent the doses of drug 1 and drug 2 alone required to produce x% effect, while D₁ and D₂ are the doses in combination that produce the same effect. The CI value provides a quantitative measure of antimicrobial interactions: **CI < 1 indicates synergy**, **CI = 1 indicates additive effects**, and **CI > 1 indicates antagonism** [6] [93].

This approach was successfully applied in a 2025 study investigating combinations of carvacrol, diallyl sulfide, and Al₂O₃ nanoparticles against VBNC *C. jejuni* [6] [93]. While time-kill assays showed only additive effects for all combinations, the CI model revealed **synergistic interactions (CI < 1) for binary and ternary combinations** containing Al₂O₃ nanoparticles, demonstrating the model's superior sensitivity for detecting synergy [93]. This highlights how mathematical modeling can uncover antimicrobial interactions that might be missed by traditional assays.

Time-Kill Assay Kinetics and Modeling

Time-kill assays provide dynamic data on bacterial reduction over time when exposed to antimicrobials, generating time-dependent curves that can be modeled mathematically. For VBNC cells, these assays measure the **reduction in viable cell counts** quantified using viability-based methods like PMA-ddPCR rather than plate counts [6] [92]. The resulting data can be fitted to various kinetic models, including:

- **First-order kinetics:** Logarithmic reduction patterns
- **Biphasic models:** Accounting for subpopulations with different susceptibilities
- **Non-linear regression models:** Capturing complex time-kill curves

These models help quantify key parameters such as **bactericidal rate constants** and **time to achieve specific reduction levels** (e.g., 99.9% reduction), providing standardized metrics for comparing antimicrobial efficacy against VBNC cells [6].

Model Selection Considerations and Limitations

When selecting mathematical models for studying antimicrobial efficacy against VBNC cells, researchers must consider several critical factors. Different stressors induce varied physiological states in VBNC cells, potentially affecting their susceptibility and requiring **strain-specific and stressor-specific model validation** [72]. The complex composition of food matrices (e.g., lipids, proteins) and environmental samples can protect bacteria and diminish antimicrobial effectiveness, necessitating **matrix-adapted models** that account for these protective effects [6]. Different detection methods (e.g., PMA-qPCR vs. DyeTox13-ddPCR) may yield varying counts of viable cells, requiring careful **calibration between quantification methods and model parameters** [8] [92].

Mathematical models in antimicrobial research face limitations including dependence on **quality and relevance of source data**, with models built from laboratory-grown bacteria potentially performing poorly when applied to environmental or clinical isolates [94]. All models face the challenge of **dissimilarity between experimental conditions and real-world applications**, particularly problematic when data from simple systems (e.g., water disinfection) is used to model complex environments (e.g., food processing plants or hospital rooms) [94].

Experimental Protocols for VBNC Antimicrobial Studies

VBNC Induction and Confirmation Protocol

Step 1: VBNC Induction

- Select appropriate stressors based on target pathogen and research context:
 - **Temperature stress:** Incubation at 4°C for 15+ days [6]
 - **Nutrient starvation:** Suspension in minimal media or phosphate-buffered saline [11]
 - **Oxidative stress:** Sublethal chlorine concentrations (e.g., 10 mg/L) [8]
 - **Osmotic stress:** NaCl solution incubation [6]
 - **Acid stress:** Exposure to mild acid solutions (e.g., pH 4.0-5.0) [6]

Step 2: Culturality Confirmation

- Perform regular plating on appropriate rich media (e.g., TSA, BHI agar)
- Use standard incubation conditions optimal for the target pathogen
- Confirm absence of colony formation over at least 2-3 times the normal incubation period
- Include positive controls (non-stressed cultures) to verify media suitability [8] [92]

Step 3: Viability Verification

- Apply viability staining (e.g., Live/Dead BacLight) with fluorescence microscopy

- Conduct metabolic activity assays (e.g., ATP measurement, CTC reduction)
- Perform viability-PCR with PMA/EMA treatment to confirm membrane integrity
- Verify presence of metabolic activity through respiratory activity measurements [\[8\]](#) [\[7\]](#) [\[92\]](#)

Time-Kill Assay Protocol for VBNC Cells

Step 1: Sample Preparation

- Prepare VBNC cell suspension in relevant matrix (e.g., process water, food homogenate)
- Adjust cell concentration to approximately 10⁵-10⁶ cells/mL using quantitative PCR methods
- Include controls: VBNC cells without antimicrobials, culturable cells with antimicrobials

Step 2: Antimicrobial Treatment

- Prepare stock solutions of individual antimicrobials and combinations
- Apply treatments at predetermined concentrations based on preliminary range-finding experiments
- Maintain appropriate contact conditions (temperature, mixing, pH)
- Include neutralizers in the protocol for chemical antimicrobials to enable precise timing

Step 3: Sampling and Quantification

- Collect samples at predetermined time points (e.g., 0, 15, 30, 60, 120 minutes)
- Immediately treat samples with appropriate dye if using viability-PCR
- Process samples through DNA extraction and qPCR/ddPCR analysis
- Analyze data to generate time-kill curves [\[6\]](#) [\[92\]](#)

Synergy Evaluation Protocol Using Combination Index

Step 1: Experimental Design

- Establish dose-effect relationships for individual antimicrobials
- Design combination ratios based on potency of individual components
- Include multiple effect levels (e.g., ED₅₀, ED₇₅, ED₉₀) for robust CI calculation

Step 2: Data Collection

- Treat VBNC cells with single agents and combinations across a concentration range
- Quantify viable cells using validated method (e.g., PMA-ddPCR)
- Calculate fraction affected (fa) at each concentration

Step 3: CI Calculation and Interpretation

- Apply CI equation for each combination at each effect level
- Classify interactions: CI < 0.9 (synergism), 0.9-1.1 (additive), >1.1 (antagonism)
- Perform statistical analysis on replicate experiments [\[6\]](#) [\[93\]](#)

Essential Research Reagent Solutions

Table 2: Key Research Reagents for VBNC Antimicrobial Studies

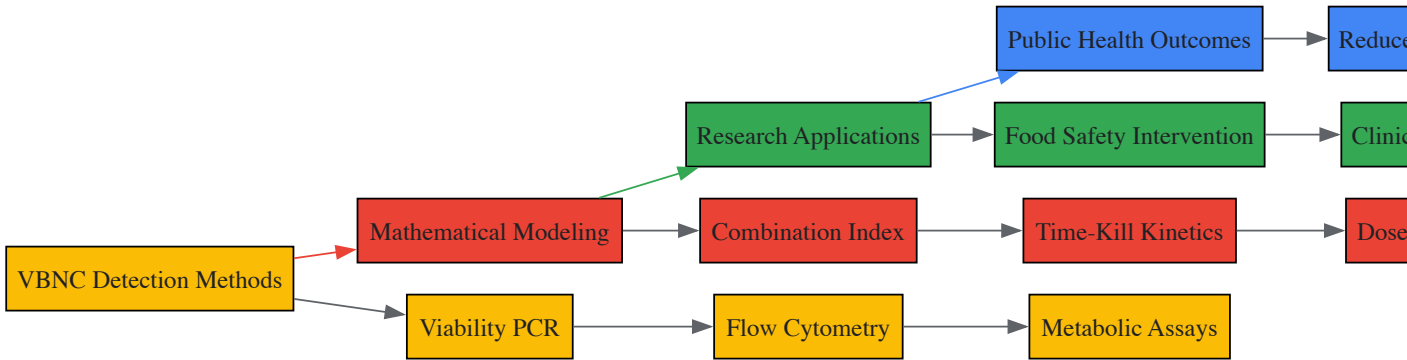
Reagent Category	Specific Examples	Function in VBNC Research	Application Notes
Viability Dyes	Propidium Monoazide (PMA), EMA, PMAx, DyeTox13	Penetrate dead cells with compromised membranes and inhibit DNA amplification	PMAx shows improved performance; DyeTox13 targets metabolic activity; concentration optimization required [8] [92]
Plant-Based Antimicrobials	Carvacrol, Diallyl Sulfide	Natural antimicrobial agents for combination studies	Require solvent (e.g., DMSO) for stock solutions; show variable efficacy against VBNC cells [6] [93]
Nanoparticles	Al ₂ O ₃ Nanoparticles (40-50 nm)	Potent antimicrobial nanomaterial against VBNC cells	Demonstrate relatively high potency alone and in combination; require aqueous stock suspensions [6] [93]
Cell Integrity Markers	SYTO 9, Propidium Iodide, CMFDA	Distinguish viable and dead cells based on membrane integrity	Used in fluorescence microscopy and flow cytometry; differentiates membrane-intact VBNC

Reagent Category	Specific Examples	Function in VBNC Research	Application Notes
			cells [8] [92]
Culture Media Components	Brain Heart Infusion (BHI), Tryptic Soy Broth (TSB)	Support growth of culturable cells for control experiments	Used for positive controls and resuscitation studies; unsuitable for VBNC cell growth [6] [92]
Neutralizing Agents	Sodium thiosulfate, Catalase, Histidine	Quench antimicrobial activity at precise time points	Essential for time-kill assays with chemical antimicrobials; validate neutralization efficacy [8] [7]

Applications and Research Implications

The integration of advanced detection methods with mathematical modeling creates powerful approaches for addressing the challenge of VBNC cells in various fields. In **food safety and quality control**, these methods enable more accurate assessment of disinfection efficacy in process wash water and on food contact surfaces, where VBNC pathogens may evade conventional monitoring [6] [8]. For **clinical microbiology and infectious disease management**, these approaches improve understanding of chronic and recurrent infections potentially linked to VBNC cells and enhance evaluation of antibiotic efficacy against dormant pathogens [91] [11]. In **antimicrobial drug development**, combination index models and time-kill kinetics against VBNC cells provide valuable data for designing effective treatments against persistent infections [6] [93].

The relationship between VBNC detection, modeling, and their applications can be visualized as an integrated research framework:



Mathematical modeling represents a transformative approach for studying antimicrobial efficacy against VBNC cells, overcoming fundamental limitations of traditional culture-based methods. The **Combination Index model** has demonstrated particular value for identifying synergistic antimicrobial combinations that would be missed by conventional assays [6] [93]. When integrated with advanced viability detection methods like **PMA-ddPCR and novel dyes**, these modeling approaches enable quantitative assessment of antimicrobial effects on non-growing bacterial populations that pose significant challenges in clinical, food safety, and environmental contexts [8] [92].

Future advancements in this field will likely focus on developing **standardized protocols** for VBNC antimicrobial testing that harmonize detection methods across laboratories [72]. There is also a critical need for **mechanistic models** that incorporate molecular understanding of VBNC induction and resuscitation pathways [91]. Additionally, **integrated models** that account for complex real-world conditions, such as food matrices and biofilm environments, will enhance practical applications of this research [6] [94]. As these modeling approaches mature, they will play an increasingly vital role in safeguarding public health against bacterial pathogens that evade conventional detection and control measures through entry into the viable but non-culturable state.

The **viable but non-culturable (VBNC) state** is a dormant survival strategy adopted by numerous bacterial pathogens when faced with environmental stressors. While cells in this state do not proliferate on conventional culture media—the standard method for detecting viable pathogens—they remain metabolically active, maintain membrane integrity, and can resuscitate when conditions improve [14]. This phenomenon poses a significant challenge to public health, clinical microbiology, and drug development, as VBNC cells **evade routine detection** yet retain **pathogenic potential**, contributing to persistent and recurrent infections [95] [72].

Understanding the true threat of VBNC pathogens necessitates direct assessment of their virulence using relevant biological models. This guide synthesizes current research on evaluating the virulence potential of VBNC bacteria, providing a technical framework for researchers and drug development professionals. We detail experimental protocols, present quantitative findings from animal and cellular models, and outline the essential toolkit for conducting these critical investigations, all within the broader context of understanding the hidden risks posed by this dormant state.

Virulence Assessment in Animal Models

Animal models provide the most holistic system for evaluating virulence, as they account for complex host-pathogen interactions, immune responses, and systemic pathology. The following models have been successfully used to demonstrate the pathogenicity of VBNC cells.

Mouse Model of Exacerbated Colitis

Background: This model is used to investigate how VBNC pathogens can influence the course of an underlying inflammatory bowel disease.

- **Organism:** VBNC *Salmonella* Enteritidis induced by chlorine treatment [74].
- **Animal:** Mice with dextran sulfate sodium (DSS)-induced colitis.
- **Experimental Groups:** Typically include:
 - DSS-only control
 - DSS + culturable *Salmonella*
 - DSS + VBNC *Salmonella*

Key Quantitative Findings: Table 1: Virulence Parameters of VBNC *S. Enteritidis* in a Murine Colitis Model

Parameter	Finding in VBNC-DSS Group vs. Controls	Implication
Clinical Severity	Exacerbated colitis severity	VBNC state does not abolish pathogenicity
Colonic Shortening	Marked reduction in colon length	Indicator of severe inflammation
Pro-inflammatory Cytokines	Elevated levels of LPS, TNF-α, and IL-6	Induction of a potent immune response
Intestinal Barrier Function	Reduced goblet cells, diminished mucus layer, downregulated MUC2 and occludin	Compromised gut barrier integrity
Gut Microbiota	Significantly elevated Firmicutes/Bacteroidota (F/B) ratio	Shift towards a dysbiotic, pro-inflammatory state

Interpretation: The data demonstrate that VBNC *Salmonella* Enteritidis retains a significant capacity to worsen inflammatory disease, compromising the intestinal barrier and driving pro-inflammatory immune responses, thereby acting as a co-factor in disease pathogenesis [74].

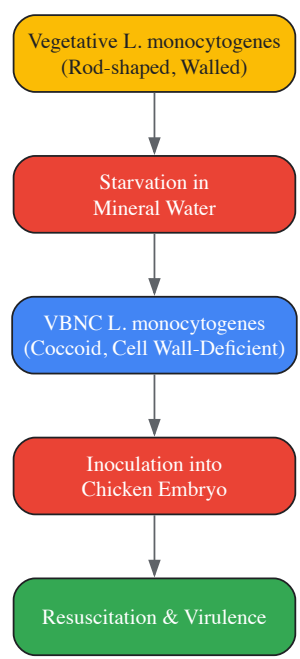
Background: The chicken embryo model is a valuable tool for studying the resuscitation of VBNC cells and their subsequent return to a virulent state in a complex, living host environment.

- **Organism:** VBNC *Listeria monocytogenes* induced by starvation in mineral water [96].
- **Model System:** Embryonated chicken eggs.

Protocol:

- **Preparation:** Incubate *L. monocytogenes* in a low-nutrient mineral water for approximately 28 days until >99% of cells are non-culturable but viable, confirmed by flow cytometry [96].
- **Inoculation:** Introduce a suspension of VBNC *L. monocytogenes* into the yolk sac or allantoic cavity of embryonated chicken eggs.
- **Incubation & Monitoring:** Re-incubate eggs and monitor for signs of infection and embryo mortality.
- **Recovery and Analysis:** Re-isolate bacteria from deceased embryos onto culture media to confirm resuscitation from the VBNC state to a culturable, vegetative form.

Key Findings: The VBNC *L. monocytogenes*, which had adopted a cell wall-deficient coccoid form during starvation, successfully resuscitated in the chicken embryo model. The resuscitated bacteria were not only culturable but also fully virulent, causing lethal infection in the embryos. This confirms that the VBNC state is a reversible dormancy and that pathogenicity is restored upon resuscitation [96].



Virulence Assessment in Cellular Models

Cellular models offer a controlled environment to dissect specific mechanisms of pathogenicity, such as invasion, intracellular survival, and cytotoxicity.

Invasion of Human Intestinal Epithelial Cells

Background: This assay tests the fundamental ability of VBNC pathogens to invade host cells, a key step in the infection process for many facultative intracellular bacteria.

- **Cell Line:** Human intestinal epithelial cells (e.g., Caco-2, HT-29) [6].
- **Organism:** VBNC *Campylobacter jejuni*.

Detailed Protocol:

- **VBNC Induction:** Induce the VBNC state in *C. jejuni* by incubation in a microcosm water at 4°C for 15 days or exposure to acid solution for 2 hours [6]. Confirm loss of culturability by plating.
- **Cell Culture Preparation:** Grow human intestinal epithelial cells to confluence in 24-well tissue culture plates.
- **Infection:** Add a suspension of VBNC *C. jejuni* to the cells at a predetermined Multiplicity of Infection (MOI). Centrifuge plates briefly (e.g., 500 x g for 5 min) to synchronize infection.
- **Incubation:** Incubate the infected cells for 1-3 hours at 37°C under appropriate CO₂ conditions to allow for invasion.
- **Antibiotic Protection:** Remove the extracellular medium and wash the cell monolayers gently with phosphate-buffered saline (PBS). Add fresh medium containing gentamicin (or another cell-impermeant antibiotic) to kill any remaining extracellular bacteria.
- **Lysis and Enumeration:** After 1-2 hours of antibiotic treatment, lyse the eukaryotic cells with a detergent like Triton X-100 or sterile water. Serially dilute the lysate and plate it onto appropriate culture media to enumerate the resuscitated, intracellular bacteria that were protected from the antibiotic [6].

Key Findings: VBNC *C. jejuni* cells maintain the ability to invade human intestinal epithelial cells. Furthermore, molecular analyses have shown that these VBNC cells continue to express critical virulence-associated genes (*flaA*, *flaB*, *cadF*, *ciaB*, *cdtA*, *cdtB*, *cdtC*), albeit at low levels, explaining their retained capacity to initiate infection [6].

The Scientist's Toolkit: Essential Reagents and Methods

Research into the virulence of VBNC bacteria relies on specialized reagents and methodologies for induction, detection, and quantification.

Table 2: Key Research Reagent Solutions for VBNC Virulence Studies

Reagent / Method	Function & Application	Specific Examples
Propidium Monoazide (PMA)	DNA intercalating dye that penetrates only membrane-compromised (dead) cells. Used with qPCR/ddPCR to selectively amplify DNA from viable (membrane-intact) cells, allowing quantification of VBNC populations. [10] [16]	Differentiation between viable and dead cells in a sample prior to molecular analysis.
Droplet Digital PCR (ddPCR)	Absolute quantification of gene copy number without a standard curve. Provides highly precise enumeration of VBNC cells when combined with PMA treatment (PMA-ddPCR). [10] [16]	Absolute quantification of viable <i>Klebsiella pneumoniae</i> cells in mouse fecal samples using single-copy genes (<i>rpoB</i> , <i>adhE</i>).
Viable qPCR (vqPCR)	A qPCR-based method that uses DNA intercalating dyes (like PMA or "Reagent D") and long gene target amplification to specifically detect viable cells, including those in the VBNC state. [21]	Detection of VBNC <i>Vibrio parahaemolyticus</i> and <i>V. cholerae</i> in seafood via <i>groEL</i> (510 bp) and <i>ompW</i> (588 bp) genes.
Flow Cytometry with Vital Stains	High-throughput method to count and characterize viable cells based on membrane integrity and enzymatic activity.	Using carboxyfluorescein diacetate (CFDA) to identify metabolically active <i>Listeria monocytogenes</i> with intact membranes. [96]
VBNC Induction Agents	Chemical or environmental stressors used to induce the VBNC state in vitro for experimental study.	Chlorine (for <i>Salmonella</i>), Lutensol AO3/Ammonium Carbonate (for <i>Vibrio</i>), starvation in mineral water (for <i>Listeria</i>). [74] [21] [96]

The collective evidence from robust animal and cellular models definitively shows that the VBNC state is not a benign persistence mechanism but a **potent threat**. VBNC pathogens retain the functional capacity to resuscitate, invade host tissues, provoke significant inflammatory responses, and cause disease. The use of advanced molecular detection methods like PMA-ddPCR and vqPCR is crucial to uncover this hidden reservoir of virulence, which standard culture methods inevitably miss. For researchers and drug development professionals, acknowledging and actively investigating the VBNC state is paramount for developing more effective diagnostic tools and therapeutic strategies to combat persistent and recurring bacterial infections.

Conclusion

The VBNC state represents a critical, yet often overlooked, survival mechanism that enables bacterial pathogens to evade conventional detection methods and antimicrobial treatments, posing a significant threat to public health and challenging current therapeutic paradigms. The synthesis of knowledge across the four intents confirms that VBNC cells are not dormant in a passive sense but are a distinct physiological state with active molecular regulation, retained pathogenic potential, and enhanced resistance. Future directions for biomedical and clinical research must focus on: 1) Developing standardized, reliable, and commercially viable detection kits for clinical diagnostics; 2) Unraveling the precise genetic switches that control entry and exit from the VBNC state to identify novel drug targets; and 3) Formulating therapeutic strategies that either force irreversible eradication of VBNC cells or prevent their resuscitation. Addressing these challenges is paramount for mitigating the hidden risks of chronic, recurrent infections and for overcoming a significant barrier in the fight against antimicrobial resistance.

References

1. Viable but nonculturable [https://en.wikipedia.org/wiki/Viable_but_nonculturable]
2. Viable but Nonculturable Bacteria: Food Safety and Public ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC3804398/]
3. The Viable but Non-Culturable (VBNC) State, a Poorly ... [https://www.mdpi.com/2076-2607/12/1/39]
4. Viable but nonculturable (VBNC) state, an underestimated ... [https://www.sciencedirect.com/science/article/abs/pii/S0966842X23001476]
5. Current Perspectives on Viable but Non-culturable State in ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC5378802/]
6. A novel mathematical model for studying antimicrobial ... [https://www.sciencedirect.com/science/article/abs/pii/S0740002025000206]
7. Quantitative Detection of VBNC State *Pseudomonas* ... [https://www.mdpi.com/2073-4441/16/2/236]
8. Detection and Quantification Methods for Viable but Non- ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC7214806/]
9. Absolute quantification of VBNC state formation and ... [https://pubmed.ncbi.nlm.nih.gov/41122686/]
10. Absolute quantification of VBNC state formation and ... [https://www.frontiersin.org/journals/cellular-and-infection-microbiology/articles/10.3389/fcimb.2025.1627943/full]
11. The importance of the viable but non-culturable state in ... [https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2014.00258/full]
12. Current Perspectives on Viable but Non-Culturable (VBNC) ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4116801/]

- 13. Exploring the impact of stress factors in inducing the viable ...
[<https://www.sciencedirect.com/science/article/abs/pii/S2212429225007783>]
- 14. Viable but non-cultivable state in oral microbiota [https://www.frontiersin.org/journals/cellular-and-infection-microbiology/articles/10.3389/fcimb.2025.1533768/full]
- 15. Induction of Viable but Non-culturable (VBNC) State in ... [https://brieflands.com/journals/jjm/articles/147096]
- 16. Absolute quantification of VBNC state formation and ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12535994/]
- 17. Detection and Quantification Methods for Viable but Non ...
[https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2020.00673/full]
- 18. Overview of VBNC, a survival strategy for microorganisms [https://pmc.ncbi.nlm.nih.gov/articles/PMC9526772/]
- 19. Wake Up! Resuscitation of Viable but Nonculturable Bacteria [https://pmc.ncbi.nlm.nih.gov/articles/PMC9818539/]
- 20. Revealing the mechanism of citral induced entry of *Vibrio* ...
[https://www.sciencedirect.com/science/article/abs/pii/S0168160524001004]
- 21. Rapid Detection and Fast Induction of Viable but ... [https://www.sciencedirect.com/science/article/pii/S0362028X25001759]
- 22. Viable but Nonculturable (VBNC) Bacteria [https://www.sigmaaldrich.com/US/en/technical-documents/technical-article/microbiological-testing/microbial-culture-media-preparation/viable-but-nonculturable-bacteria?srsltid=AfmBOooc7zT5ITvpE1UtI7u1zdG7ycTrtLuwK2MRgeWTfVVFmJpoCxV]
- 23. The Effect of the Stringent Response and Oxidative Stress ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC10932037/]
- 24. VBNC *Cronobacter sakazakii* survives in macrophages by ... [https://bmcmicrobiol.biomedcentral.com/articles/10.1186/s12866-024-03595-9]
- 25. Type II toxin-antitoxin systems as stress-responsive ... [https://pubmed.ncbi.nlm.nih.gov/40970936/]
- 26. Stress can induce the transcription of toxin-antitoxin systems ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC7368831/]
- 27. The effect of *yeaZ* on formation and resuscitation of viable ...
[https://www.sciencedirect.com/science/article/abs/pii/S0740002025001686]
- 28. UV-Induced VBNC state formation and resuscitation in *E. coli*
[https://www.sciencedirect.com/science/article/abs/pii/S0043135425012862]
- 29. Gene Expression Profiling in Viable but Nonculturable ...
[https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2015.01419/full]
- 30. Viable but Nonculturable and Persister Cells Coexist ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4598401/]
- 31. The Viable but Non-Culturable (VBNC) State, a Poorly ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC10818521/]
- 32. Relationship between the Viable but Nonculturable State ... [https://pubmed.ncbi.nlm.nih.gov/30082460/]
- 33. Viable but non-cultivable state in oral microbiota [https://pubmed.ncbi.nlm.nih.gov/40171166/]
- 34. Detection and Quantification of Viable but Non-culturable ...
[https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2019.02920/full]
- 35. Survival strategies and transition mechanisms of ... [https://www.sciencedirect.com/science/article/abs/pii/S2213343725034098]
- 36. Darkness to Discovery: A Comprehensive Mini-Review on ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC12274225/]
- 37. The importance of the viable but non-culturable state in ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4040921/]
- 38. Current Perspectives on Viable but Non-Culturable ... [https://www.mdpi.com/2304-8158/12/6/1179]
- 39. Induction of the viable but non-culturable state in bacterial ... [https://www.nature.com/articles/s41598-018-33595-5]
- 40. Current Perspectives on Viable but Non-Culturable (VBNC) ... [https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2014.00103/full]
- 41. Current Perspectives on Viable but Non-culturable State in ...
[https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2017.00580/full]
- 42. Quantitative detection of viable but nonculturable state ... [https://www.sciencedirect.com/science/article/abs/pii/S0956713520300566]
- 43. A Novel, Rapid, and Simple PMA-qPCR Method for Detection ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC7305652/]
- 44. Development of a viability digital PCR protocol for the ... [https://www.nature.com/articles/s41598-019-47976-x]
- 45. Propidium monoazide-quantitative polymerase chain ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC6582174/]
- 46. Enumeration of Viable Non-Culturable *Vibrio cholerae* ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC8592976/]
- 47. Absolute Quantification of Viable but Nonculturable *Vibrio* ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC9430983/]
- 48. How to Evaluate Non-Growing Cells—Current Strategies ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC7912045/]
- 49. VBNC bacteria: challenging detection (and a breakthrough ... [https://www.bnovate.com/post/vbnc-bacteria]
- 50. Flow Cytometry Multiplexing Used to Analyze Metabolic ... [https://www.thermofisher.com/us/en/home/life-science/cell-analysis/flow-cytometry/flow-cytometry-learning-center/flow-cytometry-resource-library/flow-cytometry-application-notes/flow-cytometry-multiplexing-used-analyze-metabolic-activity-assess-pharmaceutical-compound-toxicity-as-high-throughput-screening-tool.html]
- 51. Flow Cytometric Assessment of Membrane Integrity ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC134380/]
- 52. Population and single cell metabolic activity of UV-induced ... [https://www.sciencedirect.com/science/article/pii/S0160412019307998]
- 53. Flow Cytometry Protocol | Abcam [https://www.abcam.com/en-us/technical-resources/protocols/flow-cytometry-for-intracellular-and-extracellular-targets]
- 54. Gene Expression Profiling in Viable but Nonculturable (VBNC) ... [https://pmc.ncbi.nlm.nih.gov/articles/PMC4683178/]

- 55. Transcriptome analysis and prediction of the metabolic ... [<https://www.nature.com/articles/s41598-022-21102-w>]
- 56. Time-resolved proteomic profiling of Cupriavidus ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC11886801/>]
- 57. Bacterial culturability and the viable but non- ... [<https://www.sciencedirect.com/science/article/abs/pii/S0038071717306776>]
- 58. Proteomics and metabolic profiling of Vibrio ... [<https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2025.1689231/full>]
- 59. Viable but nonculturable (VBNC) state, an underestimated and controversial microbial survival strategy - PubMed [<https://pubmed.ncbi.nlm.nih.gov/37225640/>]
- 60. Recent Methods for the Viability Assessment of Bacterial ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC9500772/>]
- 61. The viable but nonculturable state and starvation are different stress responses of Enterococcus faecalis, as determined by proteome analysis - PubMed [<https://pubmed.ncbi.nlm.nih.gov/12426365/>]
- 62. Precision Identification of Diverse Bloodstream Pathogens in ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC6289251/>]
- 63. Longitudinal gut microbiota tracking reveals the dynamics ... [<https://www.nature.com/articles/s41467-025-66612-z>]
- 64. Induction and Resuscitation of the Viable but Non ... [<https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2019.01081/full>]
- 65. Chicken Embryo as an In Vivo Model to Revive Viable but ... [<https://pubmed.ncbi.nlm.nih.gov/40522866/>]
- 66. Resuscitation of viable but nonculturable bacteria ... [<https://www.sciencedirect.com/science/article/pii/S2090123223002060>]
- 67. Regulatory formation of VBNC state antibiotic-resistant ... [<https://www.sciencedirect.com/science/article/abs/pii/S0304389425029607>]
- 68. Disinfection-induced viable but non-culturable waterborne ... [<https://www.sciencedirect.com/science/article/abs/pii/S0304389423019490>]
- 69. Characterization and potential mechanisms of highly ... [<https://www.nature.com/articles/s41598-020-58106-3>]
- 70. A Combination of UV and Disinfectant for Inactivating ... [<https://www.mdpi.com/2073-4441/16/9/1302>]
- 71. Disinfectant-induced bacterial resistance and antibiotic cross ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC12628408/>]
- 72. Latest Developments of Research on the Viable Non ... [<https://www.mdpi.com/2076-3417/15/3/1454>]
- 73. Label-Free Flow Cytometry: A Powerful Tool to Rapidly and ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC12114458/>]
- 74. Exploring the effect of viable but non-culturable (VBNC) ... [<https://www.sciencedirect.com/science/article/pii/S277256692500240X>]
- 75. Mechanism insights into the regulation of the LuxS/AI-2 ... [<https://www.sciencedirect.com/science/article/pii/S2090123225006939>]
- 76. Investigating the physiology of viable but non-culturable ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC5738893/>]
- 77. Viable But Non-Culturable (VBNC) Bacteria: Threats and ... [<https://krakensense.com/blog/viable-but-non-culturable-vbnc-bacteria-threats-and-solutions>]
- 78. Formation and Control of the Viable but Non-culturable State ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC7308729/>]
- 79. Development of qPCR-PMAxx™ and ddPCR- ... [<https://www.sciencedirect.com/science/article/abs/pii/S0740002025002497>]
- 80. Waterborne pathogens detection technologies - PubMed Central [<https://pmc.ncbi.nlm.nih.gov/articles/PMC10708915/>]
- 81. Emerging analytical techniques for determination of ... [<https://link.springer.com/article/10.1007/s44187-025-00590-5>]
- 82. Role of viable but non culturable cells in patients with cystic ... [<https://www.sciencedirect.com/science/article/pii/S1569199324000262>]
- 83. Occurrence, predictive models and potential health risk ... [<https://www.sciencedirect.com/science/article/abs/pii/S0269749125001678>]
- 84. Detection of Viable but Nonculturable E. coli Induced by ... [<https://www.sciencedirect.com/science/article/pii/S0362028X2400214X>]
- 85. Induction of Viable but Nonculturable Escherichia coli ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC6282054/>]
- 86. Viable but Nonculturable Escherichia coli O157:H7 and ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC7082562/>]
- 87. Physiological Characterization of Viable-but-Nonculturable ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC91151/>]
- 88. The viable but non-culturable (VBNC) state of Campylobacter ... [https://sciety-labs.elifesciences.org/articles/by?article_doi=10.1101/2025.07.25.666711]
- 89. Viable but non-culturable state formation and resuscitation ... [<https://www.sciencedirect.com/science/article/abs/pii/S0043135424009114>]
- 90. An overview of the antimicrobial resistance mechanisms ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC6604941/>]
- 91. How to Evaluate Non-Growing Cells—Current Strategies ... [<https://www.mdpi.com/2079-6382/10/2/115>]
- 92. Quantitative detection and survival analysis of VBNC ... [<https://pmc.ncbi.nlm.nih.gov/articles/PMC11302299/>]
- 93. A novel mathematical model for studying antimicrobial ... [<https://pubmed.ncbi.nlm.nih.gov/39952754/>]
- 94. Mathematical Modeling of Antimicrobial Efficacy [<https://microchemlab.com/information/mathematical-modeling-antimicrobial-efficacy/>]
- 95. Significance of Viable But Non-culturable (VBNC) State in ... [<https://link.springer.com/article/10.1007/s00284-024-03947-8>]
- 96. Aquatic environment drives the emergence of cell wall ... [<https://www.nature.com/articles/s41467-024-52633-7>]